

SERIES 86

Research Analyst Qualification Examination (Part 1) Study Manual – 43rd Edition

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About the Series 86/87 Examination

The Series 86/87 Examination is a five hour and 30-minute examination which is divided into two parts. Part I (Series 86) is made up of 100 multiple-choice questions and covers fundamental security analysis and the valuation of equity securities. Part II (Series 87) consists of 50 multiple-choice questions and covers SRO and SEC rules and regulations. Candidates are given four and 30 minutes (270 minutes) to complete Part I and one hour and 45 minutes (105 minutes) to complete Part II. The Series 86 exam includes 10 additional (experimental) questions, while the Series 87 exam includes 5 additional (experimental) questions. Therefore, the Series 86 exam consists of a total of 110 questions (100 scored and 10 unscored) and the Series 87 exam consists of a total of 55 items (50 scored and 5 unscored).

Since multiple forms of the examination will be administered, the passing score will fluctuate moderately from exam to exam. (Candidates have reported a passing score of 73% on the Series 86, and 74% on the Series 87.) If a candidate passes only one part of the test, she will only be required to retake that failed section. **We recommend that you visit our website at www.stcusa.com to determine whether there have been any changes or supplemental materials created for this exam.**

The Series 86/87 Examination is designed as a qualifying examination for a Research Analyst who will be responsible for the preparation of the substance of a research report and whose name appears on the report.

In order to obtain registration as a Research Analyst, a candidate must pass both the Series 86/87 Exams and a general knowledge co-requisite—the Securities Industry Essentials (SIE) Exam. A candidate who has passed both Levels I and II of the Chartered Financial Analyst (CFA) examination may request an exemption from the Series 86 (Part I).

Topic Areas	Number of Questions
Part I – Series 86	
1. Information and Data Collection	10
2. Analysis, Modeling, and Valuation	90
Part II – Series 87	
3. Preparation of Research Reports	32
4. Dissemination of Information	18

Registering for the Examination

To register for the Series 86/87 Examinations, a person must complete FINRA's Form U4 and pay an examination fee. A Form U4 may be obtained directly from FINRA. Candidates may schedule both Series 86 and Series 87 on the same day or take them on different days. The Series 86/87 Examinations are administered on a computerized testing system.

Taking the Examination

In order to take an exam, a person will need to make a test appointment. Please use the following information to schedule an appointment and/or to learn more about the examination center:

Prometric Exam Centers
www.prometric.com/finra/
800.578.6273

The website will provide the most up-to-date information regarding “Test Center Security” and “Test Break Policies.” At the testing facility, a person will be provided with:

- A four-function calculator
- Two dry-erase boards
- Dry-erase pen

For more information regarding scheduling an exam, what to expect on the day of your exam, and what happens after your exam, please use the following link which provides information from FINRA:

www.finra.org/sites/default/files/external_apps/proctor_tutorial.swf.html

Study Manual

The study manual represents the first phase of exam preparation. Both the Series 86 and Series 87 Study Manuals each consists of five chapters.

Contained within most chapters are review questions that will test a person’s understanding of the material presented. The review questions should be taken as closed-book tests. Explanations for each question is provided at the end of the questions. A person should not proceed to the next chapter until she fully understands the explanation for any question that was answered incorrectly.

Final Examinations

The final examinations and corresponding explanations represent the most important part of your test preparation. These examinations will assist you in applying the information that you learned in the study manual to questions that are posed in the multiple-choice format and used in the Series 86/87 Examinations.

An examination should first be taken with the SHOW EXPLANATIONS turned on. As you read a question, try to answer it. However, whether your answer is correct or incorrect, read the entire explanation. You may find it helpful to highlight or take notes on any facts you didn’t know for use in future studying.

Studying each explanation is a crucial step to passing the Series 86/87 Examinations. By concentrating only on the correct response and disregarding the explanation, you run the risk of memorizing answers without fully understanding the underlying concepts.

After completing all of the examinations with SHOW EXPLANATIONS switched on, begin the process over again by retaking each examination without the explanations shown. When taking the test for the second time, you should strive to achieve a score of 85% or better to show maximum retention of the material.

Standardized Test-Taking Tips

As with any standardized test, you may be able to increase your score by employing good test-taking techniques. An efficient technique will ensure an overall understanding of the question while helping to avoid careless errors. It will also help you to stay alert throughout the entire exam. Here's a step-by-step approach that should work well for you.

Step 1: Read the question the first time through without trying to answer it. At this point, merely form an understanding of the substance of the question.

Step 2: Carefully read the four choices. Remember, since a multiple-choice examination actually gives you the answer, your job is to recognize the correct choice from among the distractors. By keeping these choices in mind when you reread the question, you will be able to pinpoint the important information and filter out any extraneous material.

Step 3: Reread the question slowly and stop at the end of each sentence to absorb the information. You may need to exaggerate this in the beginning as you get used to applying the intense concentration required to assure that you recognize all of the important facts and catch the misleading words or phrases (e.g., not, except, etc.).

Step 4: Make sure that you fully understand what the question is asking. You cannot possibly answer a question correctly before you know what it's asking. You may need to look back to the question for additional facts before you are ready to choose your answer.

Step 5: Read each choice a second time. As you read a choice, decide whether it's a possible answer. If you're able to eliminate three of the four choices, then you have your answer. However, if you only eliminate one or two, it will still help you narrow it down to your best choices. Reconsider the remaining choices by comparing their differences and decide which answer is more correct. Once you have made your decision, DON'T LOOK BACK!

NOTE: For Roman numeral questions, eliminate any clearly incorrect Roman numeral choices before trying to answer the question. Typically, Roman numeral type questions will include only two or four correct choices.

It's very important that you practice your technique so that you become proficient by the time you sit for the examination. The best place to start is in the chapter quizzes. Not only will this practice build your technique but it will also help you to identify any problem areas you may have.

Test-Taking Pitfalls

Reading the Question Too Fast Part of the battle in attempting to pass a standardized examination is determining what a question is asking. Regardless of how many times you read a question, you cannot absorb the information if you read the question too quickly.

Changing Answers Going back and changing an answer means that you are second-guessing yourself. If you employ a good test-taking strategy, there's nothing you will gain from going back to a question for a second time. If you think you are likely to obtain the information you need from another question, remember that this examination is written by expert test writers who are not going to give anything away.

Formulating an Answer Too Quickly When you are ready to answer a question, make sure to consider all four of the choices that are given. Don't formulate an answer on your own and merely look for that choice while disregarding the others. Remember, there will often be more than one correct choice and, while your choice may be right, it may not be the best response.

Making Careless Errors Don't form preconceived notions when reading a test question. Always read what's written, not what you expect to see. By keeping an open mind, reading the questions slowly, and reading them at least two times through, you should be able to avoid these types of errors.

Study Calendars

Following this Introduction, STC has included suggested study calendars. These study calendars are designed to help students in organizing their time and allowing for a manageable amount of daily study. View each calendar and choose the one that best fits your needs. Remember, these calendars are simply suggestions for how you may plan your studies. Feel free to make any modifications that you deem appropriate.

Please Note: When reviewing the online PDF Study Manual, a **keyword search** can be performed by holding down the "Control" key and clicking on the "F" key (Ctrl + F) or, for Mac users, the "Command" key and "F" key (Cmd + F).

Series 86 – *Three-Week Training*



To complete each chapter:

1. Watch the On-Demand Lecture for the chapter.
2. Read the chapter in the Study Manual
3. Build a 10-question Custom Exam for the chapter *
4. Create a custom Flashcard deck for the chapter

* To create a Custom Exam:

Log in to *my.stcusa.com*. From your Dashboard, select *Final Exams*, then scroll down and select *Create a Custom Exam*. Now, select the appropriate chapter number and, at the bottom of the screen, enter 10 in the *Number of Questions* box, and then select *Build Exam*. You can choose whether or not to have the explanation appear after each question is answered.

For the Final Exams: An examination should first be taken with the SHOW EXPLANATIONS turned ON. After completing all of the examinations with SHOW EXPLANATIONS switched on (and if time permits), begin the process over again by retaking each examination without the explanations shown (Explanations OFF).

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
WEEK 1	Complete Chapter 1	Complete Chapter 2	Complete Chapter 3 Take Progress Exams 1A and 1B (Review incorrect answers)	Complete Chapter 4 Part 1	Complete Chapter 4 Part 2
WEEK 2	Complete Chapter 5 Take Progress Exams 2A and 2B (Review incorrect answers)	Take Final Exam 1	Take Final Exam 2	Take Final Exam 3	Take Greenlight Exam 1 (Review incorrect answers from Final Exams and Greenlight 1)
WEEK 3	Take Final Exam 4	Take Final Exam 5	Take Greenlight Exam 2 (Review incorrect answers from Final Exams and Greenlight 2)	Review Weak Areas Identified by Diagnostics	Review Lecture Notes Review Crunch Time Facts

PASS THE SERIES 86!

Series 86 – *Four-Week Training*



To complete each chapter:

1. Watch the On-Demand Lecture for the chapter.
2. Read the chapter in the Study Manual
3. Build a 10-question Custom Exam for the chapter *
4. Create a custom Flashcard deck for the chapter

* To create a Custom Exam:

Log in to *my.stcusa.com*. From your Dashboard, select *Final Exams*, then scroll down and select *Create a Custom Exam*. Now, select the appropriate chapter number and, at the bottom of the screen, enter 10 in the *Number of Questions* box, and then select *Build Exam*. You can choose whether or not to have the explanation appear after each question is answered.

For the Final Exams: An examination should first be taken with the SHOW EXPLANATIONS turned ON. After completing all of the examinations with SHOW EXPLANATIONS switched on (and if time permits), begin the process over again by retaking each examination without the explanations shown (Explanations OFF).

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
WEEK 1	Complete Chapter 1	Complete Chapter 2	Start Chapter 3	Complete Chapter 3 Take Progress Exams 1A and 1B (Review incorrect answers)	Complete Chapter 4 Part 1
WEEK 2	Complete Chapter 4 Part 2	Complete Chapter 5 Take Progress Exams 2A and 2B (Review incorrect answers)	Take Final Exam 1	Take Final Exam 2	Take Final Exam 3

Series 86 – *Four-Week Training*



	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
WEEK 3	Take Greenlight Exam 1 (Review incorrect answers from Final Exams and Greenlight 1)	Take Final Exam 4	Take Final Exam 5	Take Greenlight Exam 2 (Review incorrect answers from Final Exams and Greenlight 2)	Retake Final Exam 1
WEEK 4	Retake Final Exam 2	Retake Final Exam 3	Retake Final Exam 4	Retake Final Exam 5 Review Weak Areas Identified by Diagnostics	Review Lecture Notes Review Crunch Time Facts

PASS THE SERIES 86!

CHAPTER 1

Introduction to Economics



Macroeconomic Information

The content of this chapter follows a top-down approach. First, we will look at macroeconomic measurements of the overall economy, and the attempts to predict future patterns based on the analysis of various economic indicators. This will be followed by a review of the major theories of how to influence the direction of the economy, focused primarily on the role of the Federal Reserve Board (FRB) in conducting monetary policy.

Adjusting our focus from macro to microeconomics, we will examine the key factors involved in industry analysis. We will then introduce the analysis of individual firms, a topic that will be continued throughout the rest of this text. The end of this chapter provides a listing of information sources.

Before turning our attention to look at macroeconomic measurements, we first need to review the basis of economic theory; the laws of supply and demand.

Markets and the Price System

A market is a mechanism that brings together buyers and sellers for a particular good or service. Buyers demand the item while sellers supply it. For example, in the labor market, employers act as buyers of labor; prospective employees are the sellers willing to supply it. In the economy as a whole, the participants are households, which supply resources and demand products, and business, which supply products and demand resources.

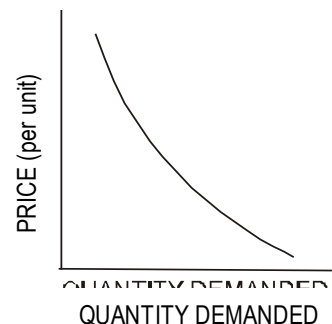
Most discussions of the theory of markets assume perfectly competitive markets, which have several qualities.

- Large numbers of independent buyers and sellers
- Ease of entry to or exit from the market
- A standardized product

While not all real markets meet these conditions perfectly, some come close (e.g., futures markets and stock exchanges). In these markets, prices are set by supply and demand.

Demand Demand for an item is defined as a schedule showing the amount of that item that buyers can and will purchase at various price levels during a given period of time. The resulting relationship is usually expressed in a graph. See Figure 1.

Figure 1



Note that the graph has negative slope. This indicates an inverse relationship between price and the quantity demanded: as price decreases, buyers will purchase greater quantities; as price increases, buyers will purchase less. This is the law of demand.

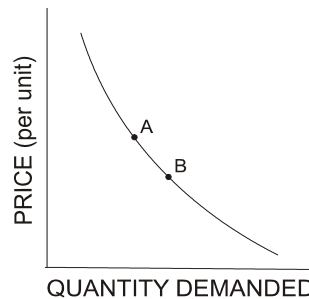
A demand curve assumes that the only factor being considered is the price of the items in question. All other factors are held constant. However, the demand curve will be affected by changes in several other factors. These are called non-price determinants, meaning factors affecting demand other than the price of the item in question. These include changes in:

- The income of buyers
- The number of buyers
- Preferences (tastes) of buyers
- Buyers' expectations regarding future events
- Prices of related items

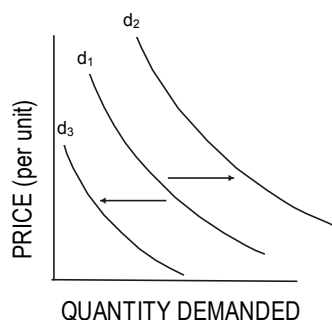
In regard to the last point, the relationship between related items may be direct or inverse. When the items are substitute or competing goods, the relationship is direct. For example, an increase in the price of airfares will increase the demand for travel by rail. Complementary goods, on the other hand, show an inverse relationship: a decrease in the price of compact discs will increase the demand for compact disc players. There are also independent goods that have no price relationship: digital cameras and golf balls, for example.

When the nonprice determinants are held constant and the price of a specific item changes, there will be a change in the quantity of the item demanded. For example, in Figure 2, the change from point A to point B indicates that as the price of the item falls, the quantity demanded rises.

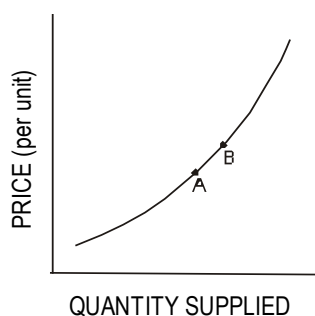
Figure 2



However, a change in a non-price factor will change the position of the entire curve. For example, in Figure 3, an increase in the number of buyers for an item might result in curve d_1 moving to position d_2 . This would indicate an increase in demand. Note that at any particular price, the quantity demanded is greater for the corresponding point on d_2 as compared to d_1 . A change from d_1 to d_3 would indicate a decrease in demand, possibly from a decrease in the number of buyers.

Figure 3

Supply Supply for an item is defined as a schedule showing the amount of that item that sellers can and will supply at various price levels during a given period of time. It can be analyzed in a similar manner to the preceding discussion of demand. For example, Figure 4 indicates a supply curve.

Figure 4

The supply curve has positive slope. This indicates a direct relationship between price and supply. For example, a change from point A to point B in Figure 4 tells us that as price increases, sellers are willing to supply greater quantities of the item. Likewise, decreases in price will result in lower supplies. This is the law of supply.

Our discussion of the law of supply assumes that non-price determinants are constant. These can include changes in:

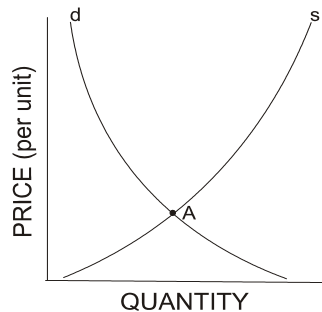
- The number of sellers
- The price of raw materials and other resources
- Technology
- Taxes and subsidies
- Sellers' expectations regarding future events
- Prices of related items

Changes in these factors can result in a change in supply, which would be indicated by a shift in the position of the supply curve. If the entire curve shifts to the right, supply has increased: at a given price, the quantity supplied is greater. A shift to the left indicates a decrease in supply.

Supply-Demand Equilibrium

Figure 5 shows the supply and demand curves for a particular item on the same graph. The intersection of the two curves is the equilibrium or market-clearing price. At this price, sellers will produce exactly as much as buyers are willing to purchase.

Figure 5



For any price above the equilibrium price, the supply curve is to the right of the demand curve, indicating sellers will produce more than buyers are willing to purchase. The resulting surplus will drive prices down to the market-clearing price. If prices are too low, the supply curve is to the left of the demand curve, meaning buyers are demanding more of a product than is available from suppliers. There will be a shortage of the item, causing prices to rise to the equilibrium point. This relationship acts as a rationing mechanism: at the equilibrium point there are no unsatisfied buyers or sellers.

Of course, it's possible that supply and/or demand could change. Either curve could shift to the left or right. If the supply curve is held constant, an increase in demand (curve shift to the right) results in price increases and quantity increases. Note that the equilibrium point would shift up and to the right. A decrease in demand would result in lower prices and quantities.

If the demand curve were held constant, an increase in supply would result in price decreases and quantity increases; the equilibrium point would shift down and to the right. A decrease in supply would be associated with higher prices and lower quantities.

Elasticity

Supply and demand curves tell us the change in quantity that will occur when the price of an item changes. However, the degree to which the quantity changes per unit price change will vary from item to item. This is known as price elasticity.

Demand Elasticity Elasticity of demand is the degree to which the quantity demanded would change for a given change in price and it's measured with the following formula:

$$E_d = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

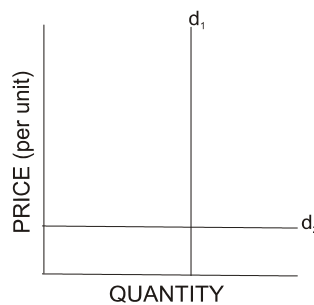
E_d is called the coefficient of elasticity. (Minus signs are ignored; the absolute value of the formula is used.) In applying the formula, different values will be obtained depending on which of two points being evaluated is used as the reference point. To avoid this difficulty, the midpoint is used as the reference in the calculation.

For example, suppose that a demand schedule shows that a \$10 unit price corresponds to a demand for 5,000 units, whereas an \$8 unit price results in a demand for 6,000 units. The difference in units demanded is 1,000. To calculate the percentage in quantity demanded, we divide by 5,500 units, rather than either 5,000 or 6,000. This results in an 18% change in quantity demanded. The percentage change in price is calculated by dividing \$2 by \$9 (22%). The coefficient of elasticity is approximately .8 (18% divided by 22%).

If $E_d > 1$, it's customary to refer to demand as being elastic or relatively elastic. This indicates that the quantity demanded will change by a larger percentage than the price change. For example, a 1% decline in price might produce a 2% increase in the quantity demanded. If $E_d < 1$, the demand is inelastic or relatively inelastic. $E_d = 1$ is called unit elasticity: the percentage change in demand will equal the percentage change in price.

The extreme cases are illustrated in Figure 6. Line d_1 represents perfectly inelastic demand. Line d_2 represents perfectly elastic demand.

Figure 6



It should be noted that, aside from the extreme cases, the coefficient of elasticity is not constant along a demand curve. Demand for an item tends to be elastic at higher prices and inelastic at lower prices.

The elasticity of an item can affect the total revenue generated from sales of that item. If $E_d > 1$, a price decrease will increase total revenue, since the lower price is more than offset by the larger percentage change in the quantity demanded. Likewise, a price increase will decrease total revenue when demand is elastic.

If $E_d < 1$, price decreases will decrease total revenue while price increases will increase total revenue. For unit elasticity, price changes have no effect on total revenue.

There are several factors that can affect elasticity of demand for a particular item, including:

- **Time** – the longer the time period examined, the more elastic demand tends to be.
- **Availability of substitutes** – the larger the number of substitute products available, the greater the elasticity.
- **Proportion of a budget devoted to the item** – the larger the percentage of a budget devoted to a particular good, the more elastic the demand will be.
- **Whether an item is a necessity or not** – items that are necessities tend to exhibit inelastic demand; demand for luxuries is elastic.

Supply Elasticity Like a demand curve, supply curves show varying degrees of elasticity. A similar price change will not result in the same percentage change in supply for all items. The main factor affecting supply elasticity is time: how quickly can a producer change production to increase or decrease the supply of a product. It will usually take longer for an automobile manufacturer to increase its output of cars as compared to the ability of a newspaper publisher to change the number of copies of the morning edition.

Effects of Changing Prices In some cases a company may need to lower the price of its product while maintaining or increasing revenue. In order to calculate the percentage increase in sales required to maintain the same revenue, use the following formula:

$$\frac{\text{Price decline percentage}}{\text{Complement of the price decrease}}$$

For example, if you lowered the price of a product by 5% and wanted to maintain the same amount of revenue, you would need to increase sales by 5.26% ($.05 \div .95 = .0526$)

To calculate the percentage increase in sales required to increase revenue, the formula is:

$$\frac{1.00 \text{ plus the revenue increase \%} - \text{Complement of the price decrease}}{\text{Complement of the price decrease}}$$

For example, if you lowered the price of a product by 5% and wanted to increase revenue by 8%, you would need to increase sales by 13.7% ($1.00 + .08 - .95 \div .95 = .137$)

The Price System

While supply and demand curves reveal the relationship between price, supply, and demand for a specific item, consideration has to be given to how prices in the overall economy are set and to how prices act as a mechanism for allocating resources. This mechanism is the price system.

If we assume that the goal of a business is to earn profits and avoid losses, it follows that business will produce goods (or services) that create profits and avoid producing those that result in losses. A profit is earned if total revenues exceed total costs. Price and quantity sold, the two topics discussed previously, determine revenues. We now turn to the analysis of costs.

Economic Costs Economic costs are payments that a business must make to secure the required amounts of productive resources. These resources include raw materials, labor, land, capital, and entrepreneurial ability. While the cost of labor (wages), land (rent), and raw materials are fairly straightforward; the cost of capital and entrepreneurial ability should be examined.

Capital refers to man-made resources that are used to produce other goods and services. Production machinery used in a factory is an example. Note that money is not capital in this sense. However, money is usually necessary to obtain capital goods. Therefore, it's helpful to distinguish between money capital and real capital (the goods themselves). Interest is the cost of money capital and therefore indicates the cost of the real capital obtained with it. As the cost of capital, interest rates ration the use of capital. If the cost of capital is less than the returns that can be earned by acquiring and using the capital, a business will borrow that capital to obtain the prospective profits.

The cost of obtaining and retaining entrepreneurial ability is called normal profit. The entrepreneur is paid for the function of organizing the business and combining the proper resources into a saleable good or service.

If the revenue earned by a business exceeds its costs, including wages, interest, cost of raw materials, rent, and normal profit, then the excess earned is called economic profit. Note that this is not the same sense of profit used by accountants. According to economists, economic profit doesn't include normal profit. Normal profit is a cost paid for entrepreneurial ability.

The Price System and Resource Allocation The presence or absence of economic profit results in the allocation of resources to the appropriate industry. If firms operating in an expanding industry are showing economic profits, new firms will enter the industry and resources will be allocated to them in an attempt to share in the economic profits available. Eventually, however, the increase of supply provided by additional entrants will drive down prices and economic profits will approach zero. At this point the industry will have achieved equilibrium size. Note that firms can still make normal profits even though economic profit is zero.

If an industry is in decline, the reverse process will occur. Firms earning losses will reallocate their resources in an attempt to earn economic profits where they're available. Again, the economic sense of earning losses could mean that the firm shows an accounting profit, but is earning less than the appropriate level of normal profits. Such firms will leave the declining industry until economic profits reach zero and equilibrium has been reestablished.

Input-Output Analysis As the price system allocates resources toward expanding industries and away from declining industries, the economy as a whole reaches general equilibrium. However, any change would now require a new equilibrium to be found. Input-output analysis is a technique that allows economists to investigate the consequences of such changes in an economy. Under this technique, the economy is divided into constituent industries. Each sector produces goods and services and consumes goods and services. These relationships are shown in tabular form, as illustrated in Figure 7.

Figure 7

Producing Sectors	Consuming Sectors				Total Output
	A	B	C	D	
A	10	50	20	20	100
B	50	30	60	10	150
C	100	300	50	200	650
D	20	80	100	40	240

In this example, Industry A has a total output of 100 units. Of that output, 10 units are consumed by Industry A itself (for example, the oil industry must consume oil in order to produce oil). The remaining output of Industry A is consumed by Industries B, C, and D.

The interdependence of the economy is illustrated by the consequences of a change in output in one industry. Suppose Industry A's output increases by 20% to 120 units. In order to accomplish this, a 20% increase in the inputs to Industry A is required. This will result in an additional 10 units of output ($50 \times 20\%$) from Industry B, 20 from C, 4 from D, and another 2 from A itself. These changes will result in further changes in the table as the increases ripple through the economy.

Cost of Production

Economic profits are earned when total revenue exceeds total economic costs, including normal profit. If resources can be allocated in a way that lowers costs, profits should improve. However, the analysis of costs will be different for different time horizons.

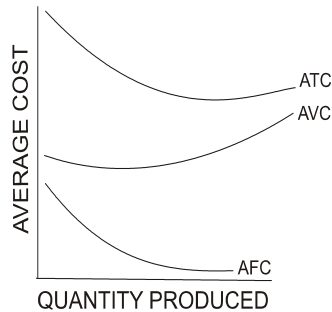
The Short Run Some types of inputs can be altered quickly, such as labor and raw materials. Other resources, such as plant and equipment, may not be so easily changed over short periods of time. Economists refer to the short run as the time period over which plant and equipment (fixed costs) cannot be changed, although they can be used in different ways. Those resources that can be changed are called variable costs.

The relationship between changes in the use of variable resources and output is given by the law of diminishing returns. It states that as more and more variable resources are combined with fixed resources, there will be a point beyond which the extra (marginal) output obtained by adding additional variable resources will decline.

The analysis of cost is often performed on a per unit or average basis. The average total cost per unit is composed of the average fixed cost per unit plus the average variable cost per unit. The average fixed cost (AFC) is found by dividing total fixed costs by output. Since total fixed costs are constant, AFC will decline as output rises. See Figure 8 below.

As with AFC, average variable cost (AVC) initially decline. However, because of the law of diminishing returns, AVC will eventually reach a minimum point and then begin to rise. Average total cost (ATC), which is the sum of AFC and AVC, will also rise after an initial decline.

Figure 8

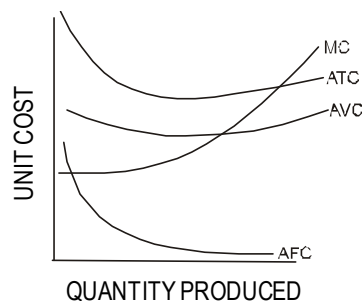


Another important concept in this type of analysis is marginal cost. Marginal cost (MC) is the additional cost of producing one more unit of output and it's calculated as follows:

$$MC = \frac{\text{Change in total cost}}{\text{Change in output}}$$

If marginal cost is less than average variable cost, AVC will continue to fall. Once $MC > AVC$, average variable cost will begin to rise. The MC and AVC curves will intersect at the minimum point on the AVC curve (see Figure 9). A similar relationship holds for MC and ATC.

Figure 9



The Long Run Over longer time horizons, firms can change their capital assets (plant and equipment) that are relatively fixed in the short run. This means that in the long run, all costs are variable. A long run average total cost curve can be constructed. This can be derived from a series of short run ATC curves. The long run ATC curve is also subject to the law of diminishing returns. After an initial decline in ATC due to *economies of scale*, the curve reaches a minimum and then begins to rise as *diseconomies of scale* begin to have an effect. The smallest level of output at which a firm can achieve minimum ATC is referred to as *minimum efficient scale* (MES).

The shape of the ATC curve can determine the structure of an industry. In some industries, diseconomies of scale are encountered quickly and MES will occur at relatively low levels of output. Such industries should contain a large number of small producers. Where economies of scale are extensive, MES would occur at higher levels of output. Such industries should contain a smaller number of large firms. In some cases, the effects of economies of scale may stop at some point, but diseconomies may not occur until somewhat higher levels of output are reached. There will be a range of outputs that show constant returns to scale. Such industries may contain many different sized firms.

Economies of Scope Minimizing the product line costs of production, marketing, and/or distribution relates to economies of scope. The marketing department of a consumer products company employs economies of scope by using the same personnel to sell different products produced by the company, often bundling products to sell as a group. Economies of scope may be useful to gain new product recognition by an established brand. If several products have been marketed as a group, but one of the products exhibits superior market potential, the additional marketing costs to promote this product represent diseconomies of scope.

Economics

The performance of an investment is influenced by the performance of the economy. An impending recession may reduce demand for equity securities. The effects of inflation or deflation may interfere with anticipated returns, or influence the value of debt securities. Therefore, economic factors must be considered when formulating an investment strategy. Two of the most significant measures of economic activity are the *Gross National Product (GNP)* and *Gross Domestic Product (GDP)*.

Gross National Product (GNP) GNP measures the total value of all goods and services produced by a national economy. For the U.S., this measure would include goods and services produced inside U.S. borders as well as those produced overseas by a U.S.-based company.

Gross Domestic Product (GDP) Gross Domestic Product has replaced Gross National Product as the most important measure of output and spending within the United States. GDP is the output of goods and services produced by labor and property located in the United States, without regard to the origin of the producer. For example, a Toyota plant in Georgetown, Kentucky is part of the U.S. GDP. Components of GDP include consumer spending, investments, government spending, and net exports. The most useful variation of GDP is *real* GDP, which is GDP adjusted for inflation using constant dollars. This is considered the key measure of aggregate economic activity. Rising GDP signifies economic growth and potential inflation. Falling GDP would likely be accompanied by job losses and easing of pricing pressures.

Inflation and Deflation

Inflation is a persistent and appreciable rise in the general level of prices. Inflation occurs when the demand for goods and services in the marketplace increases at a faster rate than the supply of these items. In other words, there's too much money chasing too few goods.

The *Consumer Price Index (CPI)* is widely considered the most important measure of inflation. The CPI measures the prices of a fixed basket of goods bought by a typical consumer. If prices, as measured by the CPI, are rising, then the economy is experiencing inflation.

The Real Interest Rate The real interest rate is the rate an investor in fixed-income securities actually receives once inflation is taken into account. For example, if a 10-year bond is yielding 7% when the inflation rate is 3%, the real interest rate is 4%. The 7% is composed of a 4% real interest rate plus a 3% inflation premium.

Inflationary periods are typically characterized by rising interest rates as investors demand higher nominal rates to earn a desired real rate of return. Interest-rate-sensitive stocks, such as those issued by utilities, react to changes in interest rates. Since utility companies are so highly leveraged, it becomes more expensive for these companies to raise money when interest rates increase. Interest charges increase, causing a drain on earnings, resulting in a decline in prices of the securities.

Historically, equity securities or other related products, such as equity mutual funds and equity ETFs, have provided the best protection against inflation. However, fixed-income securities are more vulnerable to inflation. Inflation changes represent a two-pronged risk for bondholders—interest rates rise causing the market price of holdings to fall, while the purchasing power of interest payments also decreases. Investments in gold and other commodities are considered a good hedge against inflation. Inflationary risk is also referred to as purchasing-power risk.

Deflation is a persistent and appreciable decline in the general level of prices. If the supply of goods and services is greater than the demand for those items, then producers would have to lower their prices to compete for the limited demand. If this becomes the case on a broad scale, deflation would take place.

Deflation should not be confused with *disinflation*. While deflation is a drop in prices, disinflation is a reduction in the rate of inflation.

Stagflation is a prolonged period of a high rate of inflation at the same time as a high rate of unemployment. This doesn't happen often, since high unemployment usually leads to a period of low inflation, or even deflation (falling prices) and the possibility of a recession. A period of low unemployment usually leads to rising prices and increased inflation.

The Business Cycle

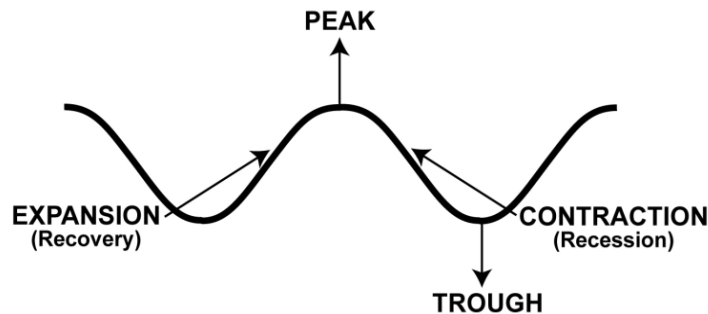
Over time, a pattern has emerged of economic ebb and flow. The business cycle represents this repetitive succession of changes in economic activity. The business cycle has four phases—expansion (recovery), peak, contraction (decline), and trough.

In the expansion phase, business activity is growing, production and demand are increasing, and employment is expanding. As this phase peaks, businesses and consumers normally borrow money to expand, causing interest rates to rise, which effectively self-limits the expansion phase.

As the cycle continues, demand for goods overtakes supply and prices rise, creating inflation. During inflationary times, there's too much money chasing a limited amount of goods. Businesses are able to charge more for products and services, causing prices to rise and reducing the purchasing power of the consumer.

As prices rise, demand slackens, which causes economic activity to decrease. The cycle then enters the contraction phase. As business activity contracts, employers lay off workers, unemployment increases, and demand slackens. Usually, these actions cause the rate at which prices are rising (inflation) to decline (*disinflation*). The situation where prices are falling in real terms is called *deflation*. The cycle enters the trough at the bottom of the economy's decline. Eventually, lower prices will stimulate demand and the economy moves into a renewed period of expansion called a recovery.

At times, the economic decline can be pronounced. By definition, a recession occurs when Real GDP declines for two successive quarters (six months). A depression occurs when Real GDP declines for a more prolonged period (two years or more).



Business Cycle Indicators

Although the economy tends to follow a general pattern, it's often difficult to determine its direction at any given time. Economists use three types of indicators that provide monthly data on the movement of the economy as the business cycle enters different phases. The three types are *leading*, *coincident*, and *lagging economic indicators*.

Leading Economic Indicators Leading economic indicators precede the upward and downward movements of the business cycle. They may also be used to predict the near-term activity of the economy. The government index of 10 leading economic indicators is released monthly. The components of the index are:

- The average workweek for production workers in manufacturing
- The average weekly initial claims for state unemployment insurance
- New orders for consumer goods and materials
- Vendor performance (companies receiving slower deliveries from suppliers)
- Contracts and orders for plant and equipment
- New building permits for private housing units
- The prices for the S&P 500 common stocks
- The Money Supply (M2)
- The change in credit outstanding for business and consumer borrowing
- Interest rate spreads, 10-year Treasury bonds less federal funds
- The index of consumer expectations

Coincident Economic Indicators Coincident indicators usually mirror the movements of the business cycle. The makeup of the coincident economic indicators includes four components.

- Employees on non-agricultural payrolls
- Personal income less transfer payments (Transfer payments represent aid for individuals in the form of Medicare, Social Security, and veteran's benefits, to list a few.)
- The Index of Industrial Production
- Manufacturing and trade sales

Lagging Economic Indicators The index of lagging indicators represents items that change after the economy has moved through a given stage of the business cycle. The index of lagging indicators should confirm the economic condition portrayed by previous leading and coincident indexes. Lagging indicators include:

- The average duration of unemployment
- The relationship of inventories to sales, manufacturing, and trade
- Labor cost per unit of output for manufactured goods
- The average prime rate charged by banks
- Commercial and industrial loans outstanding
- The relationship of consumer installment credit to personal income
- The consumer price index for services

The Effect of the Business Cycle on Securities Markets

As the economy moves through the different phases of the business cycle, the bond and equity markets react to these changes. Investors view these changes and take corresponding action, attempting to take advantage of changes in the economy.

The level and direction of interest rates will influence numerous investments and may indicate inflationary trends. Therefore, interest rates and the effects of inflation will be an important factor to consider when individuals look to make investments.

For example, bond investors are concerned that inflation will erode the purchasing power of the interest they earn. Therefore, it's important to earn a rate of return that stays ahead of the rate of inflation. An investor may wish to calculate the real interest rate on an investment. This is found by subtracting the rate of inflation from the current interest rate on a bond. For example, given a 2% rate of inflation, a bond yielding 7% would have a real rate of return of 5%.

Changes in the business cycle also affect the equities market and would have the least effect on the business of companies in a defensive industry (such as pharmaceuticals or utilities). The demand for the products produced by defensive companies would not be hurt by a downturn in the economy. Examples of defensive industries include utilities, nondurable household goods, tobacco, and food. Since people need basic services to exist, companies operating in a defensive industry are the last to be affected negatively as the economy moves through difficult periods.

The performance of cyclical stocks is often parallel to the economy of the country. If the economy is expected to prosper, then one can expect a cyclical company to prosper. If the economy is expected to decline, then one can expect a cyclical company to be less prosperous. Examples of cyclical industries include manufacturing, homebuilders, auto manufacturers, and transportation companies.

Influencing the Economy – Economic Theory

There are a number of theories as to why the economy moves in a cyclical fashion and what can be done to control that cycle. We will discuss three of the more popular economic theories: *Keynesian*, *Supply-Side*, and *Monetary*.

Keynesian Economics

Keynesian Economic Theory states that government intervention in the economy is necessary for sustained economic growth and stability. Put forth by British economist John Maynard Keynes, this theory further states that the government should use fiscal policies (i.e., budgetary and taxing policy) to combat the effects of inflation and deflation, as well as to influence economic activity.

Fiscal policy is the government's use of taxation and expenditure programs to maintain a stable, growing economy. If the economy is in a recession (trough), the government can increase its spending to stimulate demand. Alternatively, it can cut taxes, which will increase the disposable income of consumers. This action would also (indirectly) stimulate demand. If the economy is overheated (i.e., exhibiting too much demand), the government can cut its spending or increase taxes. Fiscal policy is established by Congress. Therefore, decisions in some cases may be based on political, rather than purely economic, motives. However, the main focus of fiscal policy is on economic growth and high employment.

Supply-Side Economics Supply-side economics places an emphasis on reducing taxes and the size of government to stimulate the economy. Reducing the size of government will allow for lower tax rates. By reducing taxes, individuals and corporations will have more money to invest, thereby stimulating increased economic activity.

Monetary Policy Monetary policy attempts to control the supply of money and credit in the economy. This, in turn, will affect interest rates, causing an increase or decrease in economic activity. The primary focus of monetary policy is to control inflation. The Federal Reserve System implements monetary policy in the United States.

Money Supply Money is the unit of value by which goods and services are measured and it's the medium of exchange through which business is transacted. The Federal Reserve Board (Fed/ FRB) attempts to control the money supply and credit to maintain a stable, growing economy with the aim of combating inflation. Shifts in economic conditions will influence the Fed's focus on the money supply figures. Each week the Fed compiles and publishes figures on the size of the money supply according to the M1 measure. On a monthly basis, the Fed publishes figures on M2 and M3.

DEFINITIONS OF THE MONEY SUPPLY	
<i>M1</i>	Currency in circulation + demand deposits + other checkable deposits
<i>M2 = M1</i>	+ money market deposit accounts (MMDAs) + savings and relatively small time deposits + balances at money funds + overnight repurchase agreements at banks
<i>M3 = M2</i>	+ large time deposits + term repurchase agreements at banks and savings and loans + Eurodollars (no longer published by the FRB)

Major Tools of the Federal Reserve Board (FRB)

The FRB has various tools at its disposal through which it may implement its monetary policy. These tools are:

- Setting reserve requirements
- Setting the discount rate
- Implementing open market operations
- Setting margin requirements
- Utilizing moral suasion

Reserve Requirements

Member banks are required to keep a portion of their deposits on reserve with the FRB. By adjusting the amount banks must keep on reserve at the FRB, the Fed can tighten or ease the money supply. If reserve requirements are lowered, the banks are able to extend more credit. Therefore, the money supply increases. The opposite effect occurs with an increase in reserve requirements.

After meeting its reserve requirement, a bank will strive to lend the remaining funds to borrowers. The amount of funds a bank has above the reserve requirement is called excess reserves. The money spent by the borrowers will eventually be deposited in another bank. This continues as money is deposited from bank to bank, creating a multiplier effect on deposits. In other words, the *multiplier effect* is the rate at which banks can create new money by relending deposits and, in turn, creating new deposits. Changing the reserve requirements will have an impact on the multiplier effect within the banking system. An additional factor is the velocity of money which measures the number of times a dollar is spent over a given period. The effect of these changes to the reserve requirement can be drastic; therefore, this is one of the least used tools of monetary policy.

Discount Window The FRB was originally established to aid the banking system in emergency situations by acting as a banker's bank. The FRB always stands ready to lend money to its members. It fulfills that function through its discount window. The rate charged for loans is called the *discount rate*.

When members of the Fed borrow using the discount window, new money is injected into the system (which then is expanded by the multiplier effect). The FRB can encourage or discourage use of discount window borrowing by changing the rate of interest it charges for those loans.

By decreasing the discount rate, the FRB encourages borrowing, which expands the money supply. Conversely, the money supply will contract with an increase in the discount rate. A change in the discount rate is usually taken as a very strong sign that monetary policy has shifted.

The discount rate is the only rate directly set by the FRB. Although it's largely symbolic, it acts as the basis on which other key interest rates are built, such as the federal funds rate.

Federal Funds Due to deposits, withdrawals, and loan demands, a bank may find itself with either an excess reserve position or a deficit reserve position. If a bank has excess reserves, it may lend additional funds to borrowers including commercial banks who are in a deficit reserve position.

These short-term loans of excess reserves that banks lend each other are called *federal funds*. The rate of interest charged on these loans is called the federal funds rate.

The federal funds rate is determined by supply and demand. Since federal funds are short-term (overnight), they're considered money-market instruments. Due to the short duration of the loan, the fed funds rate is normally considered to be the most volatile interest rate. The effective federal funds rate is published daily and shows the average rate charged the previous night for federal funds.

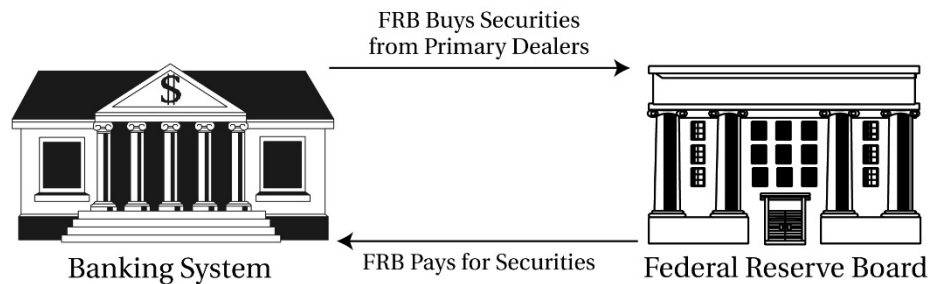
Although the FRB doesn't directly set the fed funds rate, it does set a target. The FRB's Open Market Operations are designed to maintain the fed funds rate within this prescribed target range.

Open Market Operations Unlike the previous tools (reserve requirements and the discount window) that entail very public moves on the part of the FRB, open market operations can be implemented very quietly, without disrupting the financial markets.

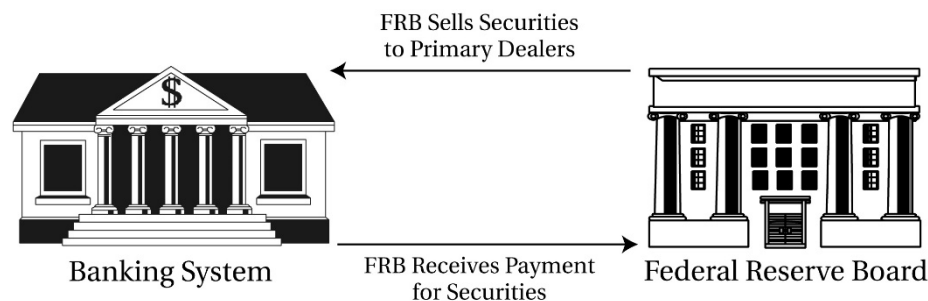
The *Federal Open Market Committee (FOMC)* oversees the FRB's buying and selling of U.S. government securities in the secondary markets. Open market operations are the most effective and frequently used tool of monetary control employed by the FRB, but it's also the most flexible tool and the easiest to reverse.

Open market operations typically involve the purchase and sale of U.S. government securities, primarily Treasury bills. The FRB, however, also trades government notes and bonds. These trades are executed through *primary government dealers*, banks, and brokerage firms appointed by the FRB.

Buying Securities If the Fed buys securities, it pays for these securities with funds that are ultimately deposited in commercial banks. This causes deposits at banks to increase and therefore adds to the funds available for loans. The result is an increase in reserves. Money becomes more available and interest rates tend to move downward. This is referred to as an easy money policy.



Selling Securities Should the Fed wish to tighten (reduce) the money supply, it will sell securities to banks and securities dealers. The banks and dealers will pay for these securities by withdrawing the money from their demand (checking) accounts. The withdrawal of money from the banks will decrease the amount of money available for loans. This will have a tightening effect on the money supply, causing interest rates to rise.



Repurchase Agreements A repurchase agreement (also known as a repo) is a contract entered into by the Federal Reserve to purchase U.S. government securities from dealers, at a fixed price, with provisions for their resale back to the dealer at the same price plus a negotiated rate of interest. When the Fed effects a repo, it's lending money and, therefore, increases bank reserves (an easy money policy).

A reverse repo (also known as a *matched sale*) occurs when the FRB sells securities to dealers with the intention of buying the securities back at a future date. This has the short-term effect of absorbing funds from the money supply (a tight money policy).

Margin Requirements

The Securities Exchange Act of 1934 gave the Federal Reserve Board the power to determine the amount of credit that could be extended to purchase securities. Regulation T applies to brokerage firms. Regulation U applies to banks and all other lenders.

By increasing margin requirements, the FRB reduces the amount brokers and banks may lend, causing the money supply to tighten. Changing the margin requirement is the least effective method that the FRB uses to control credit because it only affects securities market transactions.

Moral Suasion There are times when the Fed tries to influence bank lending policies through moral suasion (also called jawboning). The Fed exerts its influence through the public media or through the examiners sent to member banks. Its efforts to control the money supply by these means are limited by the extent to which they can elicit cooperation from these institutions.

Effects of the FRB's Activities As with any product, when the supply increases, the price of that product will decrease and if the supply contracts, the price will increase. The price of money is the interest rate that lenders charge borrowers. Therefore, as the FRB changes the supply of money, the price of money (interest rates) must also change. As interest rates change, the FRB will adjust its monetary policy in order to influence the various sectors of the economy.

In summary, Federal Reserve Board activities tend to cause the following:

Activity	Effect on Money Supply and Credit (Loan) Availability	Impact on General Interest Rate Levels
Raise bank reserve requirements	Decrease	Raise
Raise the discount rate	Decrease	Raise
Raise margin requirements	Decrease	Raise
Sell government securities in the open market	Decrease	Raise
Lower bank reserve requirements	Increase	Lower
Lower the discount rate	Increase	Lower
Lower margin requirements	Increase	Lower
Buy government securities in the open market	Increase	Lower

Any method or tool that creates additional money for the banking system is potentially inflationary. Any tool or method that shrinks the amount of money available to the banking system is potentially deflationary. The FRB will use its tools in an effort to influence inflation and deflation.

For example, to combat inflation, the FRB may look to sell securities through its open market operations. Inflation normally occurs due to an abundance of money in circulation. Since selling securities reduces the money supply, this action may curb inflation.

To curb deflation, the FRB looks to buy securities. During deflationary periods, prices are dropping. Business activity is sluggish which reduces the demand for money. This results in a drop in interest rates. By purchasing securities through its open market operations, the FRB will be injecting money into the banking system in order to stimulate investment and business activity.

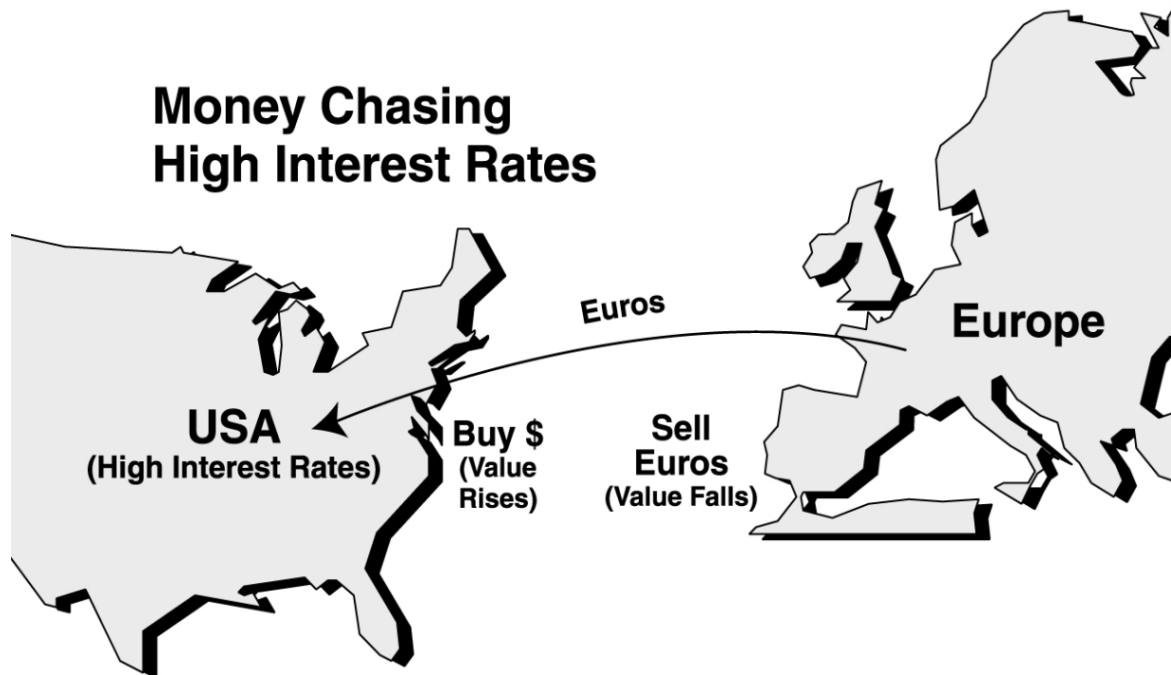
How Money is Created Money, as measured by M1, consists of currency plus checking account balances. While the FRB prints currency, the commercial banking system creates checking accounts. Banks make a profit if they can attract deposits and lend those funds at a higher rate of interest.

Intermediation and Disintermediation Banks can only create money if they're able to attract deposits and subsequently make loans. Intermediation refers to the ability of financial intermediaries (such as banks) to attract deposits and, in turn, extend credit.

Disintermediation is the process by which investors withdraw funds from banks and seek higher-yielding investments elsewhere. For example, if money-market mutual funds were yielding 4% while time deposits at banks were yielding 2%, investors may invest money in the funds with bank withdrawals. The higher mutual fund yield may affect the bank's ability to attract and keep deposits.

Foreign Exchange and Trade

Changes in interest rates affect not only the domestic economy but also international economic activity. If interest rates in the U.S. are higher than interest rates overseas, foreign investors, wishing to earn the higher rate, may want to invest in the U.S. In order to invest in the U.S., foreign investors must first convert their funds into dollars. As demand for dollars increases, the price of dollars (exchange rate) will increase. Therefore, higher U.S. interest rates compared to foreign yields may lead to a stronger dollar.



Foreign exchange rates also have an impact on foreign trade. If the U.S. dollar is strong, U.S. consumers can purchase more foreign goods. Therefore, imports into the U.S. (exports from foreign countries) will increase. Conversely, foreign consumers (with weaker foreign currencies) will be unable to purchase the relatively more expensive U.S. goods. This causes exports from the U.S. to decrease leading to a trade deficit for the U.S. (imports exceeding exports).

To correct a trade deficit, the dollar must fall. U.S. goods will become cheaper (more competitive) abroad and foreign goods more expensive in the United States. This would lead to more U.S. exports and fewer U.S. imports that should help to alleviate the trade imbalance.

Balance of Payments and Exchange Rates

The banking system also plays a key role in the market for foreign exchange (currency) and in import-export transactions. Exchange rates in the foreign currency market can have an impact on U.S. interest rates and on international trade.

Balance of Payments When American goods are sold abroad, the demand for U.S. dollars increases. The foreign importers must exchange their currency for dollars in order to pay for the transaction. A U.S. bank that facilitates the exchange will have a deposit of foreign currency that's exchanged for the deposit of U.S. dollars now owned by its customer, the exporter. Note that this results in an increase in the U.S. money supply and also makes a deposit of foreign currency available (owned by the U.S. bank) that could be sold to those who might need it.

An American importer would need to purchase foreign currency to pay for foreign goods. If the U.S. bank sells its deposit of foreign currency to the American importer, the U.S. money supply is reduced, as is the supply of foreign currency held by U.S. banks. In effect, American exports earn a supply of foreign currency with which American importers can pay for foreign goods purchased by the United States. The schedule that shows a summary of all such import-export transactions is the balance of payments.

The current account shows the results for trade transactions completed in the current year. The balance of U.S. exports of merchandise against U.S. imports of merchandise would constitute the balance of trade. Overall U.S. exports would also include exports of services and U.S. imports would show a corresponding entry for imports of services. Other current inflows and outflows would include investment income and transfers, such as gifts, to foreigners.

The capital account deals with long-term transactions, usually involving property or financial assets. For example, a U.S. corporation investing in a factory in a foreign country would represent an outflow, while the purchase of U.S. common stock by a foreign investor would represent an inflow of capital. Borrowing by U.S. businesses in foreign markets and by foreign companies in the U.S. market is also reflected.

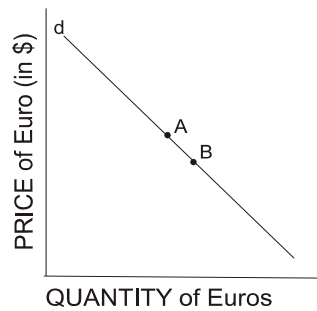
Since deficits, as in the preceding example, must be paid for in some way, a current account deficit would have to be balanced by a capital account surplus. This could occur through the sale of U.S. assets to foreigners or through borrowing. Likewise, a current account surplus is balanced by a capital account deficit. The net of the two accounts must be zero.

While the net of the current and capital accounts must be zero, a country that consistently runs a current account deficit will face difficulties. Such deficits will drain its foreign currency reserves. At that point, it may have to borrow heavily, change the value of its currency, or face tough macro-economic measures.

Exchange Rates Most major international currencies are freely convertible. The rates for conversion are set in an open market trading system called the Interbank system. The Interbank system consists of the world's major commercial banks and central banks trading directly with each other or through currency brokers. The U.S. dollar is used as the reserve currency and serves as a point of common reference for currency values.

As in any competitive market, the foreign exchange market should reflect the forces of supply and demand. A demand curve for Euros is illustrated in Figure 11a. As the price moves down from point A to point B, note that the quantity demanded will increase. This is because cheaper Euros mean that Americans will perceive goods denominated in Euros as less expensive. Since they will tend to buy more goods at these lower prices, the demand for Euros will increase (as the basis of payment for the products).

Figure 11a



A supply curve for Euros is shown in Figure 11b on the next page. If the price of the Euro (in dollars) increases in going from point C to point D, the quantity of Euros supplied to the market will increase. Since Euros are more expensive, the dollar is therefore cheaper. U.S. goods will look less expensive to some Europeans, who will tend to buy more American products. To pay for them, they will exchange more Euros for dollars, increasing the supply of Euros in the market.

Figure 11b

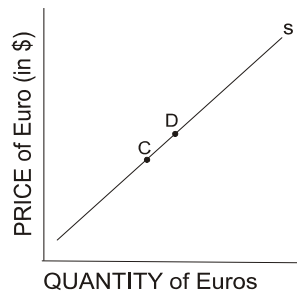
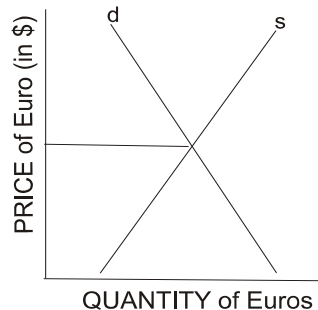


Figure 11c shows intersecting demand and supply curves. The intersection point represents the equilibrium exchange rate at these demand and supply levels.

Figure 11c



There are several factors that could alter the exchange rate by modifying demand or supply schedules, such as:

- **Changes in income** — as countries become wealthier, consumers tend to purchase more goods in general, including imported goods.
- **Changes in preferences for goods of a particular country** — if German cars become a status item, demand will increase.
- **Changes in real interest rates** — if interest rates adjusted for inflation rise in a particular country, foreign investors will wish to purchase investment in that country.
- **Price changes** — if domestic prices are rising more rapidly than the prices of foreign goods, imports will increase.
- **Speculation** — expectation of political or economic changes may result in buying or selling of a currency by speculators.

In theory, flexible exchange rates should adjust automatically to correct a trade imbalance. For example, if the U.S. has a trade deficit with Germany, there will be a greater demand for Euros. This will move the demand curve in Figure 11c to the right. That will increase the price of the Euro in dollars. German goods will look more expensive to Americans, who will import less. The dollar will be weaker against the Euro, making U.S. products look cheaper to Germans. America will export more to Germany.

Unfortunately, there may be other negative consequences from such an adjustment. The increased cost of imported German goods will translate into inflationary pressure in the U.S. economy. If U.S. export industries are already operating at near capacity, increased demand because of a weak dollar may also cause inflationary price increases in those industries.

Effects of Currency Fluctuations on a Company's Profitability Changes in currency rates can have a significant influence on the revenue, cost of goods sold, and operating profit margin of a company. For example, if a U.S. company receives revenue from Europe and the price of the dollar declines against the euro, the company will report higher revenue (since the value of the euro has increased). If a U.S. company manufactures a product in Japan and the price of the dollar declines against the yen, the company's cost of goods sold will increase (since the value of the yen has increased). The following table summarizes the relationship between currency fluctuations concerning the U.S. dollar and the euro. If we substitute other currencies for the euro, the effect is the same.

International Activities					
Headquartered	Manufacturing	Sales	Euro	Operating Profit Margin	Revenues
1. U.S.	U.S.	Europe	▲	Increases	Increase
2. U.S.	U.S.	Europe	▼	Decreases	Decrease
3. U.S.	Europe	Europe	▲	Unchanged	Increase
4. U.S.	Europe	Europe	▼	Unchanged	Decrease
5. Europe	Europe	U.S.	▲	Decreases	Decrease
6. Europe	Europe	U.S.	▼	Increases	Increase
7. Europe	U.S.	U.S.	▲	Unchanged	Decrease
8. Europe	U.S.	U.S.	▼	Unchanged	Increase
9. U.S.	Europe	U.S.	▲	Decreases	Unchanged
10. U.S.	Europe	U.S.	▼	Increases	Unchanged
11. Europe	U.S.	Europe	▲	Increases	Unchanged
12. Europe	U.S.	Europe	▼	Decreases	Unchanged

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CHAPTER 2

Introduction to Industry and Company Analysis



Overview of Industry Analysis

The next step in top-down financial analysis is to look at various industries to determine the opportunities available given the current and projected state of the economy. Stages of the business cycle, interest rate environments, fiscal and monetary policies, and the relative values of foreign currency will combine to create an environment that's more or less favorable to a given industry.

Cyclical stocks move with the business cycle. If the economy is in a period of prosperity, these companies prosper. However, as the economy falters, cyclical stocks decline. The common stock of a machine tool company is an example of a cyclical stock. As the economy expands, new orders for machinery increase and machine tool companies prosper. Other examples of cyclical stocks include basic industries (rubber, steel, etc.), construction firms, and manufacturers of durable goods.

The stocks of defensive companies react less to changes in the business cycle than cyclical stocks. Examples of defensive companies include utilities, tobacco, alcohol, cosmetics, and food companies. Since people need basic services to exist, these companies are the last to be affected negatively as the economy moves through difficult periods.

Interest-rate-sensitive stocks, such as those issued by utilities, react to changes in interest rates. Since utility companies are highly leveraged, it becomes more expensive for these companies to raise money when interest rates increase. Interest charges increase, causing a drain on earnings, resulting in a decline in prices of these securities.

Stock Price Cycles

The stock market generally falls prior to a recession and rises prior to an expansion. Therefore, the ability to predict turns in the business cycle is very desirable. However, the lag time between changes in the market and the economy can be considerable. In addition, different industry groups will show their best relative performance at different stages of the stock market cycle, as in the following examples:

Early to middle bull market

- Credit-related industries
- Construction
- Durable consumer goods
- Transportation

Middle to late bull market

- Capital goods
- Some financial stocks

Late bear market

- Non-durable consumer goods
- Utilities
- Some energy stocks

Life Cycle of an Industry

In addition to tracking the business cycle, many industries experience their own internal cycle of birth, growth, maturation, and decline. An analyst needs to determine if this pattern fits the industry being investigated and then needs to determine where in the pattern a particular industry is currently located.

The first stage of the industry life cycle, birth and early development, is characterized by high growth rates and potentially high profits. There may be many new entrants into the industry initially; however, many will fail in the competition for the available high profits. During the 1980s, the history of the hardware and software industries for the personal computer seem to fit this stage.

The second stage is marked by rapid growth. However, after the competitive struggle of stage one, consolidation often leaves fewer, larger companies in the industry. Quality improves at the same time that prices decline. An example of this is the consolidation in the computer software industry in the early 1990s.

The third stage is often difficult to differentiate from stage two. This stage is marked by mature growth which, though still significant, is not as rapid as stage two. The industry's products begin to face competition from other products and markets may start to become saturated.

Finally, an industry reaches stage four, in which the growth slows substantially. In some industries, there will be no growth at all, or an actual decline if a product becomes obsolete.

While many industries fit this pattern, the length of the life cycle can be substantially different from case to case. While the life cycle of the automobile industry has spanned many decades, some consumer products, particularly electronics, may complete a life cycle within a single decade.

Earnings Cycles

Within the life cycle of an industry, it's useful to isolate the typical pattern of changes in industry profitability.

Initially, a new industry may experience negative earnings. Once rapid growth begins, margins improve, causing earnings to turn positive and begin to grow. As the mature growth stage is reached, profit margins begin to level out and eventually decline.

Although margins begin to decline during the mature growth stage, earnings don't yet decline since overall revenues are still increasing, although at a slower rate. Once revenue growth is insufficient to offset declining margins, earnings level off and then begin to decline.

Note that the overall pattern is similar to that for the basic industry life cycle—rapid initial growth, then slower growth, followed by a leveling off and possible decline. However, increases and decreases in profits are not coincident with changes in revenues, since the change in profit margins is an intermediate variable that will affect earnings.

Analysis of Markets

There are several ways for an industry to maintain a good rate of growth including:

1. Steady penetration of rapidly growing markets
2. Accelerating penetration of markets with stable growth rates
3. Accelerating penetration of rapidly growing markets

One may assume that the most rapidly growing industries will find themselves in the third situation. On the other hand, declines in market penetration, or in the growth of the target market, or both, can lead to industry decline. To study the market for an industry, the analyst must first define “the market.” There are several ways to perform this classification, including by location, age, or income.

Output Analysis One approach is to define markets for a product by the industry that consumes the product. This enables the analyst to use input-output analysis to study an industry. This technique divides the economy into industry groups, and sales from each industry to each other industry are determined. From this information, the analyst can locate the major markets for an industry. Even more important, a comparison of these values over time can indicate whether an industry is expanding its markets in ways that will sustain growth.

Costs of Production and Pricing

Two major factors that influence the pattern of industry profits are unit costs and pricing strategies.

Unit Costs In most industries, as output increases in the early stages of growth, cost per unit declines due to economies of scale. The decline in unit cost is due to several factors. Large start-up costs are amortized over a larger number of units produced. These start-up charges include costs for new equipment, training for employees, and research and development. As employees gain more experience, they become more productive. Technological improvements to equipment and processes lead to more efficiency and higher quality.

A graph relating the decline in unit costs to the industries output is referred to as a learning curve. However, eventually the curve begins to flatten out as unit costs stop falling with increases in production. The size of very large industries may actually lead to diseconomies of scale.

Since determining unit costs is often difficult for analysts, particularly when examining data from companies that operate in several industries, unit prices are often used as a substitute for unit costs.

Unit Pricing In examining pricing strategies, significant attention is paid to those companies within an industry that have the largest market share. The assumption is that these companies are further along on the learning curve and therefore have lower unit costs. The pricing strategies adopted by the industry leaders have an impact on the profitability of the entire industry. Analysts believe that the nature of the pricing strategy can provide insights into the competitive situation in the industry and into the industry’s own view of its growth potential.

One type of pricing strategy is *umbrella pricing*. Under this approach, the market leaders set their prices just above the unit costs of the industry's highest cost producers. This allows the low-cost producers to earn an attractive rate of return, while the other producers earn only modest returns. Prices don't change very often. Since the market leaders are not attempting to gain market share through aggressive pricing, they appear to be satisfied that their current market share will produce growing profits because the market itself will grow. Since they don't seem to be afraid of price competition from new entrants, it can be assumed that the costs of entry into the market are high.

Another approach to pricing is a *learning curve pricing strategy*. With this strategy, the low-cost producers use their cost advantage by regularly lowering prices. This puts pressure on the higher-cost producers to improve efficiency in order to maintain profitability in the face of declining prices. It also makes it increasingly difficult for new competitors to enter the market. Lower prices also result in increased penetration of the target market, as the leaders seek growth in rising unit sales.

Industries that are making the transition to the final phase of their life cycle often set prices based on their target market's demand for their product and their industry's capacity to produce it. If the market for a product is flat, price increases must be relied on to maintain profitability. An industry that raises prices at every opportunity is signaling its belief that little growth in unit sales can be expected. Prices are lowered only when lower demand makes it impossible to maintain current price levels. Excessive price increases risk driving the consuming industry to consider substitute products.

Additional Factors of Industry Analysis

Products

The ultimate goal of a business is to sell goods and services at a profit. The analyst's job is to predict the reliability and extent of future profit growth. The initial discussion of market theory is the basis for this analysis, especially the factors that may impact future supply and demand. As previously described, as prices rise, buyers seek out substitute products. As demand for a particular good increases, so too should demand for complementary goods.

On the supply side, industries are considered the buyers of raw materials, labor, and capital equipment. An analysis of supplier dynamics should indicate the ability of an industry to experience continued growth. Industries that have limited suppliers are more subject to cost increases (e.g. oil companies).

Demographics Forecasted changes in population and wealth distribution can often signal future demand. The aging populations in Europe and North America will undoubtedly demand more pharmaceutical and healthcare-related products. Emerging economies in Asia will demand more automobiles and telecommunications equipment. Urban migration in Brazil has created demand for infrastructure projects. Industries that offer products to meet these needs should experience growth in the future.

Barriers to Entry

Economists typically break down industries into the following basic competitive models:

- **Pure Competition** – Many producers of a commodity
- **Monopolistic Competition** – Many producers with small market share
- **Oligopoly** – Several large producers controlling 70 to 80% of the market share
- **Duopoly** – Two producers
- **Monopoly** – Single producer
- **Monopsony** – A single buyer in the market

When moving from pure competition to pure monopoly, greater and greater barriers to entry are encountered. These barriers are the difficulties experienced by a new firm that's seeking to enter an industry. In pure competition, there are no barriers to entry and firms can easily come and go. One example is agriculture without government subsidies.

Most firms fall into the segment of monopolistic competition. As the name implies, this is between the rarely seen extremes of pure competition and monopoly. The barriers to entry will vary widely among these industries.

A duopoly occurs when two companies own all or nearly all of the market for a given type of product or service.

Oligopolies are industries that are dominated by a small number of large firms (e.g., automobiles, oil, and steel). In these industries, barriers to entry are typically related to economies of scale and/or economies of scope. As described earlier, economies of scale are obtained when the average cost of producing an item declines as the number of items produced increases. Economies of scope relate to vertical integration. The ability to control various stages of production presents an enormous barrier to new entrants.

Ownership of raw materials, such as Alcoa's historic dominance of aluminum producing bauxite or De Beers near monopoly in diamond production, also create barriers to entry. Barriers also include the possession of proprietary knowledge, such as patents and research. A final barrier is created by the existence of established firms. Brand name awareness and consumer habits make the entry of new firms difficult in most industries.

An example of a natural monopoly is a water utility. The ability for competition is virtually nonexistent given the legal monopoly granted to a municipal water system. In exchange for its legal monopoly, the water utility agrees to be subject to strict government regulation.

Capital Intensity Another barrier to entry is the amount of equipment required and, therefore, the cost of producing a given product. Industries vary based on the amount of capital required to support operations. Manufacturing typically requires a large initial investment in fixed assets (plant and equipment), as well as ongoing capital requirements for maintenance, upgrades, raw material inventory, storage, and transportation. Conversely, a software servicing company may require very little capital.

The more capital intensive an industry, the greater its sensitivity to interest rate changes. If the cost of capital (interest rates) rises, an industry that relies on borrowed capital will face increased pressure to raise prices to maintain profits. These higher prices may, in turn, reduce demand for the industry's products.

Regulatory Issues

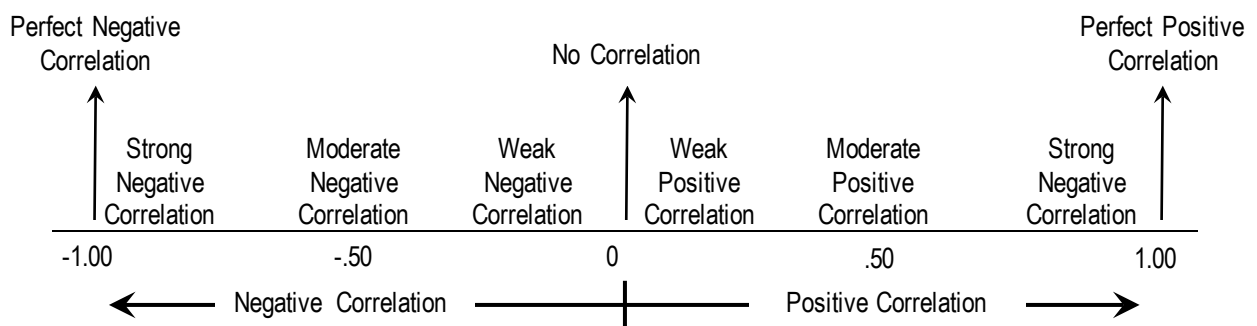
A final factor in industry analysis is government regulation. Certain industries (e.g., banks) are highly regulated and limited in the manner in which they can conduct business. Other industries, such as many Internet-based ventures, have been left virtually unregulated. Regardless of the direct regulation of a particular industry, changing federal regulations regarding tax policy, environmental laws, and labor laws impacts virtually all industries. Multinational firms face the additional problems of both international and country specific laws and regulations. Therefore, an important aspect of industry analysis is an understanding of current and potential future changes to the legal environment in which business is conducted.

Statistical Information

There's a significant amount of data produced as statistical information that may be relevant to a research analyst. In regression analysis, the emphasis is on predicting one variable from the other. Linear regression is best used when relationships tend to trend together or opposite one another. An example is retail sales, which tends to rise or fall according to the level of disposable income. Correlation is used to describe the relationship between two different sets of variables.

If both increase, there's a positive correlation. If one increases while the other decreases, there's negative correlation.

The following drawing summarizes the strength and direction of the correlation:



Correlation is a statistical technique that can show whether pairs of variables are strongly related. For example, a company's revenue and GDP are related. Companies tend to have increased revenue when the GDP is increasing. However, the relationship is not perfect. Companies of the same industry vary in capital structure, and it's easy to consider two companies where the smaller one is growing its revenue at a faster pace than a larger competitor. Correlation can indicate just how much of the variation in company revenues is related to the GDP.

Although this correlation is fairly obvious, some data may contain unsuspected correlations. Analysts may suspect that there are correlations, but are not certain which are the strongest. The exam will not test on the specific correlation calculations, but it may ask a person to determine whether correlation is positive, negative, or nonexistent between two sets of numbers.

Company Analysis

Once an analyst has selected a particular industry as an appropriate target for analysis, there remains the task of selecting a specific company from within all those in the industry. Both quantitative and qualitative methods are employed. Items that are important to a fundamental analyst include a company's financial statements (e.g., the balance sheet and income statement), details regarding the company's product line, the experience and expertise of the company's management, and the outlook for the company's industry.

Product Line Analysis

Based on the industry analysis performed in the previous step, an analyst will locate each of a company's products at its life-cycle stage. If a company produces essentially one product, that industry's potential for growth will heavily influence the growth potential of the firm. In addition, with respect to unit cost and market shares, the position of that company within the industry will affect an analyst's judgment of it.

The situation is more complicated for a company that operates in several industries. Each product line must be analyzed with respect to its industry. Then, an overall judgment about the company's prospects is reached based on a weighting of the individual product line analysis.

Since market share determines the size of an industry's potential market, company analysis requires an understanding of individual market share. Market share leaders are often price leaders. Established firms that have established brand names often have advantages over smaller competitors that may be forced to compete on price, thereby lowering their overall profits. Niche players can develop market penetration by offering products that are tailored to specific needs within the wider industry, while development of unique products is also a key to market differentiation.

Quality of Management

A company's performance is often dependent on its success in the labor market. The ability to attract and retain skilled management should create a competitive advantage. Of course, the cost of recruitment and retention has risen dramatically in recent years and may impact net profits.

A company analyst must have a solid understanding of the firm's business plan and be able to objectively determine the strengths, weaknesses, opportunities, and threats of that plan. Although the firm itself should have diligently completed this analysis, internal prejudice is unavoidable. Analysts who rely on a firm's own projections are not adding value.

Costs and Sources of Revenue and Growth

As previously mentioned, the variables of cost are directly related to bottom line profits. Although some costs can be controlled by a firm, others may not. Reliance on a small number of suppliers, scarce resources, or unstable markets may call into question the likelihood of achieving and maintaining profit growth. An examination of a firm's supply chain will begin with an understanding of the industry dynamics of suppliers, buyers, and distributors. This is followed by an examination of a particular firm's operational policies to determine how well the firm may cope with future uncertainties.

While the analysis of industry and company specific factors will focus the analyst on potential opportunities and pitfalls, these factors are often difficult to quantify. The true costs of production, sources of revenue, and uses of capital require an analysis of the firm's financial statements and a comparison of that analysis with other peer firms.

Financial Statements

Up to this point, the analyst has been dealing with both micro and macroeconomic factors that are often intangible. By examining financial statements, the analyst is given a unique look at the inner workings of a particular company. These historic records may reveal the efficiency, solvency, and value of a given firm. Additionally, when combined with an understanding of potential future economic trends, they may predict future growth potential.

Operating Leverage Operating leverage is the measurement of the degree to which a firm or project relies on fixed rather than variable costs. The higher the degree of operating leverage, the greater the potential dangers from forecasting risk. In other words, if a relatively small error is made in forecasting sales, it can be magnified into large errors in cash flow projections. If the majority of costs for a company or project are fixed, then the costs will remain high while sales are dropping.

Financial Leverage The financial leverage is the degree to which an investor or business is utilizing borrowed money. Companies that are highly leveraged may be at risk of bankruptcy if they're unable to make payments on their debt. Also, they may be unable to find new lenders in the future. However, leverage is not always bad; leverage can increase the shareholders' return on their investment and there are often tax advantages associated with borrowing.

The Balance Sheet

The balance sheet (also referred to as the *statement of financial condition*) represents the financial picture of a company as of a specific date and is divided into three major sections—*Assets*, *Liabilities*, and *Stockholders' Equity*. The document is termed a balance sheet because the total assets will always equal the sum of the total liabilities plus the stockholders' equity.

Therefore, the basic formula is:

Assets = Liabilities + Stockholders' Equity (Net Worth)

This could also be shown as:

Assets – Liabilities = Stockholders' Equity (Net Worth)

Basic Balance Sheet Equation

ASSETS	LIABILITIES
	STOCKHOLDERS'
	EQUITY

Assets represent all of the items owned by the corporation, while the liabilities section indicates all of the items owed by the corporation. The difference between the total assets and total liabilities is the stockholders' equity (also referred to as the company's net worth).

The exhibit below represents a sample balance sheet. An explanation of each area follows.

National Corporation Balance Sheet For period ending December 31, 20XX			
ASSETS		LIABILITIES	
Current Assets		Current Liabilities	
Cash	\$ 43,000	Accounts Payable	\$188,000
Marketable Securities	62,000	Interest Payable	27,000
Accounts Receivable	270,000	Dividends Payable	40,000
Inventories	330,000	Taxes Payable	72,000
Total Current Assets:	\$705,000	Total Current Liabilities:	\$327,000
Fixed Assets		Long-Term Liabilities	
Land	\$ 64,000	9% Debentures due 2035	\$300,000
Plant and Equipment	630,000		
Furniture and Fixtures	280,000		
Less: Accumulated Depreciation	(220,000)	TOTAL LIABILITIES	\$627,000
Total Fixed Assets:	\$754,000		
Intangible Assets		STOCKHOLDERS' EQUITY	
Goodwill	\$ 30,000	6% Preferred Stock, \$100 par value, 500 shares outstanding	\$ 50,000
		Common Stock, \$3.00 par value, 300,000 shares authorized, 200,000 shares outstanding	600,000
		Capital Surplus	52,000
		Retained Earnings	160,000
		Total Stockholders' Equity:	\$862,000
TOTAL ASSETS	<u>\$1,489,000</u>	TOTAL LIABILITIES AND STOCKHOLDERS' EQUITY	<u>\$1,489,000</u>

Components of the Balance Sheet: The Assets Section

The assets section of the balance sheet indicates the items owned by the corporation. There are three basic subsections to this category—*current assets*, *fixed assets*, and *intangible assets*.

Current Assets Current assets are cash and other items that may be converted into cash within a short period (usually one year). These include cash, marketable securities, accounts receivable, and inventories.

It's important to examine the method the company uses for valuing the inventory. In most cases, either LIFO (last-in, first-out) or FIFO (first-in, first-out) methods are used. Under LIFO, the cost of the last item produced is applied to the price of the first item sold from inventory. The FIFO method applies the cost of the first item produced to the money received from the first item sold.

In a period of rising prices, FIFO will result in a greater profit because a lower cost basis is used. Therefore, the company will report larger profits and pay a larger amount of taxes. On the other hand, using LIFO during an inflationary period will result in lower profits and lower taxes.

Fixed Assets Fixed assets are items that are used by the company in its day-to-day operations to create its products. This section will list the company's physical property such as land, buildings, equipment, and furniture.

With the exception of land, fixed assets lose some of their value each year due to normal use. The IRS allows a company to claim this wear and tear on assets as a deduction against income (referred to as *depreciation*). On the balance sheet, fixed assets are shown at a value of their original cost less accumulated depreciation.

Intangible Assets Intangible assets don't have any physical value, but add substantial value to a company. Some intangible assets differentiate the company from competitors and are proprietary such as patents, trademarks, franchises, and copyrights. Goodwill is another intangible asset which represents the amount that was paid above the value of the assets to acquire the company. Factors contributing to goodwill include a company's potential earning power due to its reputation in the marketplace, customer relations, and skilled staff.

The Liabilities Section

The liabilities section indicates the company's debts. Some of these debts must be paid in a short period (current liabilities), while other loans don't need to be repaid for many years (long-term liabilities). Current liabilities are those debts that will become due in less than one year and are easily identified by the word "payable." Included in this section are accounts payable (the amount a company owes for goods and services that are purchased on credit), interest payable, notes payable, and taxes payable.

Long-term liabilities are debts that are incurred by a corporation which become payable in one year or more. Examples include bonds and long-term bank loans.

Working Capital Working capital, often referred to as net working capital, equals the company's total current assets minus its total current liabilities. Working capital represents the net amount of a company's liquid resources. Changes in current assets or liabilities will effect a company's working capital.

Book Value and Book Value Per Share Book value equals a company's total assets less intangible and total liabilities. Book value per share for a company is the total equity divided by its number of shares outstanding.

The Stockholders' Equity Section

This section represents the company's net worth, which indicates the shareholders' ownership interest. Listed in this section are the different classes of stock, retained earnings, and capital surplus.

Preferred stock is listed in the balance sheet based on the par value of the outstanding shares. The typical par value for preferred stock is \$100. Common stock is listed in the balance sheet based on the par value of the company's outstanding common stock. However, the par value of common stock is only used for bookkeeping purposes and doesn't influence the market price of the stock.

Capital surplus (also referred to as *paid-in capital* or *paid-in surplus*) is the amount of premium paid by shareholders above par value for shares being sold to the public by the corporation. This doesn't include funds the company derives from business profits, which are referred to as retained earnings (also referred to as *earned surplus*). Retained earnings represents net profits that have been retained for future use by the corporation. Dividends are typically paid from a corporation's retained earnings.

The Income Statement

An income statement (also considered a *profit and loss statement*) indicates a corporation's performance during a specified period. The purpose of this statement is to detail the company's net income (or net loss), which is accomplished by offsetting the revenues by expenses. A sample income statement, along with an explanation of each component, is shown on the following pages.

National Corporation Income Statement For period ending December 31, 20XX	
Sales	\$660,000
Less:	
Operating Expenses:	
Cost of Goods Sold	240,000
Selling and Administrative Expenses	<u>120,000</u>
	300,000
Less:	
Depreciation Expense	<u>80,000</u>
Operating Income	220,000
Plus:	
Other Income	<u>30,000</u>
Earnings Before Interest and Taxes	250,000
Less:	
Bond Interest Expense	<u>27,000</u>
Earnings Before Tax	223,000
Less:	
Taxes (34% Rate)	<u>75,820</u>
Income from Continuing Operations	\$147,180
Loss from Discontinued Operations	<u>40,120</u>
Net Income	<u><u>\$107,060</u></u>

Notice that while interest expenses are deducted before taxes are paid, dividends are paid from net income.

Components of the Income Statement

Sales represent the total money received and amounts billed, but not yet collected, from the company's primary source of business. Sales are reduced by day-to-day operating expenses to arrive at operating income. Operating expenses reflect the daily costs of doing business, which include the amount claimed for the depreciation of fixed assets.

Operating income is adjusted for other income (or expense) that's not generated by normal operations, leaving earnings before interest and taxes (EBIT). Other income or expense represents income generated by investments (dividends and interest). This line also reflects nonrecurring items such as a gain or loss from the sale of an asset, severance payments, losses from storm damage, acquisition costs, plant closing costs, restructuring costs, write offs, write downs, and impairment of goodwill. These are pre-tax items that will either increase or decrease earnings before interest and tax (EBIT).

Once EBIT has been calculated, interest expense is deducted to determine the taxable income. Since interest expense is tax-deductible, it's also referred to as the corporate tax shield. Based on the company's tax rate, taxes are subtracted to derive its income from continuing operations.

If the company has discontinued operations from the sale of a group of assets, such as an entire division, the revenue and expense from those activities are separated from the revenue and expense of the continuing operations and are reported separately on an after-tax basis. Income from continuing operations plus or minus the gain or loss from discontinued operations equals net income.

Sources of Information

In order to accomplish the types of analyses described in this chapter, the analyst must have access to data about the market in general, various industries, and individual companies. Much of this information is readily available from a variety of sources. It's unlikely that an analyst will need to worry about a lack of information. Instead, deciding which of many sources to investigate is a more common problem. Some of the types of sources available are mentioned next.

Industry Analysis General references that catalog sources of industry information include publications such as *Business Information Sources*, *Data Sources for Business and Marketing Analysis*, and *Directory of Business and Financial Services*.

Government Publications The North American Industry Classification System (NAICS) is the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy. The U.S. Census Bureau also publishes the *Census of Manufacturers* and *Census of Retail Trade*. The IRS publishes the *Corporation Source Book of Statistics of Income* that provides annual financial information by industry based on corporate tax returns. The U.S. Department of Commerce issues various publications that include widely used statistics such as the Producer Price Index (PPI), the Consumer Price Index (CPI), and the U.S. Industrial Outlook that forecasts prospects for over 350 industries.

Surveys With respect to plans for spending and investment, the intentions of businesses and consumers are analyzed through the use of surveys or polls. Attitudes of businesses regarding spending on plant and equipment are assessed by organizations such as the Commerce Department, the Conference Board, and McGraw-Hill, Inc. Surveys of "consumer confidence" by the Survey Research Center at the University of Michigan assess spending intentions in the consumer sector.

Industry Studies Some industry studies include both longer book-length studies, often written by academics, and shorter reviews published by broker-dealers and research firms. However, these studies are not available for all industries.

Trade Press/Trade Associations Many industries have specialized publications that follow current developments in companies, products, and markets. These can often be identified by the staff of industry trade associations. Trade associations also frequently publish their own materials with general information about the industry. Trade associations can be located through the Encyclopedia of Associations.

Business and Financial Press There are many general business publications available including *The Wall Street Journal*, *Investors Daily*, *Barron's*, and *The Financial Times*.

Financial journals tend to emphasize the theoretical aspects of investing. Much of Modern Portfolio Theory was developed in articles in these publications, which include:

- *Journal of Finance*
- *Journal of Financial and Quantitative Analysis*
- *Harvard Business Review*
- *Journal of Financial Economics*
- *Journal of Economic Theory*

Business Information Services Standard & Poor's Industry Surveys, focuses on a comparative analysis of sales and earnings. Dun & Bradstreet's Business Rankings ranks several thousand U.S. businesses by sales and employees.

Company Analysis One specific handbook to assist an analyst in locating information about specific companies is *How to Find Information About Companies*. Other general guides include:

- *Dun's Business Identification Service*
- *Standard & Poor's Register of Corporations*
- *Thomas Register of American Manufacturers*

SEC Filings Primary sources of information about publicly traded companies are SEC filings. In general, information that's filed with the SEC is available to the public. These sources include:

1. Registration statements and prospectuses for new issues filed under the Securities Act of 1933
2. Form 10-K – an annual audit financial statement filed under the Securities Exchange Act of 1934
3. Form 10-Q – the unaudited quarterly financial statement
 - The required timing for the filing of 10-K and 10-Q reports is based on the category of the filer.
 - A *large accelerated filer* has a public float of \$700 million or more, has filed one annual report (Form 10-K), and has been subject to SEC reporting requirements for least 12 months. This type of company is required to file Form 10-K within 60 days after the end of the fiscal year and Form 10-Q within 40 days after the end of the fiscal quarter.
 - An *accelerated filer* has a public float of at least \$75 million, but less than \$700 million, has filed one annual report (Form 10-K), and has been subject to SEC reporting requirements for least 12 months. This type of company is required to file Form 10-K within 75 days after the end of the fiscal year and Form 10-Q within 40 days after the end of the fiscal quarter.
 - If a company's public float is less than \$75 million, it's defined as a *non-accelerated filer* and is required to file Form 10-K within 90 days after the end of the fiscal year and Form 10-Q within 45 days after the end of the fiscal quarter.
4. Form 8-K – a form filed only when a significant or unusual material event occurs, such as a takeover or merger proposal

5. Proxy statements – filed when a corporation intends to solicit proxies from shareholders regarding issues to be voted on at a shareholder meeting. This document will contain detailed information concerning proposed executive compensation and ownership percentage.
6. Form 13D – filed by any person that acquires more than 5% of the equity of an issuer

Company Documents In addition to required SEC filings, many companies produce additional documents to provide information to shareholders and the public. These may include press releases, product literature, company histories, and annual reports to shareholders.

Company Directories Some examples of directories include:

- *Moody's Manuals* (produced by Moody's Investor's Services) – which provides financial data and description of a company's operations and assets
- *Standard & Poor's Corporation Descriptions* – similar to Moody's Manuals
- *Value Line Investment Survey* (produced by Arnold Bernhard & Co.) – which focuses on investment characteristics of companies.

Several services provide information about the key financial ratios of a company and a comparison to industry norms. These include:

1. *Annual Statement Studies* (produced by Risk Management Association)
2. *Industry Norms and Key Business Ratios* (produced by Dun & Bradstreet)
3. *Almanac of Business Ratios and Industrial Financial Ratios* (produced by CCH)

Electronic Sources Many of the sources mentioned previously are available to computer users online. Even the SEC is moving toward gathering and disseminating information electronically through its EDGAR system (Electronic Data Gathering, Analysis, and Retrieval system). Other providers include:

- *Bloomberg Professional* – Access to market information, tools, news, and research
- *Dow Jones* – Financial news, market data, and research services
- *LexisNexis* – News, company research, and due diligence solutions
- *Moody's Analytics* – Credit research, economic and consumer credit analytics, enterprise risk solutions, structured analytics, and valuation
- *Standard & Poor's* – Credit research and analytics, financial research and analysis for stocks and bonds
- *Thompson Reuters* – Company data, financial analytics, market data, market indices, news, price and valuation data

Field Data In addition to collecting data from published sources, many analysts collect information directly by interviewing various sources and making site visits. The analyst must judge the reliability and objectivity of the information obtained. The following can be direct sources of useful information about specific industries or companies:

- Personnel working for a company being analyzed (e.g., marketing and public relations, salespersons, research staff, corporate officers, purchasing department (for information on suppliers), former employees)
- Vendors, suppliers, and service providers: In addition to suppliers and other vendors, this includes trade organizations, advertising agencies, accountants, commercial and investment bankers

- Customers
- Outside observers, such as business press, federal, state, and local governments, regulatory agencies, unions, international organizations (e.g., U.N.), non-governmental organizations, (e.g., consumer or environmental groups)

Information on Industry Sectors

This next section will examine the various industry sectors in order to determine the key industry drivers. An analyst should also be able to recognize and evaluate the impact of government and industry regulation, as well as the influences of economic, monetary, and fiscal policy on the industry sector. The culmination of these activities should enable the analyst to offer an opinion or make a recommendation about the direction of the industry sector that she's reviewing.

To begin, let's identify several industry sectors and address the key drivers within each sector. For discussion and illustrative purposes, the following seven industry sectors of the economy have been chosen:

1. Paper and Forest Products
2. Media Publishing
3. Beverages
4. Pharmaceutical Services
5. Public Utilities
6. Real Estate: Real Estate Investment Trusts (REITs)
7. Specialty Chemicals

The next step is to review some specific industries and explain certain concepts.

Paper and Forest Products

Key Drivers

1. **GDP** As the economy grows, paper and forest products are needed for advertising, industrial production, printing paper, cardboard, and tissue.
2. **Per Capita Income Growth** When income growth is linked to literacy growth, this growth leads to elevated demand for paper and forest products.
3. **Inventory Swings** As the price of paper and forest products rises, paper users increase inventory stocks to more effectively cost average the product pricing.

Product Demand and Profitability In the paper and forest products industry, product demand is sensitive to economic swings. Demand in this industry is tied to gross domestic product, but varies by region and paper grade (quality).

Key Products

1. **Timber** Fiber obtained from trees is the basis of the paper and forest products industry. Timber is a renewable natural resource. The value of timber increases throughout the trees' growth cycle.
2. **Pulp** The price of pulp is a leading indicator for paper prices and directly impacts producers' earnings and stock prices. Pulp is available from markets worldwide. The price of pulp is volatile, due in part to the high fragmentation of the industry.
3. **Newsprint** Although the demand for newsprint has declined in recent years due to greater use of electronic media (e.g., computers and smartphones), there have been significant increases in paper used for cardboard packaging, toilet tissue, paper towels, and cleaning wipes.
4. **Printing and Writing Paper** Similar to newsprint, the demand for writing and printing paper has declined due to the use of new technology (e.g., email and electronic storage); however, the decline has been more than offset by increases in other areas such as tissue and cardboard for packaging.
5. **Containerboard** Used to make corrugated shipping containers.
6. **Tissue** Proven to be a fast-growing paper business. Returns are higher than other paper grades.
7. **Building Products (e.g., lumber and structural panels)** This business is highly fragmented due to low capital costs.

Cyclical Nature of the Industry The paper and forest products industry is highly cyclical, which produces wide price fluctuations. Within the industry, the cycle generally lasts for four to six years. Within this time, paper and forest products companies may have significant earnings at the peak of the cycle; however, there can be extreme losses in the trough.

Government and Industry Regulation Environmental regulations have had a negative effect over the U.S. and Canadian paper and forest products industry for the past five decades. This as a result of the increasing cost of compliance.

Media Publishing

Key Drivers

1. **GDP** Used to evaluate trends in the advertising marketplace. Economic activity is the driver of spending by businesses on advertising.
2. **Retail Advertising Spending** Retail advertisement revenues are correlated to consumer retail spending. Retailers consider this advertising to be one of the core costs associated with reaching consumers.

The Cyclical Nature of the Industry News publishing is a cyclical business because earnings in the industry are dependent on revenue that's generated from advertisements being placed in their publications. Spending by businesses on advertising is a function of the health of the domestic economy. When the economy contracts, spending on advertising declines and results in a decline in revenue for the news publishing industry. When the economy expands, spending on advertising increases and the revenue received by news publishers' increases. Therefore, earnings in the news publishing industry are cyclical, based on the cyclical nature of advertising, which is based on expansion and contraction in the economy.

Competition Other media outlets, such as magazines, Internet companies, radio broadcasters, television, cable and satellite TV create competition. Each of these industries compete for readers and viewers with the intent of attracting advertisers to increase revenue that's earned from advertising. Internet companies have taken major market share from newsprint publishers. In an effort to compete, many newsprint publishers have established online platforms to display news content and advertising.

Government and Industry Regulation Regulatory limitations exist regarding cross-ownership of newspaper and broadcast properties in the same market; however, these limitations have been eased in recent years. The percentage of ownership permitted within markets has increased. Cross-ownership has profit potential, which could bring about significant mergers and acquisitions in the industry. As a result, media mergers and acquisitions are likely to become more common.

Consolidation Newsprint publication is on the decline as a result of competition from digital media sources. In an effort to compete, news publishers will likely attempt to expand their advertising revenues by buying competitors in the digital space, which leads to increased consolidation in the industry.

Legislative Risk There's legislative risk in this industry as a result of the frequently changing regulations. Although Congress has relaxed limitations on cross ownership of newspapers and radio/TV stations by media providers, Congress could once again tighten limitations on cross ownership if it believes that business combinations are becoming anticompetitive.

Beverages – Soft Drinks

Key Drivers The demand growth of the soft drink industry is driven by GDP growth worldwide. In fact, packaged beverages are viewed as a luxury items in developing nations. The most basic of equations may be applied to the industry, which is that a 1% increase in GDP equals a 1% of unit growth.

Beer

Profitability and Growth Capacity utilization is central to positive returns by brewers. Price increases occur based on inflation, but have also risen in recent years due to premium pricing of higher quality beers. Brewers attempt to balance market share growth with profit growth. Premium priced beers provide a higher profit percentage, but this is partially offset by higher promotional and marketing expenses. Price increases have a greater bottom line impact than volume increases.

Demographics The percentage of the population of individuals reaching beer consumption age has steadily declined since the end of the 1980s.

Short and Long-Term Trends

1. **Health** Older adults are choosing to consume vitamin enhanced and healthy drinks. Beverage makers fill this demand with ready-to-drink teas and juices with added vitamins and minerals.
2. **Convenience** Consumers prefer beverages that they can drink while traveling (e.g., driving, walking, or riding) or working out at the gym). Therefore, the industry is concentrating on single-serve juices or drinks.

Product Demand and Profitability In the U.S. beverage industry (non-alcoholic), the demand is huge and is growing. Growth in international markets is directly linked to GDP growth. However, under-developed markets cannot be readily expanded through advertising alone.

Government and Industry Regulation Government regulation of the beverage industry is administered through the Food and Drug Administration (FDA). For alcoholic beverages, regulations are maintained by the Federal Bureau of Alcohol, Tobacco, and Firearms and also at the state level.

Pharmaceutical Services

Companies in the pharmaceutical services industry provide support to the pharmaceutical industry. These companies engage in the various stages of drug development, from discovery through the commercialization process. Services provided include research and development, manufacturing, analytical testing, packaging, regulatory compliance, and distribution.

Key Drivers

1. Research and development spending is a key driver of the pharmaceutical services industry.
2. Larger capital investments by private companies in the biotechnology industry have resulted in greater expenditures for clinical research and development.
3. Industry consolidation and other external causes create the need for outsourcing of pharmaceutical services.

Cyclical Nature of the Industry Research and development (R&D) spending by pharmaceutical companies and biotechnology companies directly impacts contract research organizations (CROs). CROs are pharmaceutical services companies that specialize in providing research and development for pharmaceutical and biotechnology companies that outsource their R&D on a contract basis. Spending by outsourcing companies can increase or decrease at any time based on the need for new products in the pipeline.

Short- and Long-Term Trends One trend within the industry is the seasonal spending on research and development within the third and fourth quarters of a business year. This trend should positively affect CRO industry sales and earnings since it provides greater stability, predictive results, and resource allocation than the previous feast or famine environment.

In the biotechnology sector, the trend of new product successes and swifter regulatory approvals has led to increased outsourcing for research and development.

Product Demand and Profitability There has been a greater demand for early stage clinical services from biotechnology companies. This has led to increased outsourcing of research and development to CROs.

Government and Industry Regulation The development, testing, and marketing of prescription medications is closely monitored and heavily regulated by the Food and Drug Administration (FDA) in the U.S., but also by government agencies throughout the world. Changes in the regulation of the pharmaceutical industry tend to slow spending on research and development, which could impact the profitability of CROs.

Public Utilities – Natural Gas

Commodity Prices The demand for the commodity natural gas is influenced by weather events. An unseasonably warm winter leads to reduction in demand and increase in inventory levels. The reduction in demand causes the price of natural gas to decline. Conversely, an unseasonably cold winter increases demand and causes the price of natural gas to rise.

Product Demand and Profitability The actual industry earnings may increase if the commodity price of natural gas declines. The benefits of the decline in the price of the commodity may be offset by lower demand due to more moderate temperatures.

Government and Industry Regulation Utilities provide essential services and, without regulations, they could function as monopolies. Rates charged by companies are highly regulated, effectively controlling the earnings of the utility companies. The regulations cover rates, divestitures, acquisitions, and accounting systems.

Real Estate: REITs

The key drivers for the real estate industry are economic growth, employment growth, new business formations, and demographic changes.

Product Demand and Profitability There's a greater demand for REITs in the institutional market than in the retail market; however, the demand is cyclical. To gauge the revenue growth in a REIT, a person needs to examine the pricing and volume of its properties. In order to increase growth, a REIT needs to engage in one or all of the following actions:

- Add properties
- Increase occupancy
- Increase rents

The demand for rental real estate varies between commercial use and personal apartment use. The demand for apartments is a function of both population and employment. An increase in the population leads to an increase in the demand for apartments. Higher employment leads to less rental sharing and an elevated demand for apartments. However, the expense of home ownership is another factor. When the price of home ownership increases, people tend to rent apartments.

Real estate for commercial use is dependent on employment and business demand. Retail businesses create demand when space is needed and they have the ability to pay the rent charged by the owner of the real estate. Two factors dictate a retail businesses' need for space — 1) profitability of the property and 2) the consumers' financial viability. Both of these factors are strongly influenced by location.

Cyclical Nature of the Industry The real estate industry is cyclical and dependent on supply and demand. As demand for space decreases, vacancies increase and causes rental income to fall and real estate business activity to contract. When demand for space increases, vacancies decrease and causes rental income to rise and real estate business activity to expand. Increased demand may also initiate new property development.

Government and Industry Regulation REITs were created in 1960 to allow individual investors to indirectly own real estate. At the time of their creation, REITs were prohibited from actively leasing properties and directly managing them. A REIT was required to use an outside manager or management company to lease and manage the properties.

Later, the Real Estate Investment Trust (REIT) Modernization Act was passed to enlarge the market for this type of security. As a result, REITs now have the opportunity to branch out into other aspects of business to offer enhanced tenant services (e.g. telecommunication services and maintenance services). The Act also created separate guidelines for healthcare REITs which allowed for the operation of their properties without a lease and use of independent contractors for a maximum of six years.

Projections When making projections for the real estate industry, an analyst will examine the following aspects of a REIT:

1. **A Management Team with a Proven Track Record** Ideally, the management team has some level of ownership in the REIT (a positive performance incentive), but not enough to create a conflict of interest.
2. **Consistent Cash Flow Growth** The key to consistent cash flow is location.
3. **Return on Capital and Return on Equity** Investors require a high return on both capital and on equity. Lower levels of performance can be rationalized, but only if the prospects for substantially improved results can be demonstrated. Since the typical owner of a REIT is an institution, investor patience is measured directly through the risk-reward tradeoff.
4. **Stable and Safe Dividend** Investors require an income stream from the investment vehicle. The dividend growth rate has a measurable impact on the trading price of the REIT.

Specialty Chemicals

Specialty chemicals are used in numerous industries, such as electronic manufacturing, the manufacturing of textiles, the production of foods, the production of construction goods, and other industries. Examples include specialty polymers, construction chemicals, electronic chemicals, flavors and fragrances, specialty coatings, oil field chemicals, food additives, plastics additives, feed additives, printing inks, specialty paper chemicals, adhesives and sealants, cosmetic chemicals, lubricating oil additives, water management chemicals, textile chemicals, nutraceutical ingredients, etc.

Key Drivers

1. **The Industrial Manufacturing Business Cycle** The specialty chemical industry's performance is directly linked to the industrial business cycle. Specialty chemicals experience greater demand during periods of economic growth and less demand during recessionary periods.
2. **Feedstock** The manufacturing process for specialty chemicals often requires the use of natural gas or its derivatives, such as propylene or methanol. These are referred to as feedstocks and often comprise 70% of the product's total costs. The forecasting of natural gas prices remains key to projecting the variable costs of companies within this industry.
3. **Currency Swings** A large share of the market for specialty chemicals lies outside the United States. The strength or weakness of the U.S. dollar has a direct bearing on sales success outside of the United States.

When the U.S. dollar weakens against foreign currencies, U.S. manufacturers experience a competitive advantage. This is because the currency of foreign buyers can be exchanged into more dollars, as compared to previous periods, thereby making purchases of chemicals less expensive when buying from U.S. manufacturers. Other industry drivers include gross domestic product, consumer expenditure, and U.S. housing starts.

Product Demand and Profitability Demand for specialty chemicals is driven by the demand for products that use the chemicals. Forecasting begins with the demand for the end product and reverses to determine the anticipated demand for the specialty chemicals. The four categories are electronic chemicals, flavors and fragrances, food additives, and textile chemicals. Growth within these categories range from 1 to 5% depending on the end markets served.

Cyclical Nature of the Industry The specialty chemical markets have sector cyclicity due to volume erosion and inflation in feedstock (natural gas). The customer base (manufacturers) has become better informed and increasingly more selective due to price visibility.

Projections When making projections for the specialty chemical industry, an analyst will consider the following aspects:

1. **Innovations and R&D Focus** The future in specialty chemicals appears brightest for the companies that are able to differentiate by product or through product innovation.
2. **High-Growth Markets** Electronics (including pharmaceutical chemicals) offer greater innovation opportunity and growth potential than flavors and fragrances, food additives, and textile chemicals.
3. **High Return on Invested Capital** Investors seek companies with sound fundamentals. As this industry has matured, the focus on performance metrics and return on investment has become increasingly important.
4. **Customer Base, Product Offerings, and Service** Diversified product offerings have proven to be the best hedge in this cyclical industry. Companies with narrower product bases have sought to retain their client base through greater service offerings.
5. **Global Scale** Specialty chemical companies reach a global marketplace. The diversification of the customer base lessens the risk of geographic cyclicity.

Apply Your Knowledge

The last section will present a sample of information gathered by an analyst in the form of a fact sheet and consider some typical questions related to those that are addressed when preparing a report.

Restaurant Industry Overview

Industry revenues: \$720.4 billion; 3.95% of GDP, 25-year sales growth rate of 4.5%

Broad Industry Segments Include:

Quick Service Restaurants (QSR): mature, heavily franchised, cash businesses, low average check, heavily weighted toward lower income and youthful consumers

Full Service Dining Some sectors have double digit growth; more menu selections, heavily weighted toward baby boomers and middle income customers. Includes higher end establishments with specialized menus.

Competition Eat-in versus eat-out. Within QSF: quality, consistency, cleanliness, and value; occasional price war

Full Service Dining Similar factors plus ambiance, lifestyle trends, and variety

Revenue Drivers Long-term demographic/secular and lifestyle choices, including aging population, dual income couples, longer working hours, disposable income. Growth engine is same store sales (influenced by traffic, mix, price increases) and new unit growth.

Cost Structure Food, labor, occupancy (including energy) represent roughly equal percentages of expenses; these external forces are difficult to manage. Labor turnover a major concern; operational efficiencies important in low margin business.

Financials, Highlights Negative working capital is common, leverage is reasonable, cash flows are predictable. Cash used for capital expenditures and share buybacks. EBITDA margins mid-teens, net profit margin in high single digits for the highest tier firms.

Stock Selection and Valuation Drivers Supply/demand factor (unit growth in restaurants and same store sales or comps), P/E and Relative P/E (PEG). P/Es correlate well with sales growth and earnings growth. Growth of 1x is the norm, but disparities exist.

Industry Risks Overly rapid unit growth, food cost increases, obesity, minimum wage

Company Risks Failure to meet service standards, provide value, market saturation, and evolve through lifecycle

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CHAPTER 3

Financial Statement Analysis



GAAP Reporting

The standards set forth for corporate reporting of earnings and other business transactions are governed by an extensive set of rules, regulations, guidelines, and operational procedures known as *generally accepted accounting principles (GAAP)*. GAAP standards and procedures provide guidance for accountants, auditors, and governmental regulators and present corporate financial data so as to maintain credibility with the providers of capital to the firm. Compliance with generally accepted accounting principles indicates an accurate presentation of the company's financial position and allows anyone who examines the financial documents to accept them as meeting the necessary standards. GAAP reporting sets the basis for all of these activities since its provisions are set forth by either an authoritative entity or are widely accepted as standard practice in the industry.

In order to establish the guidelines and regulations needed to standardize the financial reporting requirements, various federal and industry agencies have been created to make and interpret the rules and disseminate information about changes that may occur in accounting for reporting of business transactions. Chief among the federal agencies are the Securities and Exchange Commission (SEC) and the Financial Accounting Standards Board (FASB). Announcements and releases issued by the FASB are normally distributed in the form of Statements of Financial Accounting Standards (SFAS). A major objective of these agencies is to establish and improve the standards of financial accounting and reporting and to provide education to the various users of financial information including auditors, issuers, and the public at large.

One of the problems that may arise, however, is that of reporting non-standard events of a significant nature that may occur within the company. Unusual or non-recurring items reported under GAAP may actually give misleading information to investors and other providers of capital. Such events as changing methods of inventory valuation, changes from LIFO to FIFO, or depreciation method changes may significantly impact one quarter during the year and may not appear in ensuing accounting periods. In such cases, the company may deem it appropriate to report pro forma earnings.

Pro Forma Reporting

Under certain conditions such as when a company has significant changes in its financial activities, it may be advantageous to report on a pro forma basis. Pro forma reporting is a statement of the company's financial status if the unusual or significant items are excluded. Earnings are restated so as to eliminate the effects of items that may impact the financials in one reporting period and no others. A question that may arise; however, is should a pro forma report of earnings be used? In a recent release published by the SEC, the commission indicated that "pro forma financial information can serve useful purposes". In its release, the commission commented that, "Public companies may quite appropriately wish to focus investor attention on critical components of quarterly or annual financial results in order to provide meaningful comparisons . . ." The intention is to provide information that may be similar to management's interpretation and discussion of its own results.

For the analyst reviewing the financial reports of the company, pro forma reporting may have significant considerations. Paramount among the concerns is the fact that the company reporting pro forma information has departed from typical accounting conventions. The significant items in its financials must be identified and adjustments must be made in order to make proper comparisons with other companies or over different time periods. It should also be noted that pro forma information is determined by selectively editing GAAP information. Selectivity may present the company in a better financial position than GAAP accounting would, which could be misleading. The analyst must take special care to ensure that the information presented doesn't mislead or unfairly present a picture of the company.

Among the main concerns of the regulators and oversight committees include the following points.

Antifraud Provisions Even though a company may use pro forma information when reporting its financial status, the antifraud provisions of the securities acts still applies. Companies must take special care so as not to mislead investors, regulators, and the public at large when presenting this type of information.

Basis for Presentation Since the company is selectively using information, it should also provide the basis for its statements. Significant or non-recurring items need to be indicated and the reason for including or excluding them should be addressed. Non-balance sheet items that are omitted from financial reporting could result in a reported profit for the period when in actuality, the company would report a loss if such items were included.

Timing of the Report The analyst must be particularly aware of the timing of the release of information. Since there are no prohibitions against pro forma reporting, companies may choose to release unaudited information before audited GAAP statements are available. The unaudited information has no basis for reconciliation and unless the company prepares a reconciliation statement for release with the GAAP reports, the analyst has no method of verification.

These basic concerns are being addressed and reviewed by the SEC and other rule making agencies in order to establish guidelines for pro forma reporting and standardization for the use of such information. In the wake of the recent collapse of some major corporations and their auditors, the SEC and FASB have issued alerts and statements of financial accounting standards concerning pro forma reporting and its potential to mislead investors. The analyst must pay particular attention to the pitfalls that may be encountered through the use of pro forma information.

Aggressive Accounting Aggressive accounting refers to a method of accounting that's used to report lower expenses and higher income, or to overstate assets while understating (not recognizing) liabilities. Accounting standards require firms to follow promulgated opinions; however, in some cases there are gray areas. Capitalizing software would defer the recognition of expenses, effectively boosting earnings. Capitalizing research and development increases the cost basis of the future product, but the costs are not taken as an expense in the short run, thereby boosting earnings.

Expensing stock options lowers earnings, and writing off goodwill through impairment (as opposed to amortizing over a lengthy period) also lowers earnings. In regard to accounting practices for defined benefit pension plans, using a low discount rate is conservative, and using a higher discount rate is aggressive. The present value calculation is based on dividing the liabilities by $(1.0 + \text{discount rate})$. Therefore, the higher the discount rate, the lower the present value of the fund's long-term liabilities.

Special Issues in Accounting

Depreciation One of the fundamental concepts of GAAP is to match costs and revenues over a specific period. Some items like labor, rent, and incidental office supplies are immediately expensed. However, fixed assets (e.g., buildings, equipment, vehicles, and fixtures) provide benefits over many years. The allocation of costs reflecting the wearing out or deterioration of assets is called depreciation, and represents a non-cash charge to the income statement. Because the calculation of depreciation is based on historical costs, profits may be overstated to the extent that replacement costs are greater than historical costs.

Straight line depreciation produces a constant depreciation expense and a constant decline in the carrying value of the asset. Accelerated depreciation allows for a faster rate of decline in the asset's value in the early years of asset ownership, leading to a higher depreciation charge. Current IRS Regulations, called the Modified Accelerated Cost Recovery System (MACRS), require that the entire cost of the asset be written off over its depreciable life (no salvage value). Whenever possible, firms will use an accelerated method for tax reporting purposes, and straight line depreciation under GAAP for reporting to stockholders. The consequences of these actions will be discussed under "Deferred Tax."

Assume that a firm buys machinery for \$20,000 with an estimated useful life of five years. Under the straight line method, the annual depreciation expense is:

$$\frac{\text{Original cost}}{\text{Estimated useful life}} = \frac{\$20,000}{5 \text{ years}} = \$4,000$$

The five-year depreciation schedule would look like this:

Year at year end	Carrying Value Expense	Depreciation
1	16,000	4,000
2	12,000	4,000
3	8,000	4,000
4	4,000	4,000
5	0	4,000

The firm's balance sheet will show the initial cost of the asset less accumulated depreciation.

With few exceptions, the only accelerated method allowed by the IRS is the double declining balance or 200% declining balance method. The straight line rate of 20% per year is doubled, and 40% becomes the multiplier for the remaining carrying value in the asset account. Below is the depreciation schedule for the purchase of \$20,000 in machinery with a useful life of 5 years.

Year	Carrying Value	Depreciation Expense	Accumulated Depreciation	Book-Value at Year-End
1	20,000	$40\% \times 20,000 = 8,000$	8,000	12,000
2	12,000	$40\% \times 12,000 = 4,800$	12,800	7,200
3	7,200	$40\% \times 7,200 = 2,880$	15,680	4,320
4	4,320	$40\% \times 4,320 = 1,728$	17,408	2,592
5	2,592	2,592*	20,000	0

* The remaining amount of depreciation will be taken in year five to fully depreciate the asset.

NOTE: The calculations are shown for illustrative purposes only; real world amounts will vary considerably from these figures because MACRS allows only a half-year of depreciation in the first year the asset was placed in service. This is often referred to as the mid-year convention.

Accounting for Investments in Stock The *cost method* is used when Company A owns only a nominal percentage (i.e., less than 20%) of Company B's outstanding shares. Company A is presumed to have little control over Company B and dividends are the only return recognized. However, if Company B's shares experience a significant or a perceived permanent decline, Company A should recognize a loss by reducing the carrying value of its investment.

The *equity method* of accounting is used when Company A owns 20% to 50% of Company B's shares. Due to its large ownership position, Company A may exert considerable influence over Company B concerning management issues and dividend policy. Company A may refer to Company B as an affiliate. When Company B reports its earnings, Company A will record its proportionate share of the P&L in its investment account.

For example, if ABC owns 30% of XYZ, and XYZ had \$120,000 of income, ABC would have reported an additional \$36,000 (30% of \$120,000), and would report this amount under other income.

Consolidated financial statements are prepared to record the results of a parent and subsidiary company; this format is required when the acquiring firm purchases more than 50% of the outstanding shares of the other entity. The method to account for an acquisition is the *purchase method*.

The principal characteristics of the purchase method are summarized below.

Purchase Method

1. Mandatory for use in a cash deal
2. Viewed as an acquisition of new assets
3. The book value used for plant, equipment and other fixed assets may be adjusted (increased) to market value.
4. Goodwill (an intangible asset) is created if the acquisition price exceeds the fair value of the assets acquired.
5. Goodwill may be charged against earnings, if an impairment has occurred.

To calculate the amount of goodwill created after the acquisition, the fair market value of the company's net tangible assets are subtracted from the offer value. The net tangible assets are found by subtracting the company's liabilities, existing goodwill, and intangibles from the total assets.

Non-Controlling Interest (Formerly Referred to as Minority Interest) When consolidated financial statements are prepared (remember, this accounting process must be employed when a concern owns more than 50% of another company), 100% of the assets and liabilities of the subsidiary are reflected on the books of the parent. Suppose that the parent owns 80% of the stock of the subsidiary, with the residual 20% owned by someone other than the parent. In this event, an account called non-controlling interests is shown in the equity section of the parent's balance sheet.

An additional entry is also made on the income statement called "Non-controlling interest in net earnings of consolidated companies." This account shows how much income in a given period belongs to non-controlling or minority shareholders (rather than the parent company).

*Company X owns 70% of Company Z and Company Z reports net income of \$200,000.
Company X and Company Z would consolidate their income statements. A deduction of \$60,000 (30% x \$200,000) is made under "non-controlling interest."*

Capitalization of Interest Generally accepted accounting principles require a company to capitalize certain costs in addition to the purchase price of the asset. This means that the depreciable base of the asset will reflect installation, transportation, legal costs, commissions paid, interest to finance the construction phase and taxes. All of these costs are necessary to prepare the asset for use.

The cruise line industry provides an example where significant amounts are borrowed and interest is capitalized. Any calculation of fixed charge coverage should include the interest that's capitalized.

FASB Interpretation # 34 requires firms to capitalize the interest incurred during the period of construction of non-current assets, such as buildings or machinery. Suppose that Riverboat Casino borrows money specifically to finance the construction of a seaworthy casino. The project takes one year to complete, and interest expense is \$500,000.

The previous information is capitalized in the following accounting entry:

	DR	CR
Riverboat vessels:	\$500,000	
Interest payable:		\$500,000

Any further interest on funds borrowed will be expensed once construction has been completed. The total amount in the riverboat vessels account includes the construction costs plus interest incurred. This forms the depreciable base for the newly constructed vessel.

During the constructing period, the interest paid is not deducted in determining the net income of the company. As a result, earnings are boosted in the short run.

Leasing Rather than purchasing equipment or facilities, businesses may find leasing more attractive. The lessee is not required to put up its own capital, nor borrow to buy the asset. This leaves its debt ratios and coverage ratios unaffected. (However, many analysts treat lease payments as a type of fixed charge when computing coverage ratios.) The lessee may not want to own the asset for its useful life and doesn't want to assume the risks associated with resale.

The owner of the asset (the lessor) may deduct depreciation, and potentially profit from any appreciation in asset values. The latter benefit is subject to market factors, such as loss in popularity and technological obsolescence (e.g., computer and communications equipment).

If a lease contract is entered into for only a portion of the asset's life, the cash payment made by the lessee to rent the use of the asset is recorded as a rent expense. This type of lease arrangement is referred to as an operating lease.

If the lessee agrees to lease an asset for most of its useful life, it's referred to as a finance lease. In a way, the lessee is actually buying the asset through installment payments. A lease qualifies as a finance lease if one of the following criteria is met:

1. The title passes to the lessee at the end of the lease term.
2. The lease contains a bargain purchase element. (It's likely that the option will be exercised because the price is attractive.)
3. The lease term is equal to or greater than 75% of the useful life of the asset.
4. The present value of the minimum lease payments is greater than 90% of the fair market value of the leased asset.

In a finance lease, the firm acquiring the lease will report the lease payment as a depreciation expense and interest expense on its statement. The interest expense is calculated on the lease obligation and is reduced over time. As a result, the lease expense will be higher in the early years and will decline in the later years. Depreciation or amortization on the leased asset is generally calculated on a straight-line basis.

Intangible Assets Besides those assets that are readily identifiable by their physical form or an accounting record that evidences value, such as accounts receivable, businesses may acquire or produce intangible assets. The difficulty in determining the future expected benefits makes this area of business evaluation problematic.

In June of 2001, the Financial Accounting Standards Board (FASB) issued Statement 142, *Goodwill and Other Intangible Assets*. The statement superseded APB Opinion No. 17, *Intangible Assets*. Its provisions became effective December of that year.

Opinion 17 presumed that goodwill and other intangible assets were wasting assets (that is, finite lived), and therefore the amounts assigned to them should be amortized in determining net income; Opinion 17 also mandated an arbitrary ceiling of 40 years for that amortization. FASB 142 doesn't presume that those assets are wasting assets. Instead, goodwill and intangible assets that have indefinite useful lives will not be amortized, but rather will be tested, at least annually, for impairment.

That Goodwill Impairment Test is a two-step approach conducted at the reporting unit level. The reporting unit is referred to as the lowest level of an entity. Business units, subsidiaries, operating units, and divisions are all examples of reporting units.

Step 1: The first step is to identify potential impairments by comparing the fair value of an operating unit to its carrying amount, including goodwill. If the fair value of the reporting unit is greater than its carrying amount, goodwill is not considered impaired.

Step 2: The second step, which is only required if there's an impairment identified in the first step, is to compare the implied fair market value of goodwill to its carrying amount. If the carrying amount of goodwill exceeds its implied fair value, an impairment loss is recognized equal to the excess and presented as a separate line item on the financial statements.

Intangible assets that have finite useful lives, such as patents, copyrights, and trademarks, will continue to be amortized over their useful lives, but without the constraint of an arbitrary ceiling. In estimating useful lives, consideration should be given to legal, regulatory, and contractual provisions. Also, the effects of obsolescence, rate of technological change, and the expected actions of competitors should be considered.

Revenue Recognition When a company agrees to complete a project by signing a contract it has two methods of recognizing revenue. Under the percentage-of-completion method revenue is reported each period as the company incurs costs to complete the project. Most long term contracts are recorded using this method. Under the completed contract method neither revenue nor expenses are recognized until the contract is completed.

It's important for an analyst to understand how a company's earnings might change if a switch is made from one method to the other while a contract is ongoing. For example, if a company has been using the percentage of completion method and switched to the completed contract method, its reported revenue will be lower until it completes the contract. The company didn't change the terms of the contract but changed the method of recognizing the revenue from the contract.

Pensions Under a pension plan, an employer agrees to provide certain benefits upon an employee's retirement. These payments occur many years in the future, representing a non-current liability to the employing organization.

In a defined contribution plan, the employer makes a set annual contribution (usually a percentage of the employee's annual salary), to a deferred account managed by a trustee. (Pension assets are separated from the assets of the firm.) The company accrues a pension expense each year and, if the contribution is made, the company is entitled to a tax deduction.

A defined benefit plan guarantees an employee a certain pension benefit, tied to the length of employment at the firm and the employee's salary. The employer has to plan for benefits payable in the future, taking into account employee turnover, retirement age, salary levels, and return on investment. Based upon several assumptions the employer accrues a pension expense. A Federal law, ERISA (Employee Retirement Income Security Act) establishes minimum amounts that must be funded each year, but the amount expensed and the amount contributed may differ dramatically. If the amount of pension fund assets are greater than the pension plan liabilities, the plan is overfunded. If liabilities exceed assets, the plan is underfunded.

Key terms associated with pension expense include the following:

1. **Normal Service Cost** This is the amount that current employees earn toward their pensions. The amount expensed by the firm is the present value of these benefits.
2. **Interest of the Projected Benefit Obligation** This is the liability associated with the accumulated normal service cost, incorporating various actuarial assumptions such as years to retirement and rate of salary increases. Interest accrues on the projected benefit obligation, which is a present value figure. This interest represents another component of pension expense.
3. **Prior Service Cost** Changes to the plan can cause an additional expense, such as an increase in the benefit payable under a defined benefit plan. Accounting standards permit the firm to amortize these changes over the life of its work force.

Accounting for Pension Plans If interest rates decrease, pension fund assets will usually increase due to the type of securities in the plan (a portion of which is invested in fixed-income securities). Pension fund obligations will rise even faster since the long-term pension obligations will cost more to fund the plan in present or current dollars. (The low interest rate environment raises the present value of pension liabilities.) If this trend continues, this could lead to unfunded pension liabilities, which will need to be paid in the future.

Conservative accounting for pension funds includes a low discount rate that increases service costs. An expected return on the pension plan assets that's close to the discount rate increases the cost to the plan. In addition, a high-assumed rate of salary increases leads to higher pension benefit obligations (PBOs) for employees. If a company uses these conservative accounting procedures, it would more accurately reflect the present expenses of a company based on the expected costs in the future.

Deferred Taxes Income reported to shareholders on financial statements can differ substantially from the income reported to the IRS. One of the principle causes is that companies choose an accelerated method of depreciation for tax purposes and straight line for financial reporting. This can create a deferred tax liability attributable to timing differences. If a company's accounting income is greater than its taxable income, a corporation has a deferred tax liability. If accounting income is less than taxable income, an asset called prepaid taxes is created.

Consider the following table for the *Accelerated Corp.*:

1	2	3	4	5	6	7	8	8
Year	Accounting Income (before depreciation and taxes)	Straight-Line Depreciation	Accounting Income	Accelerated Depreciation	Taxable Income	Differential Col. 4 – Col 6	Deferred Tax Liability (Credit) Col. 7 x Tax Br.	Deferred Taxes (Cumulative – L/T Liability)
1	\$800	\$250	\$550	\$400	\$400	\$150	\$67.50	\$67.50
2	\$800	\$250	\$550	\$300	\$500	\$50	\$22.50	\$90.00
3	\$800	\$250	\$550	\$200	\$600	(\$50)	(\$22.50)	\$67.50
4	\$800	\$250	\$550	\$100	\$700	(\$150)	(\$67.50)	\$0.00

Taxes are calculated at 45%. Accelerated Corp. would make the following entries:

Year 1	Income tax expense (\$550 x .45)	\$247.5
	Current tax (\$400 x .45)	<u>– 180</u>
	Deferred tax	\$ 67.5

Year 2	Income tax expense	\$247.5
	Current tax	<u>– 225</u>
	Deferred tax	\$ 22.5

The income tax expense is based upon accounting income of \$550; the current tax expense is 45% times taxable income. The deferred tax liability is 45% of the difference between \$550 and \$400 (\$150 x 45% = \$67.5). After two years a deferred tax liability of \$90 has been created (\$67.5 from Year 1 plus \$22.50 from Year 2). In year 3, taxable income exceeds pretax accounting income and the situation starts to reverse itself.

Year 3	Income tax expense	\$247.5
	Deferred tax	<u>+ 22.5</u>
	Current tax liability	\$270

Year 4	Income tax expense	\$247.5
	Deferred tax	<u>+ 67.5</u>
	Current tax liability	\$315

At this point the deferred tax liability has been eliminated. In actuality, deferred tax liabilities may not be so readily reversed. If additional assets are bought as the company expands, deferred liabilities will grow larger. The analyst may have to conjecture whether such liabilities will ever have to be repaid. The flip side of deferred taxes is prepaid taxes. This situation could arise if an expense is recognized for accounting purposes, but not deducted for tax purposes until later.

Magnet Appliance sells kitchen appliances and has to estimate servicing expenses. Based upon its accounting income of \$600 and its estimated servicing expense of \$100, its accounting taxable income is \$500; this latter number is used to calculate income tax expense. Assume that the actual expense is only \$20. The taxable income for *Magnet* will be \$580, which is the basis for its current income tax liability.

The following entry is made:

Income tax expense (\$500 x .45)	\$225
Prepaid tax	+ 36
Current tax liability (\$580 x .45)	\$261

The prepaid tax is 45% of the difference between \$580 and \$500 ($\$80 \times 45\% = \36). The assumption is that it will reverse in later years as servicing expense increases.

Magnet's income statement would appear as follows:

Income before tax	\$500
Tax liability	261
Less: Pre-paid tax	<u>36</u>
Income tax expense	<u>225</u>
Net Income	\$275

Cash Dividends There's a difference in accounting for cash dividends based on dividend declaration and payment of the dividend. When declaring a cash dividend, dividends payable increase (a current liability) and retained earnings (stockholders equity) are reduced. When the dividend is paid, cash (a current asset) is reduced and current liabilities (dividends payable) are also reduced. Working capital is therefore reduced when you declare a cash dividend, not when the dividend is paid.

Stock Splits and Dividends Consider the following equity capitalization for *Boggle Corporation*:

Common shares: 2,000,000 issued (par value \$2)	\$ 4,000,000
Paid-in Capital in Excess of Par	\$10,000,000

After a 2:1 stock split the balance sheet is adjusted as follows:

Common shares: 4,000,000 issued (par value \$1)	\$ 4,000,000
Paid-in Capital in Excess of Par	\$10,000,000

The shares double and par is reduced in half, but there's no change to the dollar values in the capital accounts.

Suppose Boggle declares a 5% stock dividend when its shares trade at \$25 per share. If there are two million shares outstanding prior to the dividend, the number will increase by 100,000 shares (5% of 2,000,000), causing the common stock account to rise by \$200,000 (100,000 x \$2 par). The retained earnings account will be reduced by the market value of the stock (\$25), multiplied by the increase in the number of shares (100,000), or \$2,500,000. In order to balance the accounting entry, the paid-in capital increases by the difference between the two, or \$2,300,000.

The par value of common is not reduced for small stock dividends. GAAP requires that large stock dividends (25% or more), reduce the par value instead of using the fair market value of the stock to adjust the capital account. Retained earnings would not change in a large stock dividend.

Differences between the Effects of Stock Splits and Stock Dividends		
Item	Stock Split	Stock Dividend
Total paid-in capital	No change	Increase
Total retained earnings	No change	Decrease
Total par value (common stock)	No change	Increase
Par value per share	Decrease	No change

International Activities When a U.S. based company holds units of foreign currency, there's a possibility of a foreign currency transaction gain or loss. Consider the following:

1/1/09 Ingalls Corporation holds 10,000 Swiss Francs at an exchange rate of 1.6 SF to the U.S. dollar. This represents \$6,250 ($10,000 \div 1.6$).

12/31/10 The exchange rate is now 1.3 SF to the U.S. dollar. The dollar equivalent is \$7,692 ($10,000 \div 1.3$).

Ingalls Corporation would show a gain of \$1,442 on their income statement ($\$7,692 - \$6,250 = \$1,442$).

Another adjustment may occur when a U.S. company has a payable denominated in a foreign currency. For example:

12/15/09 Concord Corporation purchases goods from its French supplier for 90,000 Euros, payable on 1/15/10. The exchange rate at the time of the transaction is 1.15 Euros to the U.S. dollar. Under GAAP, the current rate or spot rate is used to record the transaction as follows:

Inventory	78,260
Accounts Payable	78,260

At year end, the exchange rate is 1.25 Euros to the U.S. dollar. The accounts payable should be adjusted to reflect the weaker Euro. The accounts payable is now 90,000 Euros to .80 U.S. dollars or \$72,000. The difference between \$78,260 and \$72,000 (\$6,260), is the foreign exchange gain with a corresponding reduction in accounts payable.

If the Euro strengthens between year end and when the account is settled, Concord Corporation will experience a foreign exchange loss.

When a company makes a sale to a “foreign” client, similar adjustments may need to be made.

Suppose that Export Corporation sells goods to Cerrantex Corporation, a Mexican firm, for 30,000 pesos when the exchange rate is 3 pesos to the U.S. dollar. On December 10, 1993, Export Corporation records the sale as follows:

	DR	CR
Accounts Receivable:	\$10,000	
Revenue:		\$10,000

At year end, the exchange rate is 3.3 pesos to the U.S. dollar. This causes the accounts receivable to be reduced to \$9,091. The loss is \$909. The entry to record this is:

Foreign Exchange Loss 909	Accounts Receivable 909
----------------------------------	--------------------------------

This entry reflects that Export Corporation will receive fewer dollars for the pesos it receives.

Foreign Subsidiaries When a U.S. based company owns a controlling interest in a foreign subsidiary (more than 50% of the outstanding stock), the balance sheet, income statement and cash flows must be combined to make them appear as one entity.

The U.S. company must engage in a process that’s referred to as translation, whereby foreign financial statements are converted into U.S. dollars. GAAP allows two different methods of translation: the current method and the temporal method.

First, one must determine the functional currency of the firm (i.e., the currency in which the firm’s transactions are denominated). This will be either the dollar, the local currency, or some other currency. If the functional currency is the local currency, the current rate method is used in the translation process. If the functional currency is the U.S. dollar, the temporal method is used, and the term *remeasurement* is used in place of translation.

One major difference between these two methods is how a gain or loss is handled. The temporal method passes the gain or loss to the income statement. The current method passes the gain or loss to a separate owners’ equity section of the balance sheet called “Foreign Currency Translation Adjustments.”

Accounting Changes A corporation may change the way it accounts for certain items on a voluntary basis as GAAP allows a choice on some matters. From time to time the Financial Accounting Standards Board issues a new requirement which causes firms to make adjustments.

If a firm changes an accounting method, GAAP requires that the firm retro-actively change the accounting in all previous periods. Then the cumulative difference between the old and new methods must be reported as an income statement item in the year the method changes. These can be very significant items, such as changing the accounting method for income taxes or the accounting for post-retirement benefits.

Inventory Inventory is a significant current asset for manufacturing and retailing firms. It represents either finished goods available for sale or work in process, meaning that additional inputs are needed to complete the product. Businesses hope to convert inventories to more liquid assets, either accounts receivable or cash.

GAAP requires that firms use historical cost to account for the value of their inventory, which is the cost at the time of acquisition. The phrase “lower of cost or market” appears in many financial statements. Here, the term “market” means the replacement cost to reacquire the product. When analyzing an inventory system, a person should be familiar with the following terminology:

- **Cost of Goods Available for Sale** This is the sum of the inventory at the start of the period, plus the purchases made during the period.
- **Cost of Goods Sold** This is an expense item, usually the most significant expense item on a manufacturer’s or retailer’s income statement. It will be listed right below the revenue line. The costs of goods sold is determined by the inventory method selected.
- **Periodic Inventory System** No specific record is maintained to reflect the reduction in inventory when goods are sold. At some interval, the firm stops business and counts the units that are left.
- **Specific Identification Method** A system used when the inventory is very unique and the matching of products sold and invoices is easily accomplished. This system might be used in a business where luxury cars, boats, or furs are sold.
- **First In, First Out (FIFO)** An inventory method that matches first costs with the first units sold. The ending inventory is matched with the cost of the most recent purchases.
- **Last In, First Out (LIFO)** Last costs are matched with the first units sold. Ending inventory is matched with prior purchases; the earliest purchases are assumed to be available for sale.
- **Weighted Average** A blended price, or average, is computed and assigned to all units sold and to those in final inventory.

The following example will illustrate how the inventory method selected influences profitability and the carrying value of inventory. Let's assume that Dustball Corp. sells vacuum cleaners and has computed its inventory using LIFO, FIFO, and a weighted average.

	FIFO	LIFO	Weighted Average
Beginning Inventory			
Vacuum A Cost	\$32	\$30	\$32
Vacuum B Cost	\$37	\$32	\$32
Vacuum C Cost	\$30	\$34	\$32
Total Beginning Inventory	\$99	\$96	\$96
Purchases			
Vacuum D Cost	\$36	\$36	\$36
Vacuum E Cost	\$39	\$39	\$39
Total Purchases	\$75	\$75	\$75
Goods Available	174	171	171

Assume that Dustball sells two vacuums in the next period for a total of \$116.

Sales Revenue	\$116	\$116	\$116
Less: Cost of Goods Sold	\$ 69	\$ 75	\$ 68.4
Gross Margin	\$ 47	\$ 41	\$ 47.6

Depending on the method of accounting for the inventory, the carrying value for the inventory is:

Vacuum C	\$30	Vacuum A	\$30	Average	\$34.2
Vacuum D	\$36	Vacuum B	\$32		
Vacuum E	\$39	Vacuum C	\$34		
Total	\$105		\$96		\$102.6

Which method of accounting should a firm use? GAAP doesn't require that costs be matched with the physical flow of goods. The firm might be motivated to minimize its taxes and show the lowest level of profits, in that case LIFO is the clear choice. Sometimes a firm would like to show its share- holders that it's doing well and report the highest profit. In this example, the weighted average has the highest gross margin. In a period of rising prices, it's not unusual for a firm to switch its inventory method in order to minimize its tax payments, therefore, switching from FIFO to LIFO. FIFO will produce illusory profits during inflationary times because inventory must be replaced with higher cost merchandise. In any event, the firm should utilize an inventory method that accurately reflects costs.

Evaluation of Capital Projects Corporations invest capital with the expectation of receiving a benefit in the future. Therefore, the project should be evaluated in order to determine if the return is commensurate with the risks incurred. The share price of the company will ultimately reflect the success of the company in pursuing worthwhile projects.

One of the methods of evaluating projects is called the *internal rate of return (IRR)*, which is the rate of discount that equates the present value of the expected cash outflows with the present values of the expected inflows.

Mathematically, the IRR is given by:

$$A_0 = \frac{A_1}{(1+r)} + \frac{A_2}{(1+r)^2} + \dots + \frac{A_n}{(1+r)^n}$$

This is the rate that discounts the stream of cash flows A_1 through A_n to equal the initial outlay at time 0.

Let's solve for r : Polaris Corporation can make an expenditure now of \$5,027 and expects to receive \$7,000 in two years. When the number of time periods is limited, as there are only two years, we can try a manual method of trial and error. If $r = 10\%$, the present value of receiving \$7,000 in two years is \$5,785; therefore 10% is too low a discount rate. A 20% discount rate is too high, leading to a present value of \$4,860. At 18%, \$7,000 discounts exactly to \$5,027 (as shown below):

$$\begin{aligned}\text{Present value} &= \$7,000 \text{ times } 1 \div (1+r)^2 \\ &= \$7,000 \text{ times } 1 \div 1.3924 \\ &= \$7,000 \text{ times } .71818\end{aligned}$$

A present value table showing the value of a dollar due at the end of “ n ” years would give a discount factor of .71818, where $n = 2$ years and the discount rate is 18%. The conclusion of this exercise is that the corporation will pursue this project if the internal rate of return exceeds the firm's cost of capital or a minimum benchmark that the firm would like to earn.

The next chapter will examine a similar technique that can be applied in the valuation of common stocks by discounting future receipts to be earned by the investor, specifically dividends over the investor's holding period.

Statement of Cash Flows

In addition to a balance sheet, income statement, and statement of retained earnings, a corporation also prepares a Statement of Cash Flows. It's also called a Statement of Sources and Uses of Funds, and it shows various sources and uses of cash. This statement reconciles changes in a company's cash position from one year to the next by analyzing a firm's operating, investing, and financing activities. This statement is a requirement in all annual reports.

The analyst can tell what really happened at a company by sifting through changes in the working capital accounts and reviewing other transactions. Usually this statement is prepared on a comparative basis over several years. In our example, we will merely state whether a particular entry is a source or use of cash.

Assume that our hypothetical company, Razor's Edge, had a cash position of \$7,000 at the end of 2013 and \$2,400 at year end 2014. During 2014 the following occurred:

Operating Activities		Comment
Net income	\$2,500	Source of cash
Depreciation	\$3,700	A non-cash charge; add back
Reduction in deferred tax	(\$4,000)	Did not affect income; used cash
Reduction in receivables	\$5,000	Clients have paid; a source of cash
Increase in inventory	(\$3,000)	Use of cash
Accounts payable increase	\$6,000	Viewed as a source of cash. If accounts payable has decreased, this is a use of cash.
Increase in cash from operating activities	\$10,200	
Investing Activities		
Increase in plant	(\$9,000)	Company used cash to expand equipment production capability
Payment for business acquired	(\$5,000)	Funds were used to purchase a related business
Decrease in cash from investing	(\$14,000)	
Financing Activities		
Common stock issued	\$6,400	Sale of stock raises cash
Debt retirement	(\$5,000)	Use of cash
Dividends paid	(\$2,200)	Cash paid to shareholders
Use of cash from financing	(\$800)	

The net result of Razor's Edge activities is a \$4,600 reduction in the cash position (\$10,200 from operating, -\$14,000 from investing, and -\$800 from financing). The cash on the company's balance sheet is reduced by \$4,600, from \$7,000 to \$2,400. Other items on the 2014 balance sheet would also change. For example, an increase in inventory (a current asset) is \$3,000 higher in 2014 from 2013, and a decrease in account receivables (a current asset) is \$5,000 lower in 2014 from 2013.

Restructuring Charges Restructuring charges are reflected on the income statement in the operating section, as well as in footnotes of a company's financial statement. This would have a negative impact on the earning of the company. There are also journal entries on a company's balance sheet. For example, if a company is writing down the value of assets and increasing expenses due to lay off workers, it will have employee severance costs. The asset write-downs would impact the balance sheet through lower carrying values of fixed or current assets. Higher liabilities related to severance costs are reflected on the balance sheet. An adjustment would need to be made to lower the retained earnings on the balance sheet to reflect the charges. The operating cash flow will not change since the increase in accrued payables to the terminated employees and decline in the assets value are sources of cash, offset by an income statement loss (as a use of cash).

Operating Leverage This is the measurement of the degree to which a firm or project relies on fixed rather than variable costs. The greater the degree of operating leverage, the greater the potential danger from forecasting risk. That is, a relatively small error made in forecasting sales may result in large errors in cash flow projections. If the majority of costs for a company (or project) are fixed, then the costs will remain high, relative to its declining sales.

Financial Leverage This terminology describes the degree to which an investor or business is using borrowed money. Companies that are highly leveraged may be at risk of bankruptcy if they're unable to make payments on their debt, or if they're unable to find new lenders as current debt matures. Leverage is not always negative; however, borrowed funds can increase the shareholders' return on investment and will often result in tax advantages, due to the ability to deduct the interest expense.

Channel Stuffing Channel stuffing is a deceptive business practice used by a company to inflate its sales and earnings figures by deliberately sending retailers in its distribution channel more products than they're able to sell. Through channel stuffing, distributors temporarily reduce inventory and increase accounts receivables. However, retailers will eventually return the excess items to the distributor, who must then make adjustments to its accounts receivable and bottom line. Channel stuffing ultimately catches up with the company. It cannot maintain sales at the rate it's stuffing. This fraudulent technique is often used by persons seeking to overstate sales and raise the value of their stock holdings. Channel stuffing is illegal. A research analyst must guard against this practice by carefully reviewing a company's financial statements to determine whether the accounts receivable growth rate exceeds the sales growth rate and analyze any significant changes in a company's Days Sales Outstanding (DSO). DSO measures how quickly a company turns its sales into cash.

Net Operating Losses Net operating losses (NOLs) that a company has experienced in past years may be used to offset income. This provision enables a company to use past losses to lower their taxable income by carrying forward these losses. Under IRS guidelines, an NOL may be carried forward indefinitely.

Financial Statement Analysis

An understanding of the conceptual basis of the financial reporting system and the preparation of financial statements is an essential prerequisite to financial analysis. Financial ratios are used to compare the risk and return of different firms in order to help equity investors and creditors make intelligent investment and credit decisions. This chapter will examine some of the fundamental financial tests that are conducted when analyzing a company.

Turnover Ratios

- Asset turnover
- Receivables and payables turnover
- Inventory turnover

Liquidity Analysis Ratios

- Current ratio
- Working capital per dollar of sales
- Quick ratio (Acid Test)

Profitability Analysis Ratios

- Earnings per share
- Fully diluted earnings per share
- EBIT margin
- EBITDA margin
- Equity turnover
- Profit margins
- Return on assets
- Return on equity
- Day sales outstanding

Capital Structure and Capital Market Analysis Ratios

- Debt-to-total capital
- Debt-to-equity
- Debt-to-EBITDA
- Dividend payout
- Interest coverage

To illustrate these financial statement formulas, let's consider the L.R. Gamble Company, a regional manufacturing company. We will identify the key formulas and then apply them to a set of comparative financial statements. The key formulas to be used include the following:

Turnover Ratios

Asset Turnover Asset turnover is an indication of how well the firm uses its asset base in generating sales. A low asset turnover ratio might indicate that the company is tying up capital relative to the other needs of the firm, while a high asset turnover might indicate that fully depreciated or dated assets are being used to generate sales.

The asset turnover is:

$$\frac{\text{Sales}}{\text{Average Assets}}$$

Receivables Turnover An important consideration in measuring the liquidity of a firm is to determine how quickly the company can collect its receivables. A high receivables turnover ratio indicates that the company is collecting on its accounts and has a greater ability to pay its current liabilities.

The receivables turnover is:

$$\frac{\text{Sales}}{\text{Average Receivables}}$$

Payables Turnover The payables turnover ratio is an indication of how quickly the company is paying its suppliers in relation to its cost of goods. If the firm is increasing its receivables turnover ratio while maintaining its payables turnover ratio, it's actually improving its cash conversion cycle.

The payables turnover is:

$$\frac{\text{Cost of Goods Sold}}{\text{Average Payables}}$$

Inventory Turnover The inventory turnover ratio indicates how often the company sells the goods that it produces. Inventory turnover also implies a time frame for processing its goods. A longer time for processing materials indicates a greater amount of capital being tied up in the processing stage while a lower time frame would indicate that the company is selling its inventory faster. In some cases, inventory turnover is calculated by using sales as the numerator; however, a preferred method may be to use the cost of goods sold which excludes implied profits from sales.

The inventory turnover is:

$$\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Liquidity Analysis Formulas

Current Ratio One of the most often used measures of liquidity is the current ratio. The current ratio calculates the ratio of current assets to current liabilities. It indicates how many dollars of current assets are available to pay each dollar of current liabilities.

The current ratio is:

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

If asked to calculate the current ratio after two companies merge, you would need to know if the company used a line of credit. If they did, the amount must be added to current liabilities.

Working Capital Per Dollar of Sales The working capital ratio is the relationship of working capital to net sales. The formula for working capital is current assets minus current liabilities. A high working capital ratio is an indication that working capital needs are high relative to sales.

The working capital ratio is:

$$\frac{\text{Net Working Capital}}{\text{Sales}}$$

Quick (Acid-Test) Ratio The quick ratio is a refined measure of current assets to current liabilities. The thought is that when determining the current dollars available to pay current liabilities, inventories shouldn't be included since they're less liquid than the other current assets. Inventories are therefore deducted from the computation.

The quick or acid test ratio is:

$$\frac{\text{Cash} + \text{Cash Equivalents} + \text{Accounts Receivable}}{\text{Current Liabilities}}$$

Profitability Analysis Formulas

Basic EPS Basic earnings per share indicates the amount of profit generated by the company that's allocated to each share of common stock outstanding. It's calculated by subtracting preferred dividends from the net income of the company and dividing the result by the number of shares outstanding. At times, it may also use total earnings divided by the weighted average share outstanding. The basic EPS excludes adjustments for the conversion of preferred stock and bonds and doesn't consider the exercise of warrants. If these adjustments were made to the number of shares outstanding, the calculation would provide the earnings per share on a fully diluted basis.

The basic EPS is:

$$\frac{\text{Earnings Available to the Common Shareholders (Net Income} - \text{Preferred Dividends)}}{\text{Weighted Average Number of Shares of Common Stock Outstanding}}$$

Dilution A reduction in earnings per share of common stock that occurs through the issuance of additional shares or the conversion of convertible securities. Diluted earnings per share (EPS) looks at all convertible securities are exercised. In other words, diluted earnings take into account if all warrants, stock options, and convertible bonds converted in for stock. This would result in an increase in the number of shares outstanding.

Fully Diluted EPS:

$$\frac{\text{Earnings Available to the Common Shareholders (Net Income} + \text{Adjustments for CSE and OPDS)}}{\text{Weighted Average Number of Shares of Common Stock} + \text{Weighted Average CSEs and OPDS}}$$

CSE = Common Stock Equivalents

OPDS = Other Potentially Dilutive Securities

Anti-Dilution Provision This the right of current shareholders to maintain their fractional ownership of a company by buying a proportional number of shares of any future issue of common stock. A provision in an option or a convertible security, it protects an investor from dilution resulting from later issues of stock at a lower price than the investor originally paid.

Equity Turnover Equity turnover measures the sales of the company in relationship to the components of capitalization that are provided by equity issues. The components of equity capital are common and preferred stock, paid-in capital, and retained earnings.

The equity turnover is:

$$\frac{\text{Sales}}{\text{Average Equity}}$$

Earnings Before Interest and Taxes (EBIT) Margin Interest and taxes are added back to net income in calculating the EBIT margin so as to provide an indication of operating income to net sales. Variations of tax changes and interest expenses are neutralized or eliminated from the computation.

The earnings before interest and taxes (EBIT) margin is:

$$\frac{\text{EBIT}}{\text{Sales}}$$

Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA) Margin The EBITDA margin indicates operating profits as a percent of sales. Depreciation, amortization, interest and taxes are added back to net income so as to eliminate any changes in these components thereby providing a stable indication of operating income to each dollar of sales.

The earnings before interest, taxes, depreciation, and amortization (EBITDA) margin is:

$$\frac{\text{EBITDA}}{\text{Sales}}$$

Gross Profit Margin The gross profit margin is an indication of the amount of each dollar of sales that's available to cover operating expenses and profit. Gross profit is calculated by subtracting the cost of goods sold from net sales and this figure is divided by net sales to determine the margin.

The gross profit margin is:

$$\frac{\text{Gross Profit}}{\text{Sales}}$$

Net Profit Margin The net profit margin is the relationship of net income to sales. The ratio indicates the amount of income (after operating expense, interest, and taxes) available per dollar of sales.

The net profit margin is:

$$\frac{\text{Net Income}}{\text{Sales}}$$

Pre-Tax Margin The pre-tax profit margin is the relationship of the past twelve months pre-tax income to net sales. The calculation uses income from operations and interest payments as a percentage of net sales and is an indication of the profitability excluding taxes.

The pre-tax margin is:

$$\frac{\text{Earnings Before Taxes (EBT)}}{\text{Sales}}$$

Operating Profit Margin The operating profit margin is the relationship of income from operations (operating profit) to net sales. It's calculated by dividing the operating income (gross profit less administrative, selling, and other operating expenses) by net sales.

The operating profit margin is:

$$\frac{\text{Income from Operations}}{\text{Sales}}$$

Return on Assets The return on asset is an important indicator of the profits of the company in relationship to its total assets. It's calculated by adding interest and taxes to net income and dividing this amount by total assets. The calculation is an indication of the earnings generated to invested capital.

The return on assets is:

$$\frac{\text{Net Income}}{\text{Average Assets}}$$

Some analysts will use EBIT instead of net income to calculate return on assets. The exam will usually provide either one or the other if this question is asked.

Return on Equity The return on equity indicates the rate of return that's available to the equity providers of capital to the company. The calculation considers the amount of earnings available to common shareholders after all other capital providers have been paid their entitlement. The rate of return incorporates the business risk as well as the financial risk assumed by the common shareholders.

The return on equity is:

$$\frac{\text{Net Income Available to Common}}{\text{Average Common Equity}}$$

Days Sales Outstanding (DSO) Year-end trade receivables net of allowance for doubtful accounts, plus financial receivables, divided by net sales per day. A decrease in DSO represents an improvement and an increase is considered a deterioration in net sales.

The DSO is:

$$\frac{\text{Accounts Receivable}}{\text{Total Credit Sales} \div \text{Number of Days in the Period Being Analyzed (usually 360)}}$$

Capital Structure and Capital Market Analysis Formulas

Debt to Total Capital The debt to total capital ratio of the firm is a measure of the financial leverage to available capital. It indicates the percentage of the total capitalization of the company made up of debt. The ratio reflects the level that the company is trading on its equity.

The debt to total capital is:

$$\frac{\text{Debt}}{\text{Total Capital (Debt + Equity)}}$$

Debt-to-Equity Ratio The debt-to-equity ratio of the firm is a measure of the financial leverage to available equity (not total capital).

The debt-to-equity ratio is:

$$\frac{\text{Total Debt}}{\text{Total Equity}}$$

Debt-to-EBITDA Ratio The debt-to-EBITDA ratio is calculated by adding short-term and long-term debt and dividing by the earnings before interest, taxes, depreciation, and amortization (EBITDA). The resulting ratio is an indication of the company's leverage; that is, the higher the ratio, the greater the leverage. A high ratio indicates a greater likelihood that a company may default on its debt. A lower ratio is an indication that a company may have the ability to incur additional debt, without a significant increase in the risk of default. The debt-to-equity and debt-to-total capitalization ratios measure debt capital, but not a company's ability to meet its debt service obligations.

$$\frac{\text{Short-Term and Long-Term Debt}}{\text{EBITDA}}$$

Interest Coverage Ratio Interest coverage is an indication of how many dollars of earnings (before interest and taxes) are available to meet interest expenses. The ratio indicates how many times during the year the company can meet its interest payments. A low ratio would indicate a high debt burden while a high ratio indicates low levels of debt to earnings.

The interest coverage ratio is:

$$\frac{\text{EBIT}}{\text{Interest Expense}}$$

NOTE: Although some professionals use the formula of EBIT (operating income divided by interest expense) to find a company's interest coverage ratio, many investment banking professionals use EBITDA rather than EBIT as the numerator. Many bond indentures specifically define the interest coverage ratio as EBITDA divided by interest expense.

Dividend Payout Ratio The dividend payout ratio is the calculation of the percentage of earnings of the company actually paid to shareholders in the form of dividends. It provides an indication of the level of earnings that are used to support dividend payments. A dividend payout policy is an indication of a management decision to distribute a percentage of the earnings of the company.

The dividend payout ratio is:

$$\frac{\text{Dividends}}{\text{Net Income}}$$

Apply Your Knowledge

Now, let's turn our attention to the L.R. Gamble Company's balance sheet and income statement to answer some fundamental analysis questions.

L.R. Gamble Balance Sheets

Use the following financial statements to calculate the indicated ratios and margins:

	December 31, 2018	December 31, 2017
Current Assets:		
Cash	\$929,375	\$930,250
Accounts receivable	1,067,737	765,162
Inventories at cost on the last-in, first-out basis	1,495,351	1,515,983
Prepaid assets	81,514	51,178
Total current assets	<u>\$3,573,977</u>	<u>\$3,262,573</u>
Gross plant, property, and equipment	\$2,308,539	\$2,039,199
Less accumulated depreciation	<u>(945,569)</u>	<u>(760,720)</u>
Net plant, property, and equipment	1,362,970	1,278,479
Investment in subsidiaries at cost	43,132	34,401
Deferred development expenses	<u>\$180,069</u>	<u>\$116,878</u>
Total fixed and long-term assets	<u>\$1,586,171</u>	<u>\$1,429,758</u>
Total Assets	<u><u>\$5,160,148</u></u>	<u><u>\$4,692,331</u></u>
Liabilities:		
Bond sinking fund installment	\$0	\$100,000
Accounts payable (trade)	132,200	153,220
Provision for pensions	163,500	177,300
Provision for federal income tax	417,300	226,675
Other accrued liabilities	146,800	224,974
Total current liabilities	<u>\$859,800</u>	<u>\$882,169</u>
Long-term liabilities:		
Provision for deferred income tax	98,301	103,418
First mortgage bonds	<u>800,000</u>	<u>800,000</u>
Total long-term liabilities	<u>\$898,301</u>	<u>\$903,418</u>
Total Liabilities	<u>\$1,758,101</u>	<u>\$1,785,587</u>
Stockholders' Equity:		
Common stock of \$1.00 per value, 350,000 shares authorized, 295,371 and 268,022 shares issued and outstanding	295,371	268,022
Paid-in surplus	2,198,339	2,152,198
Retained earnings	908,337	486,524
Total equity	<u>\$3,402,047</u>	<u>\$2,906,744</u>
Total liabilities and equity	<u><u>\$5,160,148</u></u>	<u><u>\$4,692,331</u></u>

Income Statement	December 31, 2018	December 31, 2017
Sales (Net)	\$9,990,486	\$8,821,121
Cost of goods sold	(6,755,013)	(6,339,304)
Gross margin	\$3,235,473	\$2,481,817
Selling expense	(1,482,635)	(1,365,342)
General and administrative expenses	(595,605)	(538,517)
Total selling, general, and administrative expenses	(2,078,240)	(1,903,859)
Income from operations	1,157,233	577,958
Bond interest	(48,000)	(48,000)
Income before provision for taxes	1,109,233	529,958
Provision for federal income tax	(487,864)	(219,744)
Provision for state income tax	(65,545)	(33,499)
Net income	\$555,824	\$276,715

Note A: The weighted average number of common shares outstanding for 2018 and 2017 were 281,697 and 268,022 respectively.

Note B: In 2018, the company distributed dividends to the common shareholders in the amount of \$134,011.

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CHAPTER 4

Valuation of Corporate Stock



This chapter will focus on the various methods used to calculate valuation, including discussions of some of the strengths and weaknesses of each approach.

Valuation

When valuing a firm, two main approaches will be examined—*relative valuation* and *discounted cash flow analysis*. Relative valuation attempts to value a company based on its price as compared to its financials, such as revenues, earnings, book value, and enterprise value. Examples include the price-to-earnings ratio, price-to-sales ratio, price-to-book value ratio, and enterprise value to EBITDA. Then these ratios will be compared to other companies within the same sector (peer group members) to determine whether a firm is undervalued or overvalued.

The other methodology—discounted cash flows (DCF)—doesn't make peer-group comparisons. In this form of analysis, the target is viewed independently based on the present value of its expected future cash flows. In many cases, both methods can be applied, using a primary and then secondary method to confirm a valuation.

In merger valuations involving strategic buyers, the examination will focus on the potential market valuation and financial status of the combined entity after the proposed M&A activity is completed. Whether the proposed acquisition is accretive or dilutive to the earnings of the new entity will also be reviewed. Although this chapter will focus on the calculations involved in valuation, it's important to understand that many investors approach an acquisition using numerous other factors. Many buyers limit targets to specific businesses in which they have an expertise. As a famous investor wrote, "We try to stick to businesses we believe we understand." Other investors are less discerning as to the type of target. These buyers attempt to purchase assets they regard as underpriced, regardless of the nature of the potential acquisition's business.

Relative Valuation

Price-to-Earnings (P/E)

One of the key ratios used by investors for security valuation is the price/earnings ratio. The P/E ratio incorporates the important elements of investors' concerns—the price of the security in the market and its earnings per share—and allows comparisons from one company to another. Although the P/E ratio could be calculated using basic or diluted EPS, the more conservative method is to calculate it using diluted EPS.

The price/earnings ratio of a corporation is a comparison of its current price in the market relative to its earnings per share. It's an indication of how the market capitalizes a company's earnings, or what the market is willing to pay for each dollar of earnings. If the market value of the stock is trading at five times its earnings and the corporation's EPS is \$3.20, it's expected that the price per share will be \$16.00. An increase of one dollar in earnings would, in turn, project an increase in the share price by \$5.00. Therefore, if the earnings increase from \$3.20 to \$4.20, the share price should increase from \$16.00 to \$21.00.

In calculating the P/E ratio, the numerator (market price of the stock) is relatively easy to determine. Market data is always available and can be verified with little difficulty. However, the denominator (EPS) is subject to variables including how it's calculated, the time period used, and adjustments to earnings (accounting activities related to non-recurring items). The P/E ratio may be defined using trailing or leading figures and may need to be adjusted in order to make proper comparisons between companies.

P/E Ratio Using Trailing EPS

The trailing P/E of a stock is calculated by dividing its current market price by the trailing 12 months' earnings per share. This calculation is also referred to as the current P/E, earnings multiple, or multiple of the stock. When using the trailing P/E, some problems may occur. A research analyst must be aware of these shortcomings of the measurement in order to compare different companies using the same criteria.

First of all, when comparing two companies, the same time frames must be used when calculating the earnings per share. If differing time frames are applied to the calculation, the comparisons will not be valid or relative for two companies at a specific point in time, or for one company over comparable periods.

Secondly, there may be company-specific non-recurring items that will impact earnings. The company may change methods of accounting or inventory valuation. There may have been a sale of assets, or a revaluation of assets for accounting purposes that will appear on the financial statement in a particular quarter, and will not appear in ensuing periods. Additionally, the earnings may be affected by the cyclical nature of the business. Each of these elements must be taken into consideration by the research analyst in order to apply systematic principles for comparison purposes. Due to the effects of non-recurring items, the trailing P/E may not always be appropriate for the projection of future earnings. In such cases, it may be more pertinent to use the second method of calculating EPS called leading (forward) P/E.

P/E Ratio Using Leading EPS

The leading (forward) P/E ratio is calculated by dividing the common stock market price by the projected earnings for the next year. An advantage of using forward EPS is that there's no need to remove any non-recurring items from the calculation. Since items such as major acquisitions or divestitures may impact earnings during one period and not others, they may be removed from future projections. The main reason for projecting earnings is to focus on the underlying earnings potential of the company without the effects of outside influences.

One of the drawbacks of using the forward P/E ratio is the limitation associated with accurately projecting earnings. The data used to determine earnings may not be readily available and is subject to change over the period of the projection. Non-recurring items may not be easily identifiable, and additional research may be necessary to categorize these items and remove them from the computation.

Cyclical Effects on Earnings

The cyclical impact of an industry affects the P/E ratio computation as well. Cyclicity may occur over a longer time span than the projected period; therefore, the quarterly earnings may not accurately reflect the long-term earnings potential of the company. With companies that are particularly sensitive to business or industry cycles, the P/E may not provide an adequate reflection of the company. If the trailing P/E is used, it may be especially high at the bottom of a cycle and excessively low at the top of a cycle. The P/E ratio could be highly volatile without any change in prospects for future earnings of the company. This counter-cyclical concept is known as the Molodovsky Effect. In order to counter the problems of the Molodovsky Effect, analysts will often use normalized earnings per share. Normalized EPS refers to the level of earnings that the business could achieve under midcycle conditions. One method of normalizing EPS is to use the average EPS for the most recent full cycle.

Relative P/E Many investors use the P/E as a proxy for the earnings growth rate. If the perceived growth is less than the growth of a benchmark index, it will be reflected in the P/E. This is true for both a trailing P/E and forward-looking P/E. If the market multiple is 15, a company with an identical P/E is considered to have a relative P/E of 1.0. All companies can be compared to the market P/E. Some issues will sell for more than the relative P/E (premium), others for less (discount).

Illustrative Examples

Trailing P/E You are a research analyst and are evaluating the Doxie Manufacturing Corporation. It's November 2016 and you are calculating the P/E ratio using the four previous quarterly earnings reports—the first three quarters of 2016 and the fourth quarter of 2015. When examining the financial statements, you find that Doxie had acquisition expenses in December 2015 that resulted in a reduction in earnings per share from \$.24 to \$.16. Earnings for the first three quarters of 2016 were \$.19, \$.21, and \$.18, respectively. The current market price of its stock is \$16.40.

When calculating the trailing EPS, you would normally add the previous four quarters' earnings totaling \$.74 and divide the market price of \$16.40 by the earnings. This calculation indicates a P/E ratio of 22.16 ($\$16.40 \div \$.74 = 22.16$). However, it must be noted that \$.08 of the fourth quarter 2015 earnings was the result of a non-recurring item. When factoring this item into the mix, the earnings per share used in the valuation should be \$.82 rather than \$.74. When using the adjusted figure, the calculation shows a P/E ratio of 20.00, rather than 22.16 ($\$16.40 \div \$.82 = 20.0$).

Forward P/E When calculating the leading (forward) P/E ratio, certain assumptions must be made. In some cases, the analyst will use expected EPS for the next four quarters, while in other cases, actual EPS reported for the current period will be included along with the projections. To illustrate the latter, the following information is provided.

You are examining Diamond Point Development Corporation, a listed retirement community construction company. It's April 21, 2016 and the company has just released earnings of \$.38 for the first quarter. Its common stock is currently trading at \$22.60. Last week, Diamond Point announced that it will acquire 1,200 acres of land in Florida and is planning a new gulf coast retirement community with approximately 2,400 units. The land acquisition is expected to create a write-down of \$.12 per share in the third quarter of 2016. Increased legal costs after the acquisition are expected to reduce future EPS by \$.02 in the fourth quarter. The company projects quarterly EPS to be \$.36, \$.20, and \$.34 for the remainder of 2016.

It's obvious that taking the announced EPS plus the projections will not provide the proper valuation of the company. Adjustments must be made for the non-recurring item (the land acquisition) and the increased operating expenses over the projected period. The figures used in the calculation are:

	As Reported	Adjusted
First quarter actual	\$.38	\$.38
Second quarter projected	\$.36	\$.36
Third quarter adjusted	\$.20	\$.32
Fourth quarter adjusted	\$.34	\$.36
Total	\$1.28	\$1.42

The P/E ratio is calculated by dividing the market price of \$22.60 by the adjusted EPS of \$1.42, yielding a P/E ratio of 15.92.

Normalized earnings per share When looking at the historical data of Diamond Point Development Corporation, you see that during the past 18 months the economy has experienced a significant downturn and housing starts have declined. Prospects for a turnaround look uncertain in the near-term. Rather than using the current data and projections, you decide to normalize the earnings per share. You determine that the past five years are a good indication of a complete economic cycle. In its annual report, the following information is provided for this period:

	2015	2014	2013	2012	2011
Earnings Per Share	2.46	2.20	2.24	2.08	1.96

To normalize EPS, simply take the average of the EPS figures for the five-year period and divide the closing share price by this average.

$$\text{Normalized P/E} = \frac{\text{Market Price}}{\text{Average EPS}} = \frac{\$22.60}{\$2.19^*} = 10.32x$$

$$^*\text{Average EPS} = \$2.46 + 2.20 + 2.24 + 2.08 + 1.96 = \$10.94 \div 5 = \$2.188 \text{ or } \$2.19$$

Finding the P/E Ratio If given certain information about a company, you can find the price-to-earnings ratio.

Given the annual dividend, the dividend payout ratio, and the dividend yield, find the P/E.

Annual dividend \$1.50 Dividend payout 40% Dividend yield 2%

The earnings per share of the company will equal the annual dividend divided by the dividend payout ratio ($\$1.50 \div .40 = \3.75).

The current price is calculated by dividing the annual dividend by the current dividend yield ($\$1.50 \div .02 = \75.00).

The P/E for the company is 20 ($\$75.00 \div \3.75).

Earnings Yield One of the inherent problems in using EPS and the P/E ratio is that there are times when earnings are negative. In comparing various companies, you might use a ranking system based on the P/E ratio. The companies would be ranked from the highest P/E to the lowest, the lowest being the most economic per dollar of earnings. However, if a company has negative earnings, the ranking system would put it below the lowest positive P/E. The error in this reasoning is that positive dollars must be invested to acquire zero returns which, in fact, is more expensive than a company with a low positive P/E ratio.

In such cases, it may be more advantageous to use the earnings yield, which is the earnings per share divided by the price per share. It's also the reciprocal of the price/earnings ratio. Let's look at a hypothetical example to illustrate this concept. We will compare three companies in terms of their P/E ratio, as well as their earnings yield. They're ranked in order of earnings yield.

	Market Price	EPS	P/E	Earnings Yield
<i>Fonzone, Inc.</i>	\$32.18	\$3.52	9.14	10.94%
<i>Coldtherm</i>	\$15.82	(\$0.12)	N/A	– 0.76%
<i>Wilkin Mfg.</i>	\$11.85	(\$1.20)	N/A	– 10.12%

By using the earnings yield rather than the P/E ratio, we now rank the companies from the least expensive to the most expensive in terms of the earnings that one dollar of investment will purchase. In this case, the two companies with negative earnings may be considered more expensive per dollar of share price than the company with positive EPS.

P/E-to-Growth Ratio

In our previous discussion of the P/E ratio, we used the market value of the stock and its earnings, either trailing 12 months, or forward projections. Although the P/E ratio is a valuable tool and has been used as a historical benchmark calculation for valuation, it does have its shortcomings. Principally, the P/E is based on historical data and doesn't address expectations for continued operations of the company. Intuitive investors will typically look for the future performance of a company, as opposed to past history, and will attempt to value companies based on their growth potential and prospects for continued revenues. The ratio that's used to incorporate these prospects is referred to as the price/earnings-to-growth (PEG) ratio. This ratio indicates the relationship of price/earnings to the earnings growth rate and, when used to compare two companies, will provide a more insightful value.

To calculate the PEG ratio, the price/earnings (P/E) ratio is divided by the annual growth rate of the company. The annual growth rate can be either a forward projection or it can use trailing information. As a variation, some analysts use a one-year period and others use a five-year time frame. When comparing companies, similar periods and growth factors should be used.

In order to demonstrate how the PEG ratio is used, we will look at two companies using hypothetical earnings and projected growth. The first company is a start-up high-tech Internet distributing company that's currently selling for 70 times earnings. Its projected growth rate is 20% per year for the next three to five years. The second company is a well-established grocery chain that's trading at six times earnings and is expected to grow at a rate of 8% per year over the same period.

Typically, investors look for companies with a strong growth potential. Therefore, if we were to simply compare the two growth rates, the investor might be drawn to the start-up company with the higher potential of 20%. However, if we look at the price/earnings-to-growth ratio, a different conclusion may be apparent.

If we divide the P/E of 70 by the growth rate of 20, we see that the PEG is 3.5. In the second example, where the P/E of 6 is divided by a growth rate of 8, the PEG is .75. These calculations indicate that the start-up Internet company doesn't have sufficient growth to justify its higher price/earnings ratio. The PEG of .75 in the second company tells us that it's properly priced, or that it may be undervalued, based on its growth potential. When using the PEG, investors will normally look for a ratio of 1.0 or less.

Of course, the research analyst must take note of the weaknesses of using the PEG ratio for comparison. The assumptions of both earnings and growth rates are subject to change and adjustment. Projected results may not be achieved. When comparing two companies in different sectors as we have done, there might be other factors to consider that would justify a higher PEG. Aggressive investor interest in technology, for example, might be a significant factor. A lower cost of labor in the grocery chain might be another. Another consideration is that the PEG ratio is less often used in a large-cap company, due to slower projected growth rates. Therefore, it's important to compare a company within its own sector and against other companies of equivalent size. Additionally, an attempt to apply a PEG ratio is meaningless if the company doesn't have positive earnings.

Multiples When using the PEG ratio as a valuation tool, a research analyst would assume that the price/earnings ratio should equal the growth rate (PEG ratio of 1.0). A ratio higher than 1.0 may indicate an overpriced stock, while a ratio of less than 1.0 may suggest that the stock is undervalued. Most professionals will use the benchmark PEG of an industry for comparison to a company's PEG. An additional variable comes into play when asking, "What is an investor really willing to pay for the potential earnings and growth of the company?" If a company has historically traded at 15 times earnings, and is currently trading at 12 times earnings, it might signal a buying opportunity. Under-pinning this thought is that if investors were willing to pay up to 15 times the earnings potential, the stock price should eventually trade up to its historical multiple of 15. Likewise, if the stock is trading above 15, it could signal a sell since the stock is overvalued to historic earnings and can be expected to trade down to its historic level.

Calculating the PEG Ratio If the market capitalization, total assets, total liabilities, net income, and dividend payout ratio are given, calculate the PEG ratio.

- Market capitalization is \$440 MM and net income is \$40 MM
- Total assets of \$400 MM and total liabilities of \$150 MM
- Dividend payout ratio of 50%

The market capitalization divided by the net income equals the P/E ratio. The company has a P/E ratio of 11 (\$440 MM ÷ \$40 MM). The company has shareholder equity of \$250 MM (\$400 MM – \$150 MM). The return on equity (ROE) is 16% (\$40 MM ÷ \$250 MM). The growth rate equals the complement of the dividend payout ratio multiplied by the ROE. The growth rate is 8% [$1.00 - (50\% \times 16\%)$]. The PEG is 1.375, or rounded to 1.38.

NOTE: The P/E ratio multiplied by the net income will equal the market capitalization. $11 \times \$40 \text{ MM} = \440 MM .

Using the PEG Ratio If the PEG and certain other information concerning a company is given, the next year's price target can be calculated.

Given a PEG ratio, this year's EPS, and next year's EPS estimate, calculate next year's price target.

- PEG = 1.30
- Current EPS = \$0.85
- Next year's estimated EPS = \$0.95

$.95 - .85 = .10$, then $.10 \div .85 = 11.7\%$ growth rate

PEG 1.30 x growth rate of 11.7 = 15.2 price/earnings ratio

P/E 15.2 x next year's EPS .95 = \$14.44

Given the PEG ratio and an anticipation that earnings will double, calculate the earnings yield.

PEG = 1.60. Earnings will double in six years.

PEG 1.60 x growth rate of 12 = 19.2 price/earnings ratio

$1 \div 19.2 \text{ P/E} = 5.2\%$ earnings yield

We first need to calculate the growth rate. We are told that earnings will double in six years. To answer this question without a complex financial calculator, we can use the Rule of 72. This rule allows us to calculate the approximate compounded annual growth rate it would take for funds to double in value, given a specific number of years. In this example, if we divide 72 by 6 years, this equals 12; therefore, the annual growth rate is 12%. If we divided 72 by 12, this would indicate that funds would double in value in six years at an annual growth rate of 12%.

A research analyst may need to compare a number of companies using the PEG ratio as the appropriate valuation metric. The company with the highest PEG is the most overvalued, while the company with the lowest PEG is the most undervalued.

Compound Annual Growth Rate (CAGR)

When calculating the PEG ratio, the importance of determining the growth rate was a key to the successful use of the valuation metric. The CAGR (compound annual growth rate) is the year-over-year growth over a specified period and is calculated by taking the nth root of the total percentage growth rate, where n is the number of years for the period being considered.

Use the following data to calculate the compound annual growth rate:

	2009	2010	2011	2012	2013	2014	2015
<i>Sales</i>	100	120	115	110	125	130	120
<i>Growth</i>		20.0%	-4.2%	-4.3%	13.6%	4.0%	-7.7%

Solution:

$[(120 / 100) \text{ raised to the power of } (1 / 6)] - 1 = 0.0309 \text{ or } 3.09\%$

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\left(\frac{1}{\# \text{ of years}} \right)} - 1$$

NOTE: This cannot be calculated using the simple calculator that's provided at the testing center. A method to approximate the CAGR is to calculate the growth from every period and compute an average. Then, look at the choices that are close to the average calculated for a value that's slightly lower.

Average method

$$-7.7 + 4.0 + 13.6 + -4.3 + -4.2 + 20.0 = 21.4$$

$$21.4 \div 6 = 3.57 \text{ or } 3.57\%$$

A value slightly lower is 3.09%.

A trial-and-error method could also be used as follows:

	WRONG	CORRECT
Year 1	$100.00 \times 1.0357 = 103.57$	$100.00 \times 1.0309 = 103.09$
Year 2	$103.57 \times 1.0357 = 107.27$	$103.09 \times 1.0309 = 106.28$
Year 3	$107.27 \times 1.0357 = 111.10$	$106.28 \times 1.0309 = 109.56$
Year 4	$111.10 \times 1.0357 = 115.07$	$109.56 \times 1.0309 = 112.95$
Year 5	$115.07 \times 1.0357 = 119.18$	$112.95 \times 1.0309 = 116.44$
Year 6	$119.18 \times 1.0357 = 123.43$	$116.44 \times 1.0309 = 120.04$

Price-to-Free Cash Flow and Free Cash Flow Yield

A common valuation indicator used is the price-to-cash flow ratio. The term cash flow is used in a variety of ways. In this context, we will use it in its more narrowly defined sense of cash flow from operations less capital expenditures (CAPEX). Free cash flow to the firm represents cash flow available to the providers of capital, both bondholders and stockholders, after operating expenses have been covered. Cash flow from financing debt, preferred stock, and equity has a decidedly important impact on asset valuation.

Later in this chapter we will provide a more detailed examination and calculations of free cash flow to the firm and free cash flow to equity. Cash flow per share is different from earnings per share. Therefore, in the context of price to cash flow, we must be especially careful to define the proper elements of cash flow. Our discussion in this section will use the definition of price to free cash flow to equity (FCFE). Using free cash flow to equity provides a standard valuation of the entitlement to common equity suppliers of capital and makes comparisons between different companies viable.

In order to calculate the price to free cash flow and the free cash flow yield, we will assume a closing market price on the day of calculation of \$29.60 and the free cash flow is \$2.86.

Our price-to-free-cash-flow ratio is, therefore, \$29.60 divided by \$2.86, or 10.3. The free cash flow yield (free cash flow / price) may be stated simply as the reciprocal of the price-to-free cash flow ratio. In this case, the free cash flow yield is 9.70% ($1 \div 10.3$).

It should be noted that by using free cash flow to equity in our ratio, we have accounted for interest payments on debt and taxes paid. If we were to use free cash flow to the firm, these items would need to be accounted for net of taxes.

Price-to-Sales Ratio

The price-to-sales ratio is found by dividing the market value of the equity by the sales (revenue) of a company, or by dividing a stock's current price by its revenue per share. The price-to-sales ratio is most appropriate when evaluating companies that are unprofitable, have low profit margins, or are start-ups, since such companies don't have a useful price/earnings ratio. Additionally, the ratio is also not influenced by a firm's accounting decisions, such as depreciation, research and development expenses, and non-recurring items. *Keep in mind that the average price-to-sales ratio can vary substantially depending on the industry.*

Price-to-Book Value

The price-to-book (P/B) ratio is a measurement that looks at the value that the market places on the book value of the company. It's calculated by taking the current price per share and dividing by the book value per share.

Traditional book value is:

$$\frac{\text{Shareholders' Equity}}{\text{Number of Shares of Stock Outstanding}}$$

Price-to-Book Value is:

$$\frac{\text{Share Price}}{\text{Book Value Per Share}}$$

This number will be a simple multiple, much like the price/earnings ratio, or the price/sales ratio. If the ratio is below 1.0, it means that the company is selling below its book value and theoretically below its liquidation value. Some value investors will avoid any companies that trade above two times book value (or more).

The price-to-book value ratio uses historical balance sheet numbers and is less volatile than earnings per share, and the ratio tends to be a positive number. Additionally, book value is more stable in times of abnormal earnings and provides for comparisons over the long term. It's a good method for comparison when evaluating companies such as insurance and banking institutions, which traditionally hold high levels of liquid assets.

Like the P/E ratio, the lower the P/B ratio, the better the value. Value investors would use a low P/B ratio in their stock selection screens to identify potential purchase candidates. This valuation measure was traditionally used for industrial companies that have hard assets such as factories to back up their stock. It was primarily used with very capital-intensive companies. Currently, this method provides a fairly representative measure of valuation for financial institutions.

Enterprise Value (EV)-to-EBITDA

Another valuation method that's used is the Enterprise Value (EV) of the company in relation to its earnings before interest, taxes, depreciation, and amortization (EBITDA). We will examine this relationship in this section, but first, it's necessary to determine the two respective components and explain why they're used.

Enterprise Value The enterprise value (EV) of the company is the real, tangible price at which a company may be purchased. However, EV is not necessarily the amount that a buyer is willing to pay, or a seller is willing to accept. It's calculated by starting with the market capitalization of the common stock and preferred stock and adding the debt (long-term and short-term), capital leases, minority interest and subtracting cash and cash equivalents. This value represents the actual purchase price that someone might pay to acquire the company on the day of the calculation. Many professionals simply define EV as the equity value plus the net debt.

In return for buying a company, the purchaser will have access to the cash and investments of the company. Therefore, any cash that's carried on the books and any investments in other companies is of economic benefit to the purchaser and a reduction of the overall acquisition cost. Therefore, these items are subtracted from the real economic purchase price of the company when determining the enterprise value.

If a company has cash and no debt, or less debt than cash, the enterprise value (EV) will be less than the firm's market capitalization. If a company has no cash, the EV is the market value of the stock, plus the market value of its debt. If the company has less cash than debt, EV is greater than the company's market capitalization.

Use the following information to calculate the company's enterprise value:

(Dollar values are in millions, except stock price.)

▪ Long-term debt	\$12.50
▪ Short-term debt	\$ 3.00
▪ Cash	\$ 5.70
▪ Shares of common stock outstanding	5.80 MM
▪ Stock price	\$18.00

The formula to calculate enterprise value is market capitalization plus debt (long-term and/or short-term), minus cash and cash equivalents.

$$EV = 104.40 [(5.80 \times \$18.00) + 12.50 + 3.00] - 5.70 = \mathbf{114.20}$$

If given the enterprise value, debt, and cash, calculate the market value per share.

Using the same company, the market value per share can be determined by dividing the equity value of common shares by the number of common shares outstanding. (Equity value equals the enterprise value + cash) – (long-term + short-term debt). $(114.20 + 5.70) - (12.5 + 3.00) = 119.90 - 15.5 = 104.40$. The equity value divided by the common shares outstanding is $104.40 \div 5.80 = \$18.00$

Earnings Before Interest, Taxes, Depreciation, and Amortization To calculate the earnings before interest, taxes, depreciation, and amortization, we begin with net income from the profit and loss statement. Interest expense and taxes are added to provide preinterest earnings. This is necessary to enable comparisons between companies that have different amounts of debt and different methods of accounting for debt service. Depreciation and amortization are added back to the computation to neutralize the effects of the different calculation methods across businesses. The EBITDA method provides an easier comparison between companies that have high depreciation and amortization expenses versus companies with low expenses.

By calculating the EV/EBITDA ratio, the research analyst now has a value for the overall company, rather than a value based exclusively on common stock. When viewed against the P/E, EV/EBITDA may prove to be a more appropriate valuation. Since EPS is an after-interest figure that doesn't make adjustments for depreciation and amortization, it's difficult to compare one company to another. By using EBITDA, these elements are minimized or eliminated. In addition, there are times when EPS is a negative number making the P/E ratio invalid, while the EBITDA is still a positive figure that can be used.

When comparing the EV/EBITDA ratio of two companies, a lower ratio could signal an undervalued company, while a higher ratio could mean an overpriced company. It's one more useful tool for the research analyst when measuring valuation.

When calculating the new enterprise value of a company after an acquisition, it's important to know what method of funding was used. For example, assume a company is acquired for \$70,000,000. The purchaser's stock is trading at \$25 per share, and the company is funding the acquisition by using 50% debt and 50% stock. In order to calculate the new enterprise value, add 1,400,000 shares ($\$35,000,000 \div \25) to the current outstanding shares of the acquiring company, multiply that amount by the market price per share (to calculate the new market capitalization), add the net debt (debt minus cash) of both companies and add the new debt (\$35,000,000) that was issued to fund the acquisition.

EV-to-Net Sales Another measurement that's used when comparing companies is the enterprise value to net sales. Net sales equals revenue after trade discounts, returns and allowances, and excise taxes. Just as the EV/EBITDA ratio is a better tool to compare companies that have different interest and depreciation expenses, by using net sales rather than EBITDA, the analyst eliminates the effects of different methods of calculating (or recognizing) operating expenses such as lease accounting, inventory valuation, and capitalizing expenses. The enterprise value to net sales indicates the economic value of the company and its use of capital to generate sales. In other words, it permits a research analyst to address the question: Does the company use its available capital to generate sales volume and revenue sufficiently to maximize profits? Once again, by using enterprise value rather than market capitalization, the ratio excludes other accounting variables and makes comparisons between different companies more viable.

Discounted Cash Flow (DCF)

As previously described, DCF values a company based on the present value of the expected cash flows. In order to calculate either the value of an entire firm (both equity and debt), or the equity value of the firm using DCF, the research analyst would need to have the following information:

- An estimate of the discount rate to apply to the cash flows in order to calculate the present value
- An estimate of the cash flows for a given period
- An estimate of the ending or terminal value

There are two methods of estimating the discount rate—the *weighted average cost of capital (WACC)* and the *cost of equity*. In order to decide which discount rate to use, we need to examine the cash flows. To value an entire firm, we use the free cash flow to the firm (FCFF) and use WACC as the discount rate. The WACC is used since this evaluates the entire firm. We are also able to value the equity of a company using this method by valuing the entire firm and then subtracting the company's debt and preferred stock.

Another method is to use the free cash flow to equity (FCFE) as the cash flow and the cost of equity as the discount rate. When calculating the cost of equity, we can use two different methods—the *Capital Asset Pricing Model (CAPM)* or the *dividend growth model*. We will also review another DCF valuation method using the dividends as our cash flow, which is referred to as the dividend discount model (DDM). The next section will discuss all of these methods in detail. It's important to match the appropriate cash flows with the appropriate discount rate.

The following is a summary of the two main formulas:

$\text{Value of the Firm} = \frac{\text{Cash Flow to the Firm}}{1 + \text{WACC}}$	$\text{Value of the Equity} = \frac{\text{Cash Flow to Equity}}{1 + \text{Cost of Equity}}$
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To calculate the value of the firm or its equity, add the present value of each cash flow for the number of periods that cash flows are being estimated.

Cost of Capital

To estimate the discount rate, we first need to find the cost of capital. The cost of capital may be viewed as the required rate of return necessary to enter into and finance corporate projects. As defined in the *Dictionary of Financial and Investment Terms*, the cost of capital is the "opportunity cost of the funds employed as a result of a business decision." When corporations enter into ongoing business activities and commit working capital to finance these projects, they lose the ability to invest elsewhere. To look at it another way, the cost of capital is the required rate of return on a capital commitment that wouldn't change the earnings that are available to investors (i.e., common stockholders).

In this section, we will look at the cost of capital for each of the components of capitalization that are made up of debt, preferred stock, and common stock. We will also look at the combined cost of capital for each of these components which is referred to as the Weighted Average Cost of Capital (WACC). Formulas will be employed in order to calculate the cost of capital and to show the effect of taxation and selling expenses (flotation costs) on the required rate of return. We will also examine the impact on a company's stock price when earnings and dividend payouts vary from expectations.

Pre-Tax Cost of Capital on Debt Issues

The ERZATZ Corporation has made a decision to expand its business in order to increase sales. The company intends to issue \$5,000,000 face value bonds at a 6% coupon in order to finance this expansion. In this illustration, we will ignore the effects of taxation, selling expenses, and discounts to maturity. When the bonds are issued, the corporation must pay \$300,000 in interest expense each year. To calculate the cost of capital on this debt issue, the following formula is used:

$$\text{Cost of Debt} = \frac{\text{Interest}}{\text{Principal}} = \frac{\$300,000}{\$5,000,000} = 6.0\%$$

Therefore, the company must increase its earnings by \$300,000 in order to maintain the same earnings available to the common shareholders. If earnings are increased by more than the cost of capital, the project was beneficial for the investors. If the earnings increase doesn't cover the cost of capital, the share price may trade lower to reflect the loss of earnings to the investors. Later, we will look at the effects of taxation and flotation costs in the illustration. Without these added elements, the cost of capital on debt is simply the interest rate that's paid on the borrowed money. In our illustration, the cost of debt is 6% (\$300,000 ÷ \$5,000,000). When the company borrows money and invests it, the earnings should equal the interest expense in order to cover the cost of capital. Using the previous example, let's look at what will happen to ERZATZ in the following situation. We will assume that the company has \$20,000,000 in sales before they issue the \$5,000,000 in bonds. Their operating costs are 80% of sales. As a result of the increased capital employed by the company after the bonds are issued, its sales increased by \$1,500,000 and sales expenses increase by \$1,200,000. The income statement of the company would reflect the following change:

	Before	After
Sales	\$20,000,000	\$21,500,000
Operating costs	16,000,000	17,200,000
Earnings before interest expense	4,000,000	4,300,000
Interest expense		- 300,000
Earnings	\$ 4,000,000	\$ 4,000,000

Since the expansion has covered the cost of capital, the earnings have remained unchanged. Other factors will enter into the scenario that will impact the decision-making process. Once again, the effects of taxation, future increases in earnings from the expanded base, and improvement of the company's status in its market sector are some of these considerations.

When we introduce flotation, underwriting fees, legal, accounting, and printing costs, the cost of capital is increased to cover these added expenses. Let's assume that the underwriting syndicate charged a 2.0% spread for distributing the bonds. The other expenses will be ignored for illustration purposes. The cost of capital on the debt issue is now:

$$\text{Cost of Debt} = \frac{\text{Interest}}{\text{Proceeds}} = \frac{\$300,000}{\$4,900,000} = 6.12\%$$

In this example, if the earnings available to common stockholders increase by 6.0% as a result of the expansion, the price per share would theoretically decline since the cost of capital was not covered.

Pre-Tax Cost of Capital (Preferred Stock)

Rather than borrowing the necessary capital, a corporation may finance capital projects by issuing equity securities in the form of preferred or common stock. We will examine the effects of preferred stock first, since it has some similarities to debt instruments.

When dealing with preferred stock, we will need to make some basic assumptions. We will assume that the par value of the preferred stock is \$100 and that any dividend to be paid will be fixed as a percentage of its par value. Therefore, if a corporation issues 7% preferred, it's obligated to pay an annual dividend of \$7.00 per share. In this respect, preferred stock is similar to a bond in that both the par value and income to the investor are predetermined at the time of issuance. However, a major difference is that the interest paid on the bond is a pre-tax item on the company's income statement, while the dividend is an after-tax item. In our illustration, we will look at the effects of taxation in calculating the cost of capital.

When looking at the pre-tax cost of capital on preferred stock, we must calculate the yield on investment. The formula for the calculation is:

$$\text{Cost of Preferred} = \frac{\text{Preferred Dividend}}{\text{Price of Preferred}} = \frac{\$7.00}{\$100} = 7.0\%$$

Once again, if we incorporate flotation costs into the calculation, the cost of capital will be increased. For example, if the underwriting fee on the preferred issue is 5%, the calculation is now:

$$\text{Cost of Preferred} = \frac{\$7.00}{\$95} = 7.37\%$$

Now, let's look at the tax ramifications and compare an issue of preferred stock to issuing debt instruments. We will assume that the corporation is in the 35% tax bracket and can borrow money at 7%. If the company borrows \$1,000,000 at 7%, it will need to pay interest expense of \$70,000 per year. If it issues \$1,000,000 of 7% preferred, it will need to pay \$70,000 in dividends each year. But remember, the interest expense is a pre-tax entry while the dividend is an after-tax entry on the books of the company.

Let's see how the comparison would impact the cost of capital.

	Common Only	Debt	Preferred
Earnings before interest and taxes (EBIT)	\$4,000,000	\$4,070,000	\$4,070,000
Interest expense		70,000	
Earnings before taxes	4,000,000	4,000,000	4,070,000
Taxes at 35%	1,400,000	1,400,000	1,424,500
Preferred dividends			70,000
Earnings available to common	\$2,600,000	\$2,600,000	\$2,575,500

Notice that in the computation, the pre-tax cost of capital is 7% when financed with debt. However, when financed with an issue of preferred stock, the pre-tax requirement exceeds 7%. Let's see how we make that determination. The formula is as follows:

$$\text{Pre-Tax Cost of Capital} = \frac{\text{Preferred Dividend}}{(1 - \text{Tax Rate \%})} = \frac{7.0\%}{.65} = 10.7692\%$$

In our illustration, the required increase in EBIT is determined by multiplying the proceeds of the issue by the pre-tax cost of capital percentage (\$1,000,000 x 10.7692% = \$107,692).

$$\text{Pre-Tax Cost of Capital} = \frac{\$70,000}{\$950,000} = 7.37\%$$

It's necessary to increase our EBIT by \$107,692 to cover the cost of the \$70,000 preferred dividend. Once again, let's look at the comparison between the debt financed figures and the preferred stock financed cost of capital, but this time using the adjusted preferred stock calculation.

	Debt Financed	Preferred
Earnings before interest and taxes	\$4,070,000	\$4,107,692
Interest expense	70,000	
Earnings before taxes	4,000,000	4,107,692
Taxes at 35%	1,400,000	1,437,692
Preferred dividend		70,000
Earnings available to common	\$2,600,000	\$2,600,000

After-Tax Cost of Capital

Another way to determine the necessary increase in EBIT is to base it on the after-tax cost of capital. In fact, shareholders are usually more interested in the after-tax figures than in the pre-tax numbers. The after-tax figure represents their entitlement after the project decision has been made.

We already know what the after-tax cost of capital is on the preferred. Since the dividend is an after-tax number, the cost of capital is 7.0%. However, the interest on the bond issue is a pre-tax number and therefore the after-tax cost of capital will be something less than 7.0%. The formula to determine the after-tax cost of capital on debt is:

$$\text{After-Tax Cost of Debt} = (\text{Pre-Tax Cost}) \times (1.0 - \text{Tax Rate \%})$$

In our example, the pre-tax cost of debt is 7%. To find the after-tax cost of debt, the calculation is as follows:

$$\text{After-Tax Cost of Debt} = 7.0\% \times .65 = 4.55\%$$

Using the after-tax cost of both preferred and debt issues, let's calculate the required increase in EBIT. The necessary increase in EBIT is equal to the proceeds of the issue multiplied by the after-tax cost of the issue.

$$\frac{\text{Proceeds} \times \text{After-tax cost}}{(1.0 - \text{Tax Bracket \%})}$$

In the preferred issue, the calculation is:

$$\text{Proceeds} = \$1,000,000 \times 7.0\% = \$70,000 \div .65 = \$107,692$$

For the bond issue, the calculation is:

$$\text{Proceeds} = \$1,000,000 \times 4.55\% = \$45,500 \div .65 = \$70,000$$

Again, these calculations indicate the required increase in EBIT, but the approach is from the after-tax standpoint, rather than from the pre-tax number. Regardless of which approach is used, the required increase in EBIT is the same.

Cost of Equity Capital

The last element of our discussion on the cost of capital relates to the equity portion of the balance sheet. For illustration purposes, we will consider equity to be common stock and retained earnings of the corporation. We will ignore preferred stock since it was presented in the previous section. The illustrations presented in this section regarding retained earnings represent the current-period retained earnings figure as found on the income statement. This shouldn't be confused with the life-to-date numbers that are found on the balance sheet. Our discussion will center on the concept that equity is generated in two ways—either by increasing retained earnings or by issuing new shares of common stock.

As previously described, there are two methods commonly used. The first method being described will be the Capital Asset Pricing Model (CAPM), while the second is the dividend growth model (DGM).

Capital Asset Pricing Model

CAPM is an important tool that's used to analyze the relationship between risk and rates of return. The CAPM looks at three variables to calculate the cost of equity—the risk-free rate (usually on a Treasury security), the expected return of the market (usually the S&P 500 Index), and a company's non-diversifiable risk or beta (β). The model is based on the proposition that a company's rate of return is equal to the risk-free rate plus the risk premium of the stock.

For example, if $\beta = 1.06$, the expected market return is 9.76%, and the risk-free rate is 3.35%. Then K's expected return is calculated as follows:

$$K_S = K_{RF} + [(K_M - K_{RF}) \times \beta]$$

Where:

- K_S = required rate or return (cost of equity)
- K_M = the expected return on the market
- K_{RF} = the risk-free rate
- β = a measure of the sensitivity of the firm's stock returns relative to those of the market assuming the absence of diversifiable risk (Beta)
- $K_S = 3.35 + [(9.76 - 3.35) \times 1.06]$
- $K_S = 10.14\%$

Some market professionals use the term *risk premium* to denote the difference between the expected return of the market (usually the S&P 500) and the risk-free rate (usually the rate on a Treasury security).

Changing Variables If the risk-free rate increases, while the excess market return (expected market return minus the risk-free rate) remains stable, a company that has a beta of less than 1.0 will experience an increase in the cost of equity. This will have a negative effect on the valuation of the company since the WACC will increase. For example, if the expected return of the market was 9% and the risk-free rate was 3%, according to the Capital Asset Pricing Model, a company with a beta of .65 will have an equity cost of capital of 6.9%.

Expected return on the market:	9%
Minus risk-free rate:	<u>3%</u>
Excess market return:	6%

(Excess market return x beta) + risk-free rate

- = $(6\% \times .65) + 3\%$
- = $3.9\% + 3\%$
- = 6.9%

If the risk-free rate increases to 4%, while the expected market return remains at 9%, the equity cost of capital would increase to 7.25%. This will increase the WACC of the company and have a negative impact on the valuation of the company.

Expected return on the market:	9%
Minus risk-free rate:	4%
Excess market return:	5%

$$\begin{aligned}
 &(\text{Excess market return} \times \text{beta}) + \text{risk-free rate} \\
 &= (5\% \times .65) + 4\% \\
 &= 3.25\% + 4\% \\
 &= 7.25\%
 \end{aligned}$$

The opposite is true if the risk-free rate increases, while the excess market return remains stable and the beta is greater than 1.0. In this situation, the cost of equity capital will decrease.

Dividend Growth Model

Cost of Equity Created by Retained Earnings When investors purchase common stock in a corporation, their expectation is to receive a rate of return on their investment. The expected rate of return comes from dividends and through appreciation in their stock holdings. Later in this chapter, we will describe the valuation of common stock based on dividends. For this example, we will use the constant growth rate. Our formula for calculating the required rate of return is as follows:

$$k^* = \frac{D_1}{P_0} + g$$

In the formula:

- k^* = the required rate of return (cost of equity)
- D_1 = the dividend to be paid at the end of the year
- P_0 = the current market price (at time 0)
- g = the growth rate per year

To illustrate the formula, let's look at Micromax Manufacturing Corporation. The earnings per share figures for the year is expected to be \$4.60. Historically, the company has paid an annual dividend of \$1.15. The earnings and dividends have been growing at approximately 4% per year, and this trend is expected to continue. The current market price of the common stock is \$40.00 per share. The firm's next dividend is expected to be \$1.20 (\$1.15 x 1.04 = 1.196 rounded to \$1.20). Using this information, let's determine the expected rate of cost of equity for Micromax Corporation:

$$k^* = \frac{\$1.20}{\$40.00} = 3.0\% + 4.0\% = 7.0\%$$

The expectation for the increase in price should be from \$40.00 to \$41.60, an increase of 4.0%. If the company invests the amount that it didn't pay in dividends, and doesn't earn the required rate of return (7.0%), the value of the stock should decrease. For example, if the \$3.40 of retained earnings (\$4.60 – \$1.20) is invested and earns only 3.5%, investors will not achieve their expectations. The stock will theoretically be reduced in value as follows.

This model assumes that dividends and earnings are growing at the same rate. If the growth rate of earnings and dividends slows to 3.5% but the required rate of return (cost of equity) remains 7%, then the stock is estimated to be worth only \$34.29.

$$P_0 = \frac{D_1}{k^* - g} = \frac{\$1.20}{7\% - 3.5\%} = \frac{\$1.20}{.035} = \$34.29$$

It's important to note that when the firm acquires new equity in the form of retained earnings, it must invest that equity in operations at the required rate of return. If it doesn't earn the required rate of return, the theoretical share price will fall. Conversely, if it invests at greater than the required rate of return, the theoretical price will increase.

Cost of Equity Created by a New Issue of Stock If the corporation is going to create additional equity through a new issue of common stock, the cost of capital will be higher than through retained earnings. The reason for this is that there are underwriting fees associated with the distribution of the new issue. The cost of capital on the new issue will need to cover the flotation costs. To calculate the required rate of return on the cost of capital for a new issue, the following formula is used.

$$k^* = \frac{D_1}{P_0(1 - F)} + g$$

In this formula, F equals the underwriting fee as a percentage of the sales price and (1 – F) represents the net sales proceeds to the issuer. Once again, in our equation we are assuming a constant growth rate.

Let's assume that the new issue of stock will be sold to the public at \$40 per share and the underwriters are charging a 5% underwriting fee. Using the previous assumptions for earnings, dividends, and growth, we can determine the required rate of return as:

$$k^* = \frac{\$1.20}{\$40(1.0 - .05)} = \frac{\$1.20}{\$38} = 3.16 + 4.0\% = 7.16\%$$

The additional cost of capital of .16% (7.16% – 7.00%) is necessary to cover the cost of flotation. If the capital projects that are financed by the new issue can earn 7.16%, the dividends and growth of the company can be expected to continue and the share price shouldn't decline.

Weighted Average Cost of Capital (WACC)

As shown, the different components of capital—debt, preferred, and common stock—have different required rates of return in order to maintain projected dividend payouts and growth in the company. Since many companies will raise capital through various methods including long-term debt, preferred stock, and common stock, the weighted average cost of capital provides an appropriate valuation for all these methods.

As an example, let's assume that the cost of debt in a corporation is 5% and the cost of equity is 8%. Logic would suggest that the company should finance future capital projects through the use of debt at 5%. However, there's an inherent consideration that would make the use of debt inappropriate for all future capital projects. When the firm issues debt, the debt-to-equity ratio changes and, if the company continues to finance future projects with debt, it will at some point in time incur too much leverage and be unable to issue additional debt. The firm must then finance projects through equity. To illustrate, let's say that the company finances next year's projects by issuing 5% debt and earns 6.5% on these projects. It has now used up some of its ability to finance further projects with debt due to the increase in the debt-to-equity ratio. In ensuing years, it must begin to use equity with a cost of equity of 8%. If it can earn only 6.5% on these projects, it will not meet the required rate of return and its share price will decline. To resolve the dilemma, the company is considered to be an ongoing concern and will combine the cost of debt and equity to determine its weighted average cost of capital.

Let's look at an example to see how to determine the weighted average cost of capital. In our illustration, the following assumptions are made to simplify the calculations:

After-tax cost of capital

Debt	5.0%
Preferred stock	6.5%
Equity	7.0%

When examining the balance sheet of Mollmic Corporation, we see the debt and equity items as:

	Balance Sheet	Percent
Debt	\$19,200,000	32%
Preferred	\$4,800,000	8%
Equity	\$36,000,000	60%
Total	\$60,000	100%

To calculate the weighted average cost of capital, we multiply the component cost of capital by the percentage of existing capital as follows:

	Balance Sheet	Percent		Cost of Capital		Weighted Average
Debt	\$19,200,000	32%	x	5.0%	=	.0160
Preferred	\$ 4,800,000	8%	x	6.5%	=	.0052
Equity	\$36,000,000	60%	x	7.0%	=	.0420
Total	\$60,000,000	100%				.0632

The weighted average cost of capital to the firm is 6.32%.

The various components are now brought together in their respective elemental costs of capital to represent proper financing without a detrimental effect on investor expectations. Here's a word of caution when using the weighted average cost of capital and the calculation for the required return on debt. Remember that when calculating the required rate of return on debt, the after-tax rate must be used; therefore, the pre-tax rate must be multiplied by $(1 - \text{tax rate percentage})$ to determine the after-tax rate. Return on equity requires no such adjustment since the payments to shareholders in the form of cash dividends are after-tax returns. Most market professionals use the CAPM as the cost of equity when calculating the WACC.

In some instances, you are given the company's debt-to-equity ratio in order to calculate the WACC. In order to properly calculate the WACC, you need to know the percentage of the company's capital that's debt and equity. For example, assume a company had \$40 MM in debt and \$100 MM in equity. The debt-to-equity ratio is 40% ($40 \text{ MM} \div 100 \text{ MM}$), with the total capitalization of the company being 140 MM ($40 \text{ MM} + 100 \text{ MM}$). Next, you can calculate the percentage of debt or equity to total capital as follows:

Debt Percentage of Total Capital is $\$40 \text{ MM} \div \$140 \text{ MM} = .2857$ or 28.57%

Equity Percentage of Total Capital is $\$100 \text{ MM} \div \$140 \text{ MM} = .7143$ or 71.43%.

If the company issued additional debt, the WACC would decline since the cost of debt is generally cheaper than the cost of equity. In addition, if a company replaced debt with preferred stock, the WACC would increase since there's a tax shield (tax advantage) associated with debt. Interest on debt is tax-deductible, while preferred stock dividends will be paid with after-tax dollars. For example, if the tax rate is 35% and the pre-tax cost of debt is 8.50%, the after-tax cost of debt is 5.53% ($.085 \times [1 - .35]$).

Estimating Cash Flows

Free Cash Flow

Earlier in this chapter, we defined free cash flow as cash flow from operations less capital expenditures (CAPEX). This concept states that not all of the cash flow from operations is available for free use by bondholders and stockholders. The underlying requirement is that some of the cash flow from operations must be used to reinvest in operating activities or to purchase new equipment in order to maintain present production capacity. Without committing certain amounts of cash flow to ongoing operations, the company cannot maintain its existing capacities, nor can it take advantage of opportunities for expansion nor increase its capacities. The definition of free cash flow to the firm is, therefore, the cash flow from operations available to providers of capital (bondholders and stockholders) after operating expenses have been covered. Operating expenses include working capital (investments in inventory), fixed capital (investment in new or maintenance costs on existing equipment), and the payment of taxes.

Calculation of Free Cash Flow In the calculation of free cash flow to the firm (FCFF), depreciation and amortization are added to earnings before deducting interest and taxes (EBIT). This level of cash flow is reduced by capital expenditures. If working capital has increased, the free cash flow to the firm is reduced by this change. If working capital has declined, the change is added. In some cases, the tax rate of the company is given and, in that situation, you would need to multiply EBIT by 1.00 minus the tax rate.

Other methods to define free cash flow are: EBITDA – taxes – CAPEX $\pm$ changes in working capital, or EBIT – taxes – net CAPEX (CAPEX – depreciation) $\pm$ changes in working capital. Both of these formulas take into consideration taxes and interest expense. If a firm was unleveraged (i.e., having no debt), the formula is: net income + depreciation – CAPEX $\pm$ changes in working capital.

As a valuation tool there are two approaches that may be taken. The first is the free cash flow to the firm (FCFF) and the second is the free cash flow to equity (FCFE). Both models approach valuation in a similar manner in that the value is determined by the present value of future cash flows to the firm or to its equity shareholders. In the FCFF model, we include the debt holders since they're contributors of capital to the firm, whereas, in the FCFE model, the interest on the debt will be excluded from the model.

FCFF
EBIT x (1 – tax rate) + Depreciation and Amortization – Capital Expenditures $\pm$ Changes in Working Capital
Free Cash Flow to the Firm (FCFF)

FCFE
Net Income + Depreciation and Amortization – Capital Expenditures $\pm$ Changes in Working Capital
Free Cash Flow to Equity (FCFE)

Rusty's Medical Supply Company
Selected Financial Data (\$ millions)

EBIT	392.2
Interest Expense	82.0
Taxes	105.5
Net Income	204.7
Depreciation and Amortization	87.5
Capital Expenditures	114.7
Working Capital Changes (increase)	20.5
Tax Rate is 34%	

EBIT x (1 – tax rate)	258.9
+ Depreciation and Amortization	87.5
– Capital Expenditures	114.7
$\pm$ Changes in Working Capital	20.5
Free Cash Flow to the Firm (FCFF)	211.2

Net Income	204.7
+ Depreciation and Amortization	87.5
– Capital Expenditures	114.7
$\pm$ Changes in Working Capital	20.5
Free Cash Flow to Equity (FCFE)	157.0

Present Value of Free Cash Flow

In the free cash flow to the firm approach, we will value the firm based on the present value of future cash flows discounted at the weighted average cost of capital. Free cash flow to the firm is defined as cash flow from operations minus capital expenditures. WACC is used as our discount rate since we are using FCFF. We use the formula described earlier:

$$\text{Firm Value} = \sum_{t=1} \frac{\text{FCFF}}{(1 + \text{WACC})^t}$$

By using free cash flow to the firm and the weighted average cost of capital, we derive a value for both the debt and equity holders of the company. To calculate the free cash flow to equity holders, we will deduct the market value of debt from the firm value. Another method that's used to calculate the equity value is to divide FCFE by the cost of equity.

As an example, using the previous formula, let's calculate the free cash flow to the firm based on the following assumptions:

Matacumbe Corporation has the following information on its balance sheet:

- It's issued \$2,000,000 of 10-year bonds that have a current market value of \$1,800,000
- It has 2,000,000 shares of common stock outstanding.

Also, in reviewing the income statement, the calculation for free cash flow to the firm for the period is \$8,000,000 and the weighted average cost of capital is 6.32%.

We calculate the value of the firm in terms of the present value of the future free cash flows to the firm. In our examples, we will simply use a one-year period.

$$\text{Firm Value} = \frac{\$8,000,000}{(1 + .0632)} = \$7,524,454$$

If either the FCFF or the WACC are altered, the firm valuation will change. For example, if depreciation is increased or capital expenditures are decreased, this will cause the FCFF to increase. If the cost of equity or debt increases (due to rising interest rates or credit rating declines), the firm's WACC will increase. If the FCFF increases, or the WACC decreases, the valuation will increase. If the FCFF decreases, or the WACC increases, the valuation will decrease.

To determine the free cash flow to the equity owners of the company, we would now deduct the market value of the outstanding debt. Therefore, the value of equity is equal to:

Firm value	–	Market value of debt	=	Equity value
\$7,524,454	–	\$1,800,000	=	\$5,724,454

To find the value per share of equity, we now divide the \$5,724,454 by the 2,000,000 shares outstanding to arrive at a value per share of equity of \$2.86.

As previously mentioned, an alternative approach that should provide a similar number for the value of the equity can be derived by using the following:

$$\text{Equity Value} = \frac{\text{FCFE}}{1 + \text{Cost of Equity}}$$

Terminal Value In a discounted cash flow analysis, the cash flows are projected each year into the future (for a certain number of years). At some point, instead of projecting future cash flows, a single value representing the discounted value of all future cash flows is used. This is referred to as the terminal value. This can represent a significant component of an asset's valuation.

The value can be calculated based on the liquidation value of an asset or based on a company continuing its business as an ongoing concern. In the case of valuing common stock of an ongoing concern, a research analyst will use either a DCF valuation or a multiple of comparable firms (terminal multiple). The terminal multiple is the easier method and is based on a relative valuation, usually determined by enterprise value (EV) to EBITDA for similar companies or similar transactions in a merger analysis. Using this method, you would multiply the terminal multiple (EV/EBITDA) by the last projected cash flows. Using enterprise value is preferred since the cash flow to the firm is used, and EV takes into consideration all providers of capital.

In using a DCF approach, the formula to calculate the terminal value is:

$$\text{Terminal Value} = \frac{\text{Expected cash flow}}{(\text{Discount rate} - \text{Terminal growth rate})}$$

The discount rate is the WACC and the terminal growth rate of a company and represents a growth rate that can be maintained in perpetuity. An increase in this growth rate will result in higher terminal value and, therefore, a higher valuation.

Calculating the Firm or Enterprise Value By combining the present value of the cash flows for a certain period and adding this to the present value of the terminal value, we can calculate the firm or enterprise value. The following formula is used:

$$\text{Firm Value} = \frac{\text{Cash flows each period}}{1 + \text{WACC}} + \frac{\text{Terminal value}}{1 + \text{WACC}}$$

Let's look at the following example to calculate the equity value per share.

Free Cash Flow to the Firm is projected as follows:

Year 1	32,500
Year 2	34,700
Year 3	39,600
Year 4	43,500
Terminal Value	380,000

- The WACC is 11%.
- The company has cash of \$50,000 and debt of \$100,000.
- There are 25,000 shares of common stock outstanding.

The discounted cash flow (DCF) method is used to calculate the present value of the firm's enterprise value (EV). This is determined by using the free cash flow to the firm (FCFF) divided by 1 plus the weighted average cost of capital (1 + WACC). It's necessary to discount each period's cash flow and to add the present value of the terminal value to determine the total present value of the company. The discount rate is applied (1 + WACC), and it must be adjusted for each subsequent period. To determine the equity value of the company, cash is added to the total present value (of the company) and debt is subtracted. The equity value of the company is divided by the number of shares of common stock outstanding to determine the value per share.

	FCFF	Divided By		Present Value
Year 1	32,500	$(1 + .11)$	$= 32,500 \div 1.110 =$	29,279
Year 2	34,700	$(1 + .11)^2$	$= 34,700 \div 1.232 =$	28,165
Year 3	39,600	$(1 + .11)^3$	$= 39,600 \div 1.367 =$	28,968
Year 4	43,500	$(1 + .11)^4$	$= 43,500 \div 1.518 =$	28,656
Terminal Value	380,000	$(1 + .11)^4$	$= 380,000 \div 1.518 =$	250,329
Total Present Value				\$365,397
Plus Cash				50,000
Minus Debt				(100,000)
Total Equity Value				\$315,397

The total equity value divided by the number of shares of common stock outstanding equals \$12.61 (\$315,397 ÷ 25,000).

Since the process of discounting several years of cash flows with a simple calculator is tedious, two simplified versions that are based on the perpetual growth method may be tested on the exam. In these questions, based on the variables given, the terminal value is used as an estimate of the enterprise or company value.

Shown below are the formulas to be used:

$$\text{Enterprise Value (EV Perpetuity Method)} = \frac{\text{Cash Flow} \times (1 + g)}{\text{Discount rate} - \text{Growth rate}}$$

This method may be referred to as “stable growth” and was shown earlier as a way to calculate the terminal value. In most cases, the discount rate used will either be the WACC or cost of equity. For example, a corporation has an expected long-term growth rate of 3% and cost of capital of 8%. If the company’s cash flow is expected to be \$20 million, the EV can be calculated as follows:

$$\text{EV Perpetuity Method} = \frac{\$20 \text{ million} \times (1 + 3\%)}{(8\% - 3\%)} = \frac{\$20 \text{ million} \times 1.03}{5\%} = \$412 \text{ million}$$

The other formula is referred to as Net Present Value and is calculated as follows:

$$\text{Net Present Value} = \frac{\text{Annual Cash Flow in Perpetuity}}{\text{WACC}}$$

For example, if a corporation has \$50 million of annual cash flows in perpetuity and a WACC of 10%, it will have a net present value of \$500 million (\$50 million ÷ 10%).

Dividend Discount Model

When investors buy stock in a corporation, they become equity owners of a portion of that company. The expectation for these investors is a return on their investment commensurate with the risk assumed. As the company engages in its normal business operations, it will generate cash flows in various forms, and as owners of the company, the investors have a reasonable expectation that they will receive some of the cash flows in the form of dividends. It’s logical then to view the valuation of common stock in the form of expected dividends. However, the expected dividends must be discounted to their present value. We have already noted that various models have been developed to examine the value of future cash flows. This section will concentrate on the dividend discount model.

Discounted Dividends According to one stock valuation technique, the value of a stock is equal to the present value of the expected future dividends. This valuation model assumes that investors buy stock solely for dividends and desire a current return. If an investor anticipated holding a stock for 10 years, dividends over the next 10 years will be discounted, as well as the terminal value that will be received upon the liquidation of the shares. The general formula in this dividend model is:

$$P_0 = \frac{d_1}{(1 + k)} + \frac{d_2}{(1 + k)^2} + \dots + \frac{d_n}{(1 + k)^n}$$

Where P_0 = the worth of the stock today, $d_1, d_2, \dots, d_n$ is the expected annual dividend stream, and k = the discount rate consistent with the uncertainty of the dividend estimates. The dividend rate reflects the risk-free rate and a risk premium to account for uncertainty. In theory, this rate should rise over time.

The model can be altered to account for dividend streams that remain constant (zero growth case), for those that grow at a constant rate, and those that grow at a variable rate. The zero growth case is relevant for preferred stocks or very mature companies such as electric and gas utilities. For these issues, the market prices cannot be divorced from their dividends, and they're frequently purchased for their current yield. In this case, the formula reduces to:

$$P_0 = \frac{d}{k}$$

In the constant growth case, where g is the rate of growth, the formula for the expected total return reduces to:

$$k = \frac{d + g}{P_0}$$

If the current dividend was \$2, and was expected to grow at 5% per year, and the current market price was \$40, the expected return will be:

$$k = \frac{\$2.00}{\$40} + .05 = .10$$

In the constant growth case, the formula for the worth of the stock can be simplified to:

$$P_0 = \frac{d_1}{(k - g)}$$

What should the price of a common stock be now if the projected dividend is \$3, $k = 10\%$, and $g = 7\%$? Applying the above formula, \$3 divided by 3% (.03) = \$100. We use the projected dividend one year forward.

This application is also referred to as the Gordon Growth Model. The present value of the stock (P_0) is also considered the intrinsic value of the security. Consider the following application:

The Morris Electric Company

	Current
Dividends	\$1.10
EPS	\$2.20
Return on equity	8.0%
Equity cost of capital	7.5%

The requirement to define the variables in the denominator presents another challenge. Therefore, to determine the intrinsic value of the company, it's first necessary to determine the company's growth rate using the formula:

$$g = b \times \text{ROE}$$

where g = the dividend growth rate
 b = the earnings retention rate (the complement of the dividend payout ratio)
 ROE = the return on equity

The dividend payout ratio is 50% ($\$1.00 \div \2.00) and the retention rate is 50%.

$$g = 50\% \times 8.0\%$$

$$g = 4.0\%$$

$$d_1 = D_0 \times (1 + g)$$

$$d_1 = \$1.10 \times (1.00 \times 4\%)$$

$$d_1 = \$1.14$$

Next, the price can be determined by using the dividend discount model.

$$P_0 = \frac{d_1}{(k - g)}$$

where P_0 = present price
 d_1 = the dividend in the next period
 k = the equity cost of capital
 g = the growth rate

$$P_0 = \$1.14 \div (7.5\% - 4.0\%)$$

$$P_0 = \$1.14 \div 3.5\%$$

$$P_0 = \$32.57$$

Since a current price has been projected, the P/E ratio could also be estimated. $\$32.57 \div \$2.00 = 16$.

As you can see, employing the dividend discount model correctly presents various challenges to the analyst. Chief among these are:

1. What dividend discount model should be used for a given company?
2. What method will be used to forecast dividends?
3. What rate of return should be used to discount cash flows to the present?

In addition, if the analytical approach considers earnings distributions in the form of dividends to be the primary reason for investment, what about earnings that are not distributed? Corporate earnings that are not distributed (retained earnings) are the basis for a future expansion of the company and provide for ongoing payments of dividends. Therefore, the dividend discount model must consider all future dividend payments including those based on reinvested earnings. Due to this consideration, the dividend discount model may be less sensitive to short-term fluctuation of value than other discounted cash flow models. Consequently, the dividend discount model is considered to reflect longer-term values.

Additional DCF Applications

Free cash flow analysis may also be used to calculate the amount of additional leverage or borrowing a company can incur. This valuation technique is especially useful to financial buyers such as leveraged buyout (LBO) firms. Since free cash flow to the firm equals the cash a company has available after paying its operating expenses and capital expenditures, the present value for the company's cash flows for the payback period may be used as a proxy to estimate the additional amount of debt the company can incur. The after-tax cost of debt is used as the discount rate. A company with high free cash flow has the capacity to issue additional debt.

Economic Profit

When making decisions to commit capital to produce revenue, an important tool in the decision-making process is economic profit. Economic profit differs from accounting profit in that it takes into consideration the cost of capital (i.e., not only profit from operations), but also profit above the expected returns on the capital expenditure. Therefore, economic profit may be seen as adding value to the corporation rather than destroying value. In addition to operating profit, economic profit incorporates the market expectation and opportunity cost risk that's assumed when capital is committed to any project. Internal (rather than external) changes within a company may increase the economic profit and increase the quality of earnings. An example of the difference is a reduction of bad-debt expenses. The reduction of the expense due to a higher reliability of accounts receivable will (all else being equal) improve the rate of return for investors. External changes, such as rising revenues due to inflation, don't improve the quality of earnings. Let's look at a hypothetical example of the decision-making process by comparing accounting profit to economic profit.

Let's examine the idea of required return for assuming risk. The required cost of capital will necessarily incorporate the return on risk assumed above any risk-free return. For example, if Treasury bills are currently yielding five percent, any allocation of capital would require a return exceeding five percent since T-bills are considered riskless investments. A return of six or seven percent may be appropriate; however, it still must be compared to other investments in terms of opportunity cost. If a capital investment is made in a start-up company versus a well-established company, the expected return for the start-up company would logically be higher than for the established company. If the return on both was 11%, the decision should be made in favor of the established company, thereby reducing risk for the same return.

Therefore, it should be noted that any decision to commit capital to a project must cover the operating costs as well as the risks associated with investing in the project (e.g., the economic profit). If the economic profit is greater than the cost of capital, value will be added to the corporation and, by the same token, if the economic profit is less than the cost of capital, value will be destroyed. In some cases, a capital commitment may produce accounting profit, but may have an economic loss, in which case the decision to commit capital was a poor one. Let's look at such an example.

Lapp'ale, Inc. is a small boutique brewery that produces and sells approximately 20,000 barrels of beer annually. It sells to the custom market and its wholesale price is \$80.00 per barrel. Costs of production and distribution are \$56.00 per barrel and the brewery has fixed costs of \$150,000 per year. Shareholder investment in the company is \$3,000,000. What's the return on equity?

Current Production

Revenue	\$1,600,000
Sales = 20,000 bbls x \$80.00	
Cost of goods	
Sunk costs*	\$ 150,000
Variable costs (\$56.00 x 20,000)	\$1,120,000
	\$1,270,000
Accounting profit	
	\$ 330,000
Unit cost per barrel = \$1,270,000 ÷ 20,000 = \$63.50	
Return on equity = \$330,000 ÷ \$3,000,000 = 11.0%	

*In this example, sunk costs refer to capital that has already been expended and doesn't enter into the decision-making process.

Due to the increased popularity of its product, the company has the opportunity to expand its business and double production. However, in order to facilitate the increased production, a capital expenditure of \$500,000 in the next year must be made in additional plant facilities and equipment. The question is posed to management by the sales department. Is this a wise economic decision for the suppliers of capital to the firm?

Expanded Production

Revenue	\$3,200,000
Sales = 40,000 bbls x \$80.00	
Cost of goods	
Sunk costs*	\$ 150,000
Variable costs (\$56.00 x 40,000)	\$2,240,000
	\$2,390,000
Additional capital contribution	
	\$ 500,000
Accounting profit	
	\$ 310,000
Unit cost per barrel = \$2,890,000 ÷ 40,000 = \$72.25	
Return on equity = \$310,000 ÷ \$3,500,000 = 8.86%	

Let's look at the change in economic value to the producers of capital.

	Expanded Operations	Current Operations	Change
Change in Revenue	\$3,200,000	\$1,600,000	\$1,600,000
Change in Expenses	\$2,890,000	\$1,270,000	\$1,620,000
Change in Economic Value	\$310,000	\$330,000	(\$20,000)

When looking at the prospect of expanding the business, management might take the view that, since next year's operational costs are still below the sales price of the product, the business is still generating accounting profit and, therefore, the expansion is a good decision. The company can increase its market and still generate profits. However, from a cost-of-capital point of view, the decision to expand production generates no economic value to the producers of capital and therefore shouldn't be undertaken. This is evident if we look at the marginal revenue and marginal costs involved.

	Marginal Revenue
Marginal Revenue	MR
Units of Production	P
Total Revenue	TR

In the formula

$$MR = \frac{(TR_1 - TR_0)}{(P_1 - P_0)} = \frac{(\$3,200,000 - \$1,600,000)}{(40,000 - 20,000)} = \frac{\$1,600,000}{20,000} = \$80.00 \text{ per bbl}$$

	Marginal Cost
Marginal Cost	NC
Total Cost	TC

$$MC = \frac{(TC_1 - TC_0)}{(P_1 - P_0)} = \frac{(\$2,890,000 - \$1,270,000)}{(40,000 - 20,000)} = \frac{\$1,620,000}{20,000} = \$81.00 \text{ per bbl}$$

The increased marginal cost actually caused economic value to be destroyed, even though the company was still recording an accounting profit. The cost of capital to the shareholders increased by \$1.00 per barrel, thereby reducing the return on equity from 11.0% to 8.86%. If other investments of capital produce returns in excess of 8.86%, the opportunity cost of expansion doesn't provide an adequate return for the additional risk warranted by the expansion. The decision to expand shouldn't be undertaken in this situation.

Sum-of-the-Parts Analysis

As we examine valuation techniques from the viewpoint of the providers of capital, we have noted that the most important question to be answered is—What's the return on the investment? Whether we use discounted cash flow, price ratios, free cash flow, or enterprise value, the investor is always seeking an appropriate return for the risk assumed. Additional questions arise in regard to opportunity cost (e.g., can the assets to be allocated to a particular project be better used elsewhere?) An alternative source of return on investment may be in private equity, rather than in the public domain. In a well-diversified corporation, certain levels of asset allocation can be committed to investment in non-public related activities. Private equity relates to this type of allocation in the form of venture capital or managed buyouts.

When viewing such companies in order to make asset allocation decisions, the research analyst must be aware of the various business segments within the corporate structure. Each of these segments must be valued separately and then combined into a sum-of-the-parts analysis. The idea in this type of analysis is that the separate segments of the company may be worth more in and of themselves than the combination of all segments that create the company. In other words, if the company were sold today, the individual business segments may be sold for more value than the total corporate value. Unprofitable lines of business will be identified and can be divested, whereas the more profitable lines may be enhanced through additional capital commitment. We will look at a hypothetical company to illustrate an example of the sum-of-the-parts concept.

Knight/Reader, LLC is a small publisher and distributor of textbooks and audio/video training materials. The company has also entered into an agreement with Wildworld Productions, a company that produces rock concerts and sports events for distribution of Wildworld's products on a national scale. Knight/Reader has recently expanded its business into the potentially lucrative general readership market for novels and periodicals. As an adjunct to its audio/video line of business, it has created a consumer electronics division selling AV products through Internet-based outlets. As an analyst reviewing Knight/Reader's business plan and financials for possible investment by your firm, you are faced with several questions before a recommendation can be made. Among these questions are:

1. What method of valuation should be used for this company?
2. What's the value of each business segment of the firm?
3. What valuation metric should be used for each segment?
4. What's the per-share value of the firm?

Since Knight/Reader is a private company with various business lines—some of them established and some of them new—and each growing at a different projection rate, you decide that the best approach to valuation is the sum-of-the-parts analysis. A review of their financials provides the following information:

Total Net Income	\$3,000,000
Common Shares Outstanding	2,000,000

Business Line	Percentage of Net Income	Projected Growth
Textbook sales	70%	15%
Audio/video sales	15%	20%
Media distribution	8%	10%
General market sales	5%	35%
Electronics sales	2%	30%

The required rate of return on equity for capital projects of your company when investing in non-public ventures is 18%. Knight/Reader is offering to sell the company at \$25.00 per share.

To calculate a value for the company, we must first project the growth of each of the business segments and then discount the amount of projected income back to present values. We will assume a constant growth rate over the next five years for each segment at the percentages previously indicated.

Business Line	Current Income		Projected Income	Percentage of Total
Textbook sales	\$2,100,000 (1.15) ⁵	=	\$4,223,850	64%
AV sales	450,000 (1.20) ⁵	=	1,119,744	17%
Media distribution	240,000 (1.10) ⁵	=	386,522	6%
General market sales	150,000 (1.35) ⁵	=	672,605	10%
Electronics sales	60,000 (1.30) ⁵	=	222,776	3%
Total	\$3,000,000		\$6,625,497	100%

When looking at other businesses in the market sector of this company, research indicates that they typically sell for around 15 times their net income and, therefore, if we take the projected income of \$6,625,497 times 15, we get a projected value for the company after five years of \$99,382,455. However, this figure must be discounted back to its present value at our required rate of return. The calculation for this discount is:

V_0 = Value as of the date of calculation
 T = Total income
 k^* = Required rate of return

$$V_0 = \frac{T^5}{(1 + k^*)^5} = \frac{\$99,382,455}{(1.18)^5} = \$43,440,989$$

Therefore, the per-share value of the sum of the parts equals:

$$\frac{\$43,440,989}{2,000,000} = \$21.72$$

The conclusion of our sum-of-the-parts analysis is that, by business line, some segments of the company produce positive values, while others produce negative values. Audio/visual sales and general market sales produced positive results and could be worth more than their projected values. Although consumer electronics produced a positive result, the amount was not significant in light of its growth projections. This may be the result of low profit margins and a highly competitive marketplace. Further investigation is required to determine if this product line is worth the cost of capital. Since textbook sales and the media production segments produced negative results, a closer look should be given to these two product lines. The media production segment is of special concern since the company doesn't control production of the product being distributed. If this line doesn't produce the required rate of return, the company should either renegotiate a higher fee with the producer or get rid of the product line. Textbook sales, although the mainstay of the business, seem to be flattening out. Perhaps the market is becoming saturated at this point. Closer scrutiny should be given to this segment.

In addition, opportunity costs should be closely examined. Since the company is willing to sell at a share price of \$25.00 and the value per share is only \$21.72, the offering price is too high for the cost of capital at this point. Before committing capital to this project, the purchasing company should seek to invest elsewhere as well as internal expansion before making the decision to invest in this private equity venture.

Different Valuations When using a sum-of-the-parts analysis it may be necessary to use different valuation metrics for each business unit. For example, you might use 6.0 times EBITDA for one unit, 1.5 times revenue for another unit, and 13.0 times earnings for a third unit. By adding the value of all three units, subtracting the net debt, and allowing for any non-operating adjustments, the research analyst could calculate the equity value of the firm. In some instances, you would use a DCF analysis with some businesses and a relative valuation with others.

Categorizing Valuation Multiples

Multiples of Earnings

- Equity earnings multiples: Price/earnings ratios and variants
- Operating earnings multiples: Enterprise value-to-EBITDA or EBIT
- Cash earnings multiples

Multiples of Book Value

- Equity book multiples: Price-to-book equity or tangible book value
- Capital book multiples: Enterprise value-to-book value of capital

Multiples of Revenues

- Price-to-Sales
- Enterprise Value-to-Sales

INTUITIVE APPROACH		
Sector	Multiple Used	Rationale
Cyclical	Relative P/E	Use with normalized earnings
High Tech, High Growth	PEG	When there are big differences across firms
High Tech, No earnings	Price/Sales or EV/Sales	Assume future margins will be good
Heavy Infrastructure	EV / EBITDA	Firms in sector have losses in early years and reported earnings can vary depending on depreciation method
REIT	Price/Cash Flow Funds from Operations	Generally no CAPEX Investments come from equity earnings
Financial Services	Price/BV, EV to Book Value Price/Net Tangible Book Value	Book value often marked to market
Retailing	Price to Sales	If leverage is similar across firms
Retailing	EV / Sales	If leverage is different across firms

WHAT TO LOOK FOR

- | | |
|----------------------------|---|
| 1. <i>Multiple</i> | Determining Variables |
| 2. <i>Price/Earnings</i> | Growth, Payout, Risk |
| 3. <i>Price/Book Value</i> | Growth, Payout, Risk, ROE |
| 4. <i>Price/Sales</i> | Growth, Payout, Risk, Net Margin |
| 5. <i>EV/EBIT</i> | Growth, Some Reinvestment Needs, Leverage, Risk |
| 6. <i>EV/EBITDA</i> | Growth, Greater Reinvestment Needs, Leverage, Risk |
| 7. <i>EV/Sales Ratio</i> | Growth, Net Capital Expenditure Needs, Leverage, Risk, Operating Margin |
| 8. <i>EV/Book Value</i> | Growth, Leverage, Risk and Return on Capital |
| 9. <i>EV/EBITDAR</i> | Companies that depend on leases (Restaurants and Retail) |

WHAT WORKS BEST GIVEN CERTAIN CIRCUMSTANCES**DCF**

1. Cash flows are currently positive.
2. Can be estimated with some reliability for future periods
3. When the benchmark for risk, i.e., 5-year, 10-year, and 30-year Treasury bonds, can be considered to have a positive correlation to the stock price

DDM

4. Dividends are currently positive.
5. Can be estimated with some reliability for future periods

M&A Valuation**Exchange Ratios**

When one company agrees to purchase another, the acquiring company may use cash, its common stock, or a combination of the two. If the acquisition is made in stock, a research analyst would need to understand the concept of the exchange ratio.

	<u>ABC</u>	<u>XYZ</u>
EPS	\$3.00	\$1.25
P/E Ratio	16	20

Based on the earnings per share and the price/earnings ratio, we can determine the market price of the shares—ABC is trading at \$48 and XYZ is trading at \$25. If ABC is considering making an acquisition of XYZ using its common stock, and offers a premium of 20%, what's the exchange ratio? The exchange ratio is found by dividing the offer price (not the market price) of the target company by the market price of the acquiring company. The exchange ratio is .625 [$\$30 (\$25 \times 1.20) \div \$48$].

Accretive or Dilutive To calculate if the acquisition is accretive or dilutive, a research analyst would add the combined net incomes of the two companies and divide by the total number of shares of the acquiring company, including the additional shares issued to fund the transaction.

	ABC	XYZ
Outstanding Shares	2,000,000	900,000
Net Income	\$6,000,000	\$1,125,000

The total acquisition cost is \$27,000,000 ($\$30 \times 900,000$ shares outstanding). This will require 562,500 ($\$27,000,000 \div \48 per share of ABC) additional shares of ABC to be issued, bringing the total shares outstanding to 2,562,500. The new level of earnings available to common for ABC is \$7,125,000 (ABC \$6,000,000 + XYZ \$1,125,000). $\$7,125,000 \div 2,562,500 = \2.78 EPS. The EPS of ABC before the acquisition was \$3.00. The acquisition has resulted in an earnings reduction of \$.22 per share; therefore, the activity has been dilutive to the shareholders of ABC.

Percentage Ownership To determine XYZ's percentage ownership after the merger, it's necessary to determine the exchange ratio and calculate the new number of shares issued. The new number of shares to be issued is added to XYZ's outstanding shares to find the total number of shares of the combined company. The new shares issued are divided by the combined company total. The exchange ratio previously calculated was .625, multiplied by the percentage of stock offered (100%), which equals an exchange ratio of .625. The new shares issued equal 562,500. The total number of shares of the combined company is 2,562,500. XYZ will own 22% of the combined company ($562,500 \div 2,562,500$).

Remember, many transactions are a combination of cash and stock and, in order to find the percentage ownership, you need to multiply the exchange ratio by the percentage of the offer in stock.

The previous example doesn't take into consideration a cash acquisition. If the cash used for the acquisition was borrowed funds (bank borrowing and/or a debt issue), the additional interest expense needs to be deducted from EBIT in order to calculate the revised net income. A research analyst may present various financing alternatives to the acquirer in order to create an accretive result for earnings.

Comparing a Stock versus Cash Transaction

	Company G	Company I
Revenue	1,000	900
Operating Income	98	92
– Interest Expense	10	7
– Taxes	34	32
Net Income	54	53
Shares Outstanding	60	30

Stock Transaction Company I has agreed to be acquired by Company G. The terms of the acquisition require the issuance of 1.5 shares of Company G common stock for each share of Company I. What's the effect on the EPS of Company G following the acquisition?

Based on the terms of the acquisition, an additional 45 million shares of Company G common stock will be issued ($1.5 \times 30 \text{ MM}$). The total number of shares outstanding following the acquisition will be 105 MM ($60 \text{ MM} + 45 \text{ MM}$). Company G's net income following the acquisition is \$107 MM (Company G's net income = \$54 MM + Company I's net income of \$53 MM). Prior to the acquisition, Company G had \$.90 EPS ($\$54 \text{ MM} \div 60 \text{ MM}$). The EPS following the acquisition is \$1.02 ($107 \text{ MM} \div 105 \text{ MM}$), a \$.12 accretive effect.

What if the terms of the transaction were cash in lieu of stock?

Cash Transaction If Company G agrees to acquire Company I for \$250 MM cash, raised by issuing 7% debentures, what's the accretive effect on Company G's EPS, assuming no change in the tax rate for the company?

If Company G issues \$250 MM of debt to fund the acquisition, the interest expense will increase by \$17.5 MM ($250 \times .07$).

	Company G	Company I	Combined Co.
Revenue	1,000	900	1,900.0
Operating Income	98	92	190.0
Interest Expense	10	7	34.5
Income before Tax			155.5
Taxes (assumed rate 38.6%)	34	32	60.0
Net Income			95.5
Shares Outstanding	60	30	

Since no additional shares were issued, the EPS will be \$1.59 ($\$95.5 \text{ MM} \div 60 \text{ MM}$). The acquisition has a \$.69 accretive effect.

These results are not meant to favor cash versus stock transactions since these are hypothetical numbers.

It's important to understand that these results may only provide a short-term analysis of the effect of the merger. The acquisition is made to create value and be accretive to shareholders. An acquisition is considered accretive to shareholder value if the return on invested capital (ROIC) exceeds the weighted average cost of capital (WACC).

Enterprise Value

Previously, this chapter examined enterprise value and mentioned that when a company completes an M&A transaction, its enterprise value will be affected. We have included some further examples to review this concept.

Before Transaction

	MST	RMM
Common Shares Outstanding	2,400,000	2,000,000
Market Value	\$20.00	\$11.00
Debt Outstanding	\$52,000,000	\$10,000,000
Cash and Equivalents	\$16,500,000	\$5,000,000

MST agrees to purchase RMM for \$20,000,000. The terms of the acquisition are: \$6,000,000 MST common stock (based on a market value of \$20 per share); \$4,000,000 of newly issued MST 6.0% debentures; \$10,000,000 cash to RMM shareholders, obtained from MST's revolving credit line; and \$10,000,000 in assumed debt of RMM. What is the new enterprise value if MST stock is trading at \$23.00 after the acquisition is complete?

Step 1: Take the common shares outstanding and multiply by the current market price. MST's 2.4 million shares plus 300,000 (\$6 MM ÷ \$20) additional shares issued for the transaction.

Step 2: Take the debt from MST plus the debt from RMM and add the \$4,000,000 of newly issued debt plus \$10,000,000 of debt incurred from the use of the revolving credit line.

Step 3: Subtract the total cash and equivalents of MST and RMM—\$16,500,000 and \$5,000,000, totaling \$21,500,000.

The following table illustrates the effects of the transaction:

After Transaction

	MST
Common Shares Outstanding	2,700,000
Market Value	\$23.00
Debt Outstanding	\$76,000,000
Cash and Equivalents	\$21,500,000
To Calculate EV:	
Market Cap	\$62,100,000 (2,700,000 x \$23.00)
Plus Total Debt	\$76,000,000
Less Cash	\$21,500,000
EV	<u>\$116,600,000</u>

Comparable Company and Transaction Multiple Analysis

When reviewing a potential M&A transaction, a common valuation metric is arrived at by comparing recent transactions in a sector, based on enterprise value to EBITDA, or EV to revenue. Let's examine the process and review certain adjustments.

Effects of Employee Stock Options When working on the analysis of a potential M&A transaction, a research analyst may be asked to adjust enterprise value. You will need to recalculate the number of outstanding shares based on the expected exercise of employee stock options. Most companies will use the proceeds from the exercised options to repurchase shares in the market at the current price. Any differential in the number of shares will be made up through the treasury stock account.

Current Outstanding Shares	6,000,000
Number of Options	300,000 at a strike price of \$25.00
	200,000 at a strike price of \$50.00
Current stock price is \$30.00 per share	

The expected exercise of employee stock options is limited to those that have intrinsic value (where the market value exceeds the strike price). The options that have a strike price of \$25.00 will be exercised. The options with a strike price of \$50.00 will not be exercised. Therefore, the exercised options will generate proceeds to the company of \$7,500,000 ($300,000 \times \25.00). This would permit the company to repurchase 250,000 shares from the market ($\$7,500,000 \div \30.00). The difference between the number of shares repurchased and the number of shares granted through the exercise of the options, equals 50,000 shares ($300,000 - 250,000$). The additional shares will be issued from the company's treasury stock account, bringing the number of shares outstanding to 6,050,000 ($6,000,000$ previously outstanding + $50,000$ shares from treasury stock).

Offer Value This is very important in calculating the offer price for a target company in an acquisition. It's assumed that any employee with an in-the-money option will exercise the option. In taking this example one step further, if this company was offered \$45.00 per share, the offer value would be \$272,250,000 ($6,050,000 \times \45.00).

Transaction Value In order to calculate the transaction value (enterprise value), we need to add the target company's debt and subtract the cash (add the net debt). If we assume the net debt is \$50,000,000, the transaction value is \$322,250,000. The difference between the offer value and the transaction value is the net debt.

Implied Transaction Multiple In order to calculate the implied transaction multiple, divide the enterprise value based on the offer price by the appropriate financial statement figure (e.g., revenue, EBIT, EBITDA). If the appropriate valuation metric is EV to EBITDA and the EBITDA is \$21,000,000, the multiple will be 15.3 ($\$322,250,000 \div \$21,000,000$).

Calculating Goodwill after the Acquisition is Complete

To calculate the amount of goodwill created after the acquisition, the company's net tangible assets are subtracted from the offer value. The net tangible assets are determined by subtracting the company's liabilities, existing goodwill, and intangibles from the total assets. The research analyst can find this information on the balance sheet of the target company.

TCB Corporation (\$ in millions)

Total assets	\$1,800
Total liabilities	\$700
Existing goodwill	\$250
Intangibles	\$180
Shares outstanding	25 MM
Current stock price	\$74
Offer Price	\$85

The net tangible assets are \$670 MM ($1,800 - [700 \text{ MM} + 250 \text{ MM} + 180 \text{ MM}]$). The offer value is \$2,125 MM ($25 \text{ MM} \times \$85$). The amount of goodwill created is equal to \$1,455 MM ($2,125 \text{ MM} - 670 \text{ MM}$). This amount of goodwill will be added to the acquirer's existing goodwill to calculate the new total goodwill.

Exchange-Rate Risk

Strong U.S. Dollar A strong U.S. dollar (correlated with weakening foreign currencies) will make foreign companies more attractive takeover targets for U.S.-based buyers. U.S. investors will be able to exchange their higher-priced currency for a greater amount of a foreign currency and will have more purchasing power as the dollar rises. In a strong-dollar environment, U.S. companies will be less attractive takeover candidates, since foreign buyers will have less buying power.

Weak U.S. Dollar A weak U.S. dollar (correlated with strengthening foreign currencies) will make U.S. companies more attractive takeover targets for foreign buyers since they're able to convert their higher-priced currency into a greater number of U.S. dollars. This weak-dollar environment would give foreign buyers greater purchasing power when acquiring U.S.-based assets. Foreign companies will be less attractive takeover candidates for U.S. buyers, since these potential acquirers would have less buying power.

Risk After the Deal is Announced If a company announces that it will acquire a foreign company and, at the time of the closing of the acquisition, the target company's currency has declined in value, the cost of the acquisition will have declined. For example, if the euro was currently trading at 1.24 and a German company announced a purchase of the U.S. company for \$21,000,000, this equals €16,935,483 ($\$21,000,000 \div 1.24$). At the time of the closing, if the euro was trading at 1.35, the U.S. dollar has weakened and the cost of the acquisition to the German company is now €15,555,555 ($\$21,000,000 \div 1.35$). The cost to purchase the U.S. company would have declined by €1,379,928. The reverse is true if the U.S. dollar strengthened against the euro. In this example, there's exchange-rate risk for the acquirer, not for the target company, since the offer was made in U.S. dollars.

If an offer consisted of stock or a portion in stock, there would also be exchange-rate risk. For example, if a U.S. company offered .50 shares of stock (trading at \$60) to a British company and the exchange rate is currently 1.65 dollars, the stock portion of the acquisition is £18.18 ($\$60 \times .50 = \$30 \div 1.65$). If sterling strengthens to an exchange rate of 1.75, the stock portion of the acquisition would now be valued at £17.14 ($\$60 \times .50 = \$30 \div 1.75$). In this example, the target company would receive a lower value if the offer is based on the U.S. company dollar value, and the dollar declines against sterling.

This is in addition to the risk associated with the acquirer's stock price fluctuations in a fixed exchange ratio transaction. If, in the previous example, the market price of the U.S. company declines to \$56.00 a share, the offer would decline in value even if the exchange rate remained unchanged. For example, if a U.S. company offers .50 shares of stock (now trading at \$56) to a British company and the exchange rate is currently 1.65 dollars, the stock portion of the acquisition is now £16.97 ($\$56 \times .50 = \$28 \div 1.65$). If the U.S. dollar declines and the share price of the U.S. company declines, this would have a double negative impact on the value of the transaction to the British company.

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CHAPTER 5

Examination and Review of Specific Companies



I The Analysis of Lyons Bearings & Motors Co.

The analysis of a company will begin with a breakdown of the company's financial statements. In this chapter, the applicable ratios and analysis of liquidity, capitalization, and performance will be extracted from a model company called Lyons Bearings and Motors Co. (LBM).

LBM manufactures and distributes industrial motors, drilling equipment, and pump equipment. They have acquired and divested more than a dozen different companies in the past 18 years. LBM is one of the largest companies within the machine-tool industry.

The machine-tool industry is mature. High capital investment in plant and equipment is required. Distribution of consumer goods through company-operated stores has declined during the past decade. Sales by way of the Internet have increased from zero ten years ago to 3.2% of net sales for the industry in 2015. The profit margins for goods sold over the Internet are higher than the profit margin for goods sold in company stores.

LBM receives a very small percentage of sales through the Internet. These are primarily replacement parts. LBM conducts its sales through a dedicated sales force. They have historically been loyal to the company and have helped maintain the strength of pricing by LBM due to brand identity. LBM machines are known for durability and reliability. This is crucial in the machine-tool industry due to the high price of products. The drivers of the machine-tool industry are cost of goods sold and strength of product pricing.

Due to the cyclical nature of the machine-tool industry, the ability to maintain its prices during a market downturn has allowed LBM to weather storms that have caused other members of the industry to incur substantial losses. Historically, LBM has had a good relationship with its employees' unions. In the past, LBM has been more likely to sell operating businesses as units rather than closing down factories. However, in 2015, one factory was closed down.

The Balance Sheet, Income Statement, Earnings per Share Data, and Statement of Cash Flows for LBM are illustrated in exhibits 1, 2, 3, and 4.

Exhibit 1: *Lyons Bearings & Motors Co.*

	2015	2014	2013
Assets			
1. Cash & Cash Equivalents	927.46	150.00	121.40
2. Short-Term Investments	192.20	192.20	
3. Accounts Receivable	1,374.50	1,405.30	1,482.90
4. Inventories	1,237.05	1,189.80	1,295.30
5. Prepaid Expenses, Deferred Tax	<u>381.10</u>	<u>381.10</u>	<u>288.20</u>
6. Current Assets	4,112.31	3,318.40	3,187.80
7. Property, Plant & Equipment	519.90	1,279.90	1,633.00
8. Goodwill and Other Intangibles	<u>4,404.40</u>	<u>4,896.40</u>	<u>5,689.30</u>
9. Total Assets	9,036.61	9,494.70	10,510.10
Liabilities			
10. Notes Payable	355.50	1,155.50	563.70
11. Accounts Payable	687.25	730.30	761.70
12. Accrued Liabilities/Other	<u>1,912.30</u>	<u>1,912.30</u>	<u>1,526.30</u>
13. Current Liabilities	2,955.05	3,798.10	2,851.70
14. Long-Term Debt	2,092.10	2,092.10	2,900.70
15. Accrued Post Retirement Benefit Expense	<u>1,123.50</u>	<u>1,123.50</u>	<u>920.40</u>
16. Total Liabilities	6,170.65	7,013.70	6,672.80
Stockholders Equity			
17. Common Stock	170.40	169.20	168.00
18. Paid-In Capital	367.30	364.70	324.20
19. Retained Earnings	2,841.36	2,460.20	3,666.50
20. Treasury Stock	<u>(513.10)</u>	<u>(513.10)</u>	<u>(321.40)</u>
21. Total Stockholders' Equity	2,865.96	2,481.00	3,837.30
22. Total Liabilities & Stockholder Equity	9,036.61	9,494.70	10,510.10

Footnote: There's 587.52 million dollars of excess cash.

Exhibit 2: *Lyons Bearings & Motors Co. Income Statement 2015*

23. Net Sales	9,163.34
24. Cost of Goods Sold	<u>– 7,240.40</u>
25. Gross Profit	1,922.94
26. Selling, General & Administrative	– 619.50
27. Depreciation & Amortization	– 260.00
28. Research & Development	– 142.67
29. Other Operating Expenses	<u>– 20.80</u>
30. Operating Income	879.97
31. Unallocated Expenses	<u>– 116.60</u>
32. EBIT	763.37
33. Interest Expense	<u>– 214.70</u>
34. Pre-tax Income	548.67
35. Taxes	<u>– 76.79</u>
36. Net Income	<u>471.88</u>

Exhibit 3: *Lyons Bearings & Motors Co.
Per Share Data - Year ending December 31, 2015*

37. EPS - Operations	2.79
38. Average Shares Outstanding	169,300,000
39. Common Dividend	\$0.66

Liquidity Measurements

The ability of a company to meet its financial obligations is determined by evaluating the cash flow of the company, its working capital, and ratios such as the current ratio and acid test ratio (quick assets ratio).

The working capital of a company is determined by subtracting the current liabilities (13) from the current assets (6).

$$\text{Current Assets} - \text{Current Liabilities} = \text{Working Capital}$$

For 2015 (in \$ millions) Working Capital equals:

$$4,112.31 - 2,955.05 = 1,157.26$$

The current ratio is determined by dividing the current assets (6) by the current liabilities (13).

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

For 2015 (in \$ millions) the current ratio equals:

$$4,112.31 \div 2,955.05 = 1.39$$

The acid test ratio is a more stringent measurement of liquidity than the current ratio. The liquid current assets (cash, cash equivalents, and accounts receivable) are divided by the current liabilities.

$$\text{Acid Test Ratio} = \frac{[\text{Cash (1) + Cash Equivalents (1) + Accounts Receivable (3)}]}{\text{Current Liabilities (13)}}$$

For 2015 (in \$ millions) the Acid Test equals:

$$[927.46 + 1,374.50] \div 2,955.05 = 0.779$$

Cash flow measurements are described for operating activities, investing activities, and financing activities. The cash flow statements of LBM are found in Exhibit 4. The changes in the cash flow are based on the sources and uses of cash. Various turnover ratios will be used to assess the benefits and disadvantages of changes in cash flow.

Exhibit 4: *Lyons Bearings & Motors Co. Cash Flow Statement 2015*

Cash Flow Statement 2015 (\$ millions)	
Cash Flows from Operating Activities	
40. Net Income from Operations	471.88
41. Non-recurring Items	20.80
42. Depreciation and Amortization	<u>260.00</u>
43. Cash Flow from Operations	752.68
Changes in Operating Assets and Liabilities	
44. Accounts Receivable	30.80
45. Inventories	(47.25)
46. Accounts Payable	<u>(43.05)</u>
47. Change in Working Capital	(59.50)
Cash Flow from Investing Activities	
48. Sale of Patents	492.00
49. Capital Expenditures	(200.00)
50. Sale of Assets	<u>700.00</u>
51. Net Cash Flow Investing Activities	992.00
Cash Flow from Financing Activities	
52. Increase in Common Stock	3.80
53. Change in Short Term Borrowing	(800.00)
54. Dividend	<u>(111.52)</u>
55. Net Cash from Financing Activities	(907.72)
Net Change in Cash & Cash Equivalents	
56.	777.46

Net cash flow from operating activities (items 40 – 43) begins with net income of \$471,880,000 (40). LBM sold properties and production equipment for \$20,800,000 less than its book value. The accounting loss on the income statement also results in a source of cash on the cash flow statement (41). This is because the proceeds from the sale of assets are recorded as an investing cash flow. Although cash proceeds are flowing in, the proceeds are less than their book value. The accounting loss on the income statement would result in double counting if there was not an offsetting positive cash flow in the statement of operating cash flows.

There's also a source of cash from depreciation and amortization expense of \$260,000,000 (42). Although depreciation and amortization are deductible expenses, no cash has been expended. Therefore, depreciation and amortization expense must be added back to cash to negate the non-cash charge to earnings that was previously deducted on the income statement. The result is a positive cash flow from operating activities of \$752,680,000.

Cash Flows from Operating Activities		
40. Net Income from Operations	471.88	(source of cash)
41. Non-recurring Items	20.80	
42. Depreciation and Amortization	<u>260.00</u>	(source of cash)
43. Cash Flow from Operations	752.68	

Changes in the operating assets and liabilities of LBM reflect the changes to inventories, accounts receivable, and accounts payable. An increase in accounts receivable is a use of cash, while a decline is a source of cash. In 2015, the cash flow change in accounts receivable indicates an increase of \$30,800,000 (44). This source of cash is due to the decline in accounts receivable. The inventory changes created a cash flow use \$47,250,000 (45). This indicates that LBM has increased its inventory level. Accounts payable shows a cash flow usage of \$43,050,000 (46). LBM has reduced the level of accounts payable by that amount. The changes in receivables, payables, and inventories have negative cash flow of \$59,500,000.

Changes in Operating Assets and Liabilities

44. Accounts Receivable	30.80	(source of cash)
45. Inventories	(47.25)	(use of cash)
46. Accounts Payable	<u>(43.05)</u>	(use of cash)
47. Change in Working Capital	(59.50)	(use of cash)

Significance of the Changes

Change in Accounts Receivable LBM experienced a decline in accounts receivable of \$30.8 million from \$1,405.3 million in 2014 to \$1,374.5 million in 2015. This results in a \$30.8 million source of cash on the 2015 cash flow statement. This is a source of cash because less money from credit sales is tied up in the collection phase. It indicates that the company is using its assets more efficiently. However, is the decline in accounts receivable because customers are paying their credit charges more quickly, or is LBM's business slowing, resulting in less sales on credit? We can gain some insight by calculating the accounts receivable turnover ratio.

$$\text{Accounts Receivable Turnover Ratio} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

$$\text{Average Accounts Receivable} = \frac{(\text{Opening Accounts Rec. Bal.} + \text{Closing Accounts Rec. Bal.})}{2}$$

The accounts receivable turnover ratio is used to compare a company's ability to collect cash from one year to the next. In this way it's possible to recognize trends. By observing the trend, it can be determined if the company's ability to collect cash is increasing or decreasing. It can also be compared to the industry average to determine if the company is collecting cash faster, slower, or in line with the industry average.

(\$ millions)

2015 Net Sales (Income Statement item 23): 9,163.34

Average Account Receivable

(Balance Sheet item 3: years 2014 & 2015)

$$[1,405.30 + 1,374.50] \div 2 = 1,389.90$$

To perform the calculation, it's necessary to obtain information from the Balance Sheet (Exhibit 1) and from the Income Statement (Exhibit 2).

Account Receivable Turnover for 2015

$$9,163.34 \div 1,389.90 = 6.59$$

Assume the following financial information:

The Average Accounts Receivable for 2014 = 1,444.10

Net Sales in 2014 were 10,156.00

$$\text{The Accounts Receivable Turnover for 2014} = 10,156.00 \div 1,444.10 = 7.03$$

The accounts receivable turnover ratio has declined from 7.03 in 2014 to 6.59 in 2015. This indicates that LBM's ability to collect cash has weakened. The company is receiving payments more slowly than in the previous year.

The decline in accounts receivable turnover from one year to the next can be measured by calculating the rate of change. Rate of change in accounts receivable = (Accounts Receivable Turnover Ratio 2015 – Accounts Receivable Turnover Ratio 2014) ÷ (Accounts Receivable Turnover Ratio 2014) = (6.59 – 7.03) ÷ 7.03 = - 0.0626 or - 6.26%. This represents a 6.26% decline in the company's ability to turn over receivables.

The 2015 industry average for the accounts receivable turnover ratio was 6.7. In 2014 the average was 7.15. The decline in the industry average accounts receivable turnover ratio from 2014 to 2015 was 6.3%. LBM's decline appears to be in line with the change in the industry average. This may not indicate a weakness specific to LBM, but may indicate weakness in the industry in general.

Change in Inventory LBM has increased its inventory by \$47.25 million from \$1,189.8 million in 2014 to \$1,237.05 million in 2015. This is a use of cash, because more money is tied up in the production phase, resulting in less available cash. The inventory turnover ratio is used to examine how efficiently inventory is being used. If inventory is increasing, is the buildup of inventory a result of a slowdown in sales, or is the buildup because of additional inventory purchases necessary to keep up with greater demand?

Inventory Turnover Ratio = $\frac{\text{Cost of Goods Sold (COGS)}}{\text{Average Inventory}}$
Average Inventory = $\frac{(\text{Opening Inventory} + \text{Closing Inventory})}{2}$

2015 COGS (Income Statement item 24): 7,240.40

Average Inventory

(Balance Sheet item 4: years 2014 & 2015)

$$[1,189.80 + 1,237.05] \div 2 = 1,213.43$$

Inventory Turnover Ratio 2015

$$7,240.40 \div 1,213.43 = 5.97$$

Assume the following financial information:

The Average Inventory for 2014 = 1,242.55

COGS in 2014 was 8,124.80

The Inventory Turnover Ratio for 2014 = 6.54

The inventory turnover ratio is a tool to measure how efficiently a company is managing its inventory. The inventory turnover ratio is used to compare a company's ability to move inventory and generate cash from one year to the next. In this way it's possible to recognize trends. By observing the trend, it can be determined if the company's ability to move inventory is increasing or decreasing. It can also be compared to the industry average to determine if the company is moving its inventory faster, slower, or in line with the industry average.

To perform the calculation, it's necessary to obtain information from the Balance Sheet (Exhibit 1) and from the Income Statement (Exhibit 2).

LBM's inventory turnover ratio has declined from 6.54 in 2014 to 5.97 in 2015. The slowdown in the inventory turnover may be an indication that LBM's ability to sell its products has weakened, resulting in a buildup of inventory.

The decline in inventory turnover can be measured by calculating the rate of change.

Rate of change in inventory turnover = $[\text{Inventory Turnover Ratio 2015} - \text{Inventory Turnover Ratio 2014}] \div \text{Inventory Turnover Ratio 2014} = [5.97 - 6.54] \div 6.54 = -0.087$ or - 8.7%. LBM's inventory turnover ratio has declined by 8.7% from 2014 to 2015.

We can compare the decline in LBM's inventory turnover ratio to the industry's inventory turnover ratio average to gain some insight. The industry inventory turnover ratio was 6.85 in 2014 and 6.70 in 2015. $[6.70 - 6.85] \div 6.85 = -0.022$ or - 2.2%

LBM is moving its inventory less rapidly in 2015 as compared to 2014, as evidenced by an 8.7% decline in inventory turnover. In comparison, the inventory turnover ratio average for the industry has declined by only 2.2%. The rate of decline in the inventory turnover ratio is significantly greater for LBM as compared to that of the industry average. This may indicate that LBM's inventory is building faster than the industry's, which might be a sign that LBM's business is declining at a faster rate than that of its peers.

Accounts Payable Turnover Ratio LBM reduced its accounts payable from \$730.3 million in 2014 to \$687.25 million in 2015. This \$43.05 million decrease in accounts payable is a use of cash from operating activities.

How much leverage does LBM have with its suppliers? The accounts payable turnover ratio provides a picture of LBM's stature with its suppliers. A decrease in the accounts payable turnover ratio is a positive indicator. It indicates that the company has more leverage with its suppliers, allowing the firm to pay for its purchases more slowly, thereby slowing its use of cash.

$\text{Accounts Payable Turnover Ratio} = \frac{\text{Cost of Goods Sold (COGS)}}{\text{Average Accounts Payable}}$ $\text{Average Accounts Payable} = \frac{(\text{Opening Payables} + \text{Closing Payables})}{2}$

2015 COGS (Income Statement item 24): 7,240.40

Average Accounts Payable

(Balance Sheet item 11: years 2014 & 2015)

$[730.30 + 687.25] \div 2 = 708.78$

Accounts Payable Turnover Ratio 2015

$7,240.40 \div 708.78 = 10.22$

Assume the following financial information:

The Average Accounts Payables 2014 = 746.00

COGS in 2014 was 8,124.80

The Accounts Payable Turnover Ratio for 2014 = 10.89

LBM's accounts payable turnover ratio was 10.89 in 2014 and it declined to 10.22 in 2015. The decline in the accounts payable turnover ratio can be measured by calculating the rate of change.

Rate of change in accounts payable turnover = $(\text{Accounts Payable Turnover Ratio 2015} - \text{Accounts Payable Turnover Ratio 2014}) \div \text{Accounts Payable Turnover Ratio 2014} = (10.22 - 10.89) \div 10.89 = -0.0615$ or - 6.15%

LBM's accounts payable turnover ratio declined by 6.15% from 2014 to 2015. In contrast, the change in the industry average accounts payable turnover was insignificant, declining by only 0.44% from 11.50 in 2014 to 11.45 in 2015. This indicates that LBM has more leverage with its suppliers than its peers. Perhaps LBM has renegotiated payment terms with its suppliers allowing the firm to pay its cost of goods more slowly, reducing the speed at which it uses cash.

Asset Turnover Ratio The efficiency in the use of LBM assets is measured through the asset turnover ratio of LBM.

$\text{Asset Turnover Ratio} = \frac{\text{Net Sales}}{\text{Average Assets}}$ $\text{Average Assets} = \frac{(\text{Assets Beginning of Period} + \text{Assets End of Period})}{2}$
--

2015 Net Sales (Income Statement item 23): 9,163.34

Average Assets

(Balance Sheet item 9: years 2014 & 2015)

$[9,494.70 + 9,036.61] \div 2 = 9,265.66$

Asset Turnover Ratio 2015

$9,163.34 \div 9,265.66 = .99$

The asset turnover ratio for LBM is slightly below 1.0. A higher asset turnover ratio indicates a more efficient use of assets. In 2014, the average assets were \$10,002.40 million. Net Sales for 2014 were \$10,156.00 million. The asset turnover ratio for 2014 was 1.02. Therefore, there were minor changes in the efficiency of assets used by LBM during 2015. In comparison to the industry average of 1.05, it would appear that LBM under performs the industry in terms of assets used efficiently (0.99 vs. 1.05).

For a more refined view, compare LBM's asset efficiency to the highest and lowest companies within the industry sector:

	Asset Turnover Ratio	Return on Assets	Return on Equity
<i>Metric Tooling, Inc.</i>	1.16	.687	.2104
<i>Moore Engines Co.</i>	.98	.491	.1275

At 0.99, LBM is near the lower end of the industry and slightly above Moore Engines Co. This could be due to outdated means of production and a lag in the technology applications due to its size and scope.

Cash Flow from Investing Activities The cash flow for LBM from investing activities is based on the investments made in other companies, capital expenditures for property, plant and equipment, and the sale of patents for \$492,000,000 (48) at their amortized cost basis. In 2015, LBM invested \$200,000,000 (49) in plant modernization, which represents a use of cash. LBM sold assets and equipment for \$700,000,000 (50), which was a source of cash.

Cash Flow from Investing Activities	
48. Sale of Patents	492.00
49. Capital Expenditures	(200.00)
50. Sale of Assets	<u>700.00</u>
51. Net Cash Flow Investing Activities	992.00

For 2015 the Cash Flow from Investing Activities equals \$992,000,000. LBM expects the plant modernization and sale of assets to improve the efficiency of its assets. LBM has announced plans to increase the level of capital expenditures for plant upgrades.

Cash Flow from Financing Activities includes the issuance of equity and debt (source of cash), the retirement of debt (use of cash), payment of dividends (use of cash), and repurchase of stock (use of cash).

Cash Flow from Financing Activities	
52. Increase in Common Stock	3.80
53. Change in Short Term Borrowing	(800.00)
54. Dividend	<u>(111.52)</u>
55. Net Cash from Financing Activities	(907.72)

LBM paid a dividend of 66-cents per share to 169 million outstanding shares of common stock. LBM paid off debt totaling \$800 million. It didn't issue debt during 2015.

Net Cash Flow LBM had positive net cash flow of \$777,460,000 in 2015.

This is calculated as follows:

Cash flow from operations, 752.68 (43) – change in working capital, 59.50 (47) + net cash flow investing activities 992.00 (51) – net cash from financing activities, 907.72 (55)

Net Change in Cash & Cash Equivalents	
56.	777.46

Free Cash Flow to the Firm

It's also necessary to calculate the free cash flow to the firm for LBM. Free cash flow to the firm is essentially the cash that's available to all of the providers of capital, both the debt and equity stakeholders. It will influence the target price for the company in view of the free cash flow of its peer group.

Calculation of Free Cash Flow (\$ millions)	
EBIT (32) x (1 - tax rate)	656.50
Depreciation & Amortization (27)	+260.00
Capital Expenditures (49)	– 200.00
Net Change in Working Capital (47)	– 59.50
Free Cash Flow to the Firm (FCFF)	<u>657.00</u>

LBM has free cash flow to the firm of \$657 million dollars.

In many instances of analysis, net operating profit after taxes (NOPAT) is used as a starting point to calculate FCFF, rather than the more conventional EBIT x (1 - tax rate) as shown above. NOPAT is calculated by starting with net income and adding back interest expense, non-recurring costs, tax paid on investment and interest income, and subtracting investing and interest income, as well as the tax shield from interest expense. As you can see, it's quite involved. Therefore, for the purposes of simplification in future estimates of NOPAT, we will continue to use the formula EBIT x (1-tax rate). This is a proxy for the true calculation of NOPAT, and is preferred by many in the industry because of its simplicity.

Earnings Measurements

The net income of LBM for 2015 (36) is \$471,880,000. The earnings before interest and taxes (EBIT) and earnings before interest, taxes, depreciation, and amortization (EBITDA) will now be considered.

Net sales – operating expense = earnings before interest & taxes (EBIT), which is also referred to as operating profit. For 2015, EBIT is \$763,370,000 (32).

EBIT excludes income and expenditure from discontinued operations. Revenue and expense from discontinued operations are separated from the revenue and expense of continuing operations and reported separately on an after-tax basis. To report gains or losses from discontinued operations, the gain or loss must involve a major structural change, like the sale of an entire division. Income or loss from discontinued operations is found on the income statement after income from continuing operations. The net of income or loss from continuing operations and income or loss from discontinued operations is reported as net income. In 2015, LBM is not reporting any gains or losses from discontinued operations; therefore, income from continuing operations and net income are equal.

If the activity that produced the income or loss is not the result of a major structural change, it's reported pre-tax and is included in EBIT. Therefore, the calculation for EBIT includes income and loss from non-recurring items, such as the sale of an asset at a gain or loss, income from investments, severance costs, impairments, write offs, write downs, restructuring charges etc. When projecting net income, adjustments must be made for these non-recurring items.

The term extraordinary items (formerly an after-tax item) is no longer applicable. The Financial Accounting Standards Board (FASB) disallowed the use of extraordinary items in generally accepted accounting principles (GAAP) effective December 15, 2015. The reason for the change is because, for an item to be considered extraordinary, it had to be both unusual and infrequent, and it's difficult to assess what's both unusual and infrequent. It's also difficult to determine if the preparer of the statements evaluated the item appropriately.

In the case of a company with minimal depreciation and amortization expense, EBIT is an important liquidity barometer for creditors. EBIT represents the amount of cash that such a company will be able to use to pay off creditors. It's distinct from free cash flow, which is the cash available to shareholders.

For 2015 \$ millions: $763.37 + 260.00 = 1,023.37$

EBITDA = [EBIT (32)] + [Depreciation + Amortization (27)]
--

EBITDA can be used to analyze the profitability between companies and industries and it eliminates the effects of financing and accounting decisions. EBITDA represents the earnings of a company before the deduction of interest expenses, taxes, depreciation, and amortization. Although sometimes referred to as operational cash flow, EBITDA doesn't represent cash earnings. EBITDA is a good metric to evaluate profitability, but not cash flow.

Return on Investment

The analysis of Lyons Bearings & Motors Co. now turns toward performance measurement. The return on capital invested, the weighted average cost of capital, the return on assets, and return on equity will be calculated. LBM's gross profit margin, net profit margin and operating margin, will be measured.

Exhibit 5: *Key Ratios of Machinery Industry and Lyons Bearings & Motors Co.*

Machine-Tool Industry Averages

ROA	ROE	ROIC	Net Profit Margin	Net Operating Margin	LT Debt-to-Total Cap	Interest Coverage
.0546	.1688	.074	4.75%	7.91%	.51	3.17

Lyons Bearings & Motors Co.

ROA	ROE	ROIC	Net Profit Margin	Net Operating Margin	LT Debt-to-Total Cap	Interest Coverage
.0509	.1765	.1122	5.15%	8.33%	.42	3.56

The return on assets indicates profit as a percentage of operating assets. This is used as a measurement of the company's strength in allocating its resources.

$$\text{Return on Assets (ROA)} = \frac{\text{Net Income (36)}}{\text{Average Assets}}$$

$$\text{Average Assets} = \frac{(\text{Assets Beginning of Period} + \text{Assets End of Period})}{2}$$

Average Assets

(Balance Sheet item 9: years 2014 & 2015)

$$[9,494.70 + 9,036.61] \div 2 = 9,265.66$$

$$\text{For 2015 \$ millions: } 471.88 \div 9,265.66 = .0509 \text{ or } 5.09\%$$

LBM's ROA is in line with industry averages.

The return on equity determines the company's investment return, based on capital invested by shareholders.

$$\text{Return on Equity (ROE)} = \frac{\text{Net Income (36)}}{\text{Average Equity}}$$

Average Equity

(Balance Sheet item 21: years 2014 & 2015)

$$[2,481.00 + 2,865.96] \div 2 = 2,673.48$$

$$\text{For 2015 \$ millions: } 471.88 \div 2,673.48 = .1765 \text{ or } 17.65\%$$

The ROE for LBM is higher than the industry average.

An important metric is the return on invested capital (ROIC). Invested capital represents the debt and shareholder investment in the company.

$$\text{Return on Invested Capital (ROIC)} = \frac{\text{Net Profits After Taxes}}{\text{Invested Capital}}$$

$$\text{Invested Capital} = \text{Total Assets (9)} - \text{Non-Interest-Bearing Current Liabilities (11 + 12)} - \text{Excess Cash (in footnotes)}$$

Net Operating Profits after Taxes (NOPAT): measures the cash generated by operating activities. In some applications it excludes items such as investment income and interest expense, which are included in net income, but are non-operational in nature.

Net operating profit after taxes = 656.50 (see calculation of free cash flow)

Invested capital = 9,036.61 – (2,599.55 + 587.52) = 5,849.54

$656.50 \div 5,849.34 = .1122 = 11.22\%$

Profit Margin

The gross profit margin divides gross profit (sales or revenues – cost of goods sold) by the revenues (net sales). A low gross profit margin indicates a weak or weakened price structure. A high profit margin indicates the company's ability to reduce prices in an economic downturn while remaining competitive and profitable.

Gross Profit Margin = $\frac{\text{Gross Profit (25)}}{\text{Net Sales (23)}}$

For 2015 \$ millions: $1,922.94 \div 9,163.34 = .2099$ or 21%

The gross profit margin is useful for assessing the general competitiveness of the company. In this case, the industry average gross profit margin is 18.7%; this has trended downward from 19.1% over the past five years. LBM has increased its gross profit margin from 20.25 to 21% during the same period. LBM's product selling price is rising faster than its cost of goods. This indicates strength in the marketplace for its pricing structure. This will be considered later in the creditworthiness of LBM and their ability to further leverage through the credit markets.

The net operating margin measures the percentage of revenue that becomes operating income. The operating expenses of the company: selling, general and administrative expenses, rents, depreciation, amortization, real estate taxes, and research & development are all considered in the operating margin.

Net Operating Margin = $\frac{\text{Net Operating Income (32)}}{\text{Net Sales (23)}}$
--

$763.37 \div 9,163.34 = .0833$ or 8.33%

The net operating margin in the industry is 7.91%. The higher operating margin of LBM is a sign of efficiency in the operating expenses of the company.

The net profit margin evaluates net income divided by sales. Operating income minus interest expense and taxes equals net income.

$$\text{Net Profit Margin} = \frac{\text{Net Income (36)}}{\text{Net Sales (23)}}$$

$$471.88 \div 9,163.34 = .0515 \text{ or } 5.15\%$$

The compounded average growth rate (CAGR) of net profit margin for LBM in the past five years is 3%. The average net profit margin of the machine-tool industry is 4.75%. The CAGR for net profit margin in the industry, exclusive of LBM is .8%. LBM has shed acquisitions that have proven marginally profitable, while strengthening core business and expanding profitable business acquisitions. The trend of LBM has been to perform more efficiently than the machine-tool industry over this period.

Summary of Operational Ratio Analysis

LBM has a higher operating and net profit margin than the industry average. These higher ratios indicate that LBM has greater control over the price of its products than the industry average. The return on assets of LBM equals the industry averages. There's room for improvement in the production of LBM's products; additional plant modernization should improve the ROA in the long run.

Debt Structure Evaluation

The evaluation of the debt structure of LBM begins with the debt to total capitalization Ratio. This will be evaluated by considering total debt and also long-term debt exclusively.

$$\text{Long-Term Debt-to-Total Capitalization Ratio} = \frac{\text{Long-Term Debt (14)}}{[\text{Stockholders' Equity (21)} + \text{Long-Term Debt (14)}]}$$

$$\text{For 2015: } 2,092.10 \div [2,865.96 + 2,092.10] = .42$$

The previous years (2014 & 2013) had ratios of .46 and .43 respectively.

LBM has great variation in short-term borrowing year to year, while its long-term borrowing practices are stable. The long-term debt to total capitalization ratio is lower than the industry average of .51. LBM has a credit rating of single A on its long-term debt. LMB could issue additional long-term debt of \$500,000,000 and still remain on the lower side of the industry average.

This is an important consideration based on the following:

- During the past two years LBM has sold a number of small business units that had proven to be marginally profitable.
- The management of LBM has committed to capital investments within the existing business units of \$150,000,000 for the upcoming year.
- LBM has reduced its short-term debt substantially and has significant free cash flow for acquisition purposes.
- LBM is at the low end of the industry debt to total capitalization ratio. It can readily issue additional debt, if the weighted average cost of capital is attractive.

The total interest expense in 2015 was \$ 214,700,000. Almost 1/2 of the interest expense was associated with short-term debt. The interest expense for LBM in 2015 for long-term debt was \$108,800,000. The weighted average interest expense is 5.2%. To determine if LBM can effectively issue more debt, it's prudent to calculate the interest coverage ratio both for 2015 and for an assumed debt issue level.

$\text{Interest Coverage Ratio} = \frac{\text{Earnings Before Interest and Taxes (32)}}{\text{Interest Expense (33)}}$
--

$763.37 \div 214.70 = 3.56$. LBM's interest coverage ratio is the second highest within the industry group of 12 machinery manufacturers. The average interest coverage ratio for the industry is 3.17

If LBM issues additional long-term debt, the current debt environment would allow 10-year bonds to be issued at par with an interest rate of 5.5%. To reach the average debt to total capitalization ratio of the industry, LBM could increase its long-term debt outstanding by \$550,000,000 (using the formula for long-term debt to total capital ratio: $[2,092 + 550] \div [2,865.96 + 2,092.10 + 550] = .48$

The additional annual debt service from the issue of bonds would equal: $.055 \times \$550,000,000 = \$30,250,000$

The existing debt service requirement \$214,700,000 + the anticipated debt service from newly issued bonds \$30,250,000 ($\$550,000,000 \times 5.5\%$) = \$244,950,000.

The new Interest Coverage Ratio: $763.37 \div 244.95 = 3.12$

The Interest Coverage Ratio is now in line with the industry average of 3.17

The Weighted Average Cost of Capital

The next step in the evaluation of LBM is to calculate the weighted average cost of capital (WACC). Considering each source of capital, it can be determined how much interest the company must pay for each dollar that it borrows.

Exhibit 6: *The Weighted Average Cost of Capital*

E = the value of the company's equity = 2,865

D = the value of the company's debt = 2,092

$V = E + D = 2,865 + 2,092 = 4,957$

$E \div V$ = the percentage of financing that's equity = .58

$D \div V$ = the percentage of financing that's debt = .42

R_e = the cost of equity = .085*

R_d = the cost of debt = .052

T_c = the corporate tax rate = .14

$WACC = [.58 \times .085] + [.42 \times .052 \times .86] =$
 $.0493 + .0188 = .06810 = 6.81\%$ Weighted Average Cost of Capital

In the event additional bonds were issued, bringing LBM closer to the industry average for long-term debt-to-total capitalization, the average cost of debt would increase slightly to 5.3%. The WACC would decline to 6.60%, due to the lower cost of capital for the debt. It's more attractive for LBM to issue debt than to issue equity at this time.

* The equity cost of capital was calculated using the capital asset pricing model, based on the following:

Expected return of the market = 8.7%

Risk-free rate = 3.4%

Beta = .96

The equity cost of capital = $[(.087 - .034) \times .96] + .034 = .085$ or 8.5%

Return on Invested Capital-to-Weighted Average Cost of Capital Spread

In and of itself, the ROIC is not an effective valuation tool; however, when compared to another ratio, such as WACC, it can be used to determine if the company is generating economic value. The ROIC to WACC spread is a measure of the efficiency and effectiveness of a company's use of capital. The calculated difference between ROIC and WACC is considered the economic value added (EVA).

$\text{ROIC-to-WACC Spread} = \text{ROIC} - \text{WACC}$
--

The ROIC to WACC spread determines if the firm is creating value, experiencing no value creation, or destroying value. If ROIC exceeds WACC economic value is being added, because returns exceed the cost of capital. If the ROIC to WACC spread is zero, then the company is breaking even on its capital costs, but not adding value. If the company's ROIC is less than its WACC, the company is destroying value.

For example, let's assume there are three companies and they each have approximately the same WACC of 8%.

Company A	Company B	Company C
ROIC – WACC = EVA 12% – 8% = 4% 4 cents of value per \$1 invested	ROIC – WACC = EVA 8% – 8% = 0% 0 cents of value per \$1 invested	ROIC – WACC = EVA 6% – 8% = -2% 2 cents of value lost per \$1 invested

Company A is generating 4 cents of value per \$1 invested because its ROIC exceeds its WACC by 4% (12%-8%). Company B is not generating any value, because its ROIC of 8% is just meeting its 8% cost of capital (8%-8%). Company C is destroying 2 cents of value per \$1 invested, because its ROIC is 2% less than its WACC (6%-8%).

EVA can be measured in terms of a firm's capital, using the following formula:

- $EVA = NOPAT - (WACC \times \text{Invested Capital})$

or

- $EVA = (ROIC - WACC) \times \text{Invested Capital}$

The market value added (MVA) is the present value of EVA. Therefore, the enterprise value of a firm can be estimated by calculating the expected EVA for a series of years, finding their present value and summing the results. Equity value can be calculated from enterprise value by subtracting the firm's debt and adding the firm's cash.

The following is a calculation to determine whether Lyons Bearings & Motors is adding value to the contributors of capital:

- $NOPAT = \$656.50$
- $WACC = 6.81\%$
- $\text{Invested Capital (as previously calculated)} = \$5,849.54$
- $EVA = NOPAT - (WACC \times \text{Invested Capital}) = \$656.50 - (0.0681 \times \$5,849.54) = \$258.14$

Lyons Bearings & Motors Co. has generated \$258.15 of EVA in the last 12 months.

Alternatively:

- $ROIC = 11.22\%$
- $EVA = (ROIC - WACC) \times \text{Invested Capital} = (0.1122 - 0.0681) \times \$5,849.54 = \$257.96$

Lyons Bearings & Motors Co. has generated \$257.96 or approximately \$258 of EVA in the last 12 months.

$\text{ROIC-to-WACC Ratio} = \frac{ROIC}{WACC}$

The ROIC-to-WACC ratio is ROIC divided by WACC. A ratio of greater than 1.0 is an indication that the company is creating value. A ratio of less than 1.0 is an indication that the company is destroying value.

For 2015, the ROIC-to-WACC ratio was $11.22\% \div 6.81\% = 1.65$. In 2015, LBM created value for its equity shareholders. This doesn't necessarily translate directly into appreciation of stock prices, since the price of the shares are based upon the expectation of future performance.

The ROIC-to-WACC ratio can be used to estimate the enterprise value (EV) of a company. The model assumes that the ratio of enterprise value to invested capital is equal to the ratio of ROIC-to-WACC.

$$\frac{EV}{IC} = \frac{ROIC}{WAC}$$

Therefore,

$$EV = IC \times \frac{ROIC}{WAC}$$

Enterprise Value = $5,849.54 \times 1.65 = 9,651.74$

Equity value can be calculated from enterprise value by subtracting long-term debt of 2,092.10, short-term notes of 355.50, and adding cash of 927.46.

Equity value = $9,651.74 - 2,092.10 - 355.50 + 927.46 = 8,131.60$

Price per share can be calculated by dividing equity value by shares outstanding.

$$\text{Price Per Share} = \frac{\text{Equity Value}}{\text{Shares Outstanding}}$$

Shares outstanding in millions (as of December 31, 2015) = 169

$8,131.60 \div 169 = \$48.12$ per share

From Exhibit 7, the closing price on December 31, 2015 was \$56.67. LBM closed at 118% of the estimated equity value per share ($\$56.67 \div \$48.12 = 1.18$ or 118%). This means LBM is trading at an 18% premium to what this valuation model would predict. LBM appears to be overvalued relative to this model. Other valuation methods may yield different results.

Exhibit 7: *The Machine-Tool Industry*

Company	Market Cap (\$ million)	P/E	Dividend Yield	Price (12/31/15)
Cromwell Corp.	26,681	17.5	1.8%	76.89
Brock Machine Co.	23,961	20.1	1.2%	77.22
Metric Tooling Co.	16,588	16.3	1.3%	65.85
Lyons Bearings & Motors	9,577	19.5	1.2%	56.67
Titan Trucks	7,175	16.6	1.8%	59.79
Polo Precision	6,911	24.4	1.3%	58.23
Teknetics, Inc.	2,449	23.1	1.4%	18.50
Hydraulic Brake, Inc.	2,212	14.8	2.3%	53.31
Hardwick Pump Co.	2,101	20.1	0.4%	58.35
Warfield Engineering	1,570	11.8	NA	18.28
Moore Engines Co.	1,536	21.8	1.6%	42.43
Global Manufacturing, Co.	919	20.2	2.1%	19.15

Earnings Per Share

Earnings per share data is shown in **Exhibit 3**

LBM had 169,600,000 shares outstanding on December 31, 2014. The company had 169,000,000 shares outstanding December 31, 2015.

The formula for simple earnings per share is:

$$\text{EPS} = \frac{(\text{Net Income [36]} - \text{Preferred Dividends})}{\text{Average Number of Common Shares Outstanding}}$$

For 2015: The average number of common shares outstanding = [shares outstanding for 2014 + shares outstanding for 2015] ÷ 2

EPS 2015: \$471,880,000 ÷ ([169,600,000 + 169,000,000] ÷ 2) = \$2.79

Dividend Yield The dividend yield percentage is calculated by dividing the annual dividend by the current market price.

$$\text{Dividend Yield Percentage} = \frac{\text{Annual Dividend (39)}}{\text{Current Market Price of Common Stock}}$$

LBM paid an annual dividend of .66 cents and the closing price is \$56.67.

$$.66 \div \$56.67 = .012 \text{ or } 1.2\%$$

Enterprise Value

Enterprise value is determined by calculating the market capitalization of the common and preferred stock, adding the value of all interest-bearing debt, and subtracting cash and equivalents.

In millions

Market capitalization of common stock: $169 \times \$56.67 =$	\$ 9,577.23
Short-term notes (10)	+ 355.50
Long-term debt (14)	+2,092.10
Cash and equivalents (1)	– 927.46
Enterprise value	\$11,097.37

The Effects of Additional Debt Issue

If LBM were to borrow additional capital based on a 5.5% interest rate, the WACC would decline to 6.60%. The use of this additional capital for plant modernization or acquisition of a company would ideally have at least the same effect on the revenue of the company as its existing operations. This has not always been the case in the past with LBM. Part of the company's present strength is based on management's decisions to sell off underperforming assets. This has enabled LBM to generate a higher than industry average return on invested capital.

What increase in value can LBM achieve if the debt issued can produce an increase in net sales consistent with the present performance of the company? There are several steps needed to calculate a new ROIC-to-WACC ratio based on this assumption.

- The total debt of LBM would increase by \$550 million, from \$2,447.60 million to \$2,997.60 million.
- LBM's debt as a percentage of sales was .2671 in 2015; therefore, the projected net sales (as a multiple of debt) will be $\$2,997.60 \text{ million} \div .2671 = \$11,222.76 \text{ million}$.
- LBM'S NOPAT as a percentage of sales in 2015 was .7164; therefore, projected NOPAT (as a percentage of sales) will be $\$11,222.76 \text{ million} \times .7164 = \804.00 million
- Invested capital would increase by \$550 million, from \$5,849.54 million to \$6,399.54 million.
- $\text{ROIC} = \text{NOPAT} \div \text{Invested capital}$: $\$804.00 \text{ million} \div \$6,399.54 \text{ million} = .1256$
- The WACC based on a \$550 million debt issue = .0660 or 6.60%
- $\text{ROIC-to-WACC ratio} = .1256 \div .0660 = 1.90$

A significant increase in the ROIC-to-WACC ratio may be obtained through an additional offering of debt. The competitive advantage period of the positive ROIC-to-WACC ratio (the period during which the ratio is positive) is influenced by the low interest rate environment in the example. This environment will not continue indefinitely. The question also remains whether LBM can achieve the same level of performance through capital expenditures as is currently produced.

A Possible Acquisition

LBM is considering a tender offer for Global Manufacturing. Global Manufacturing produces fire trucks and would complement LBM's drilling and pump equipment division. The company has 48,000,000 shares outstanding and it's likely that the shareholders of the company would accept a tender offer for \$22 per share. The \$1,056 million (48 million x \$22) could be raised through the issue of \$550 million debt and \$506 million FCF. The existing debt of Global Manufacturing is at a higher average interest cost than LBM. After the acquisition, LBM's weighted interest expense would increase to 5.60%.

What's the benefit of the acquisition? Global Manufacturing has a ROIC of 7.5% and a WACC of 8.1%. The company's ROIC-to-WACC ratio is .9259. This is far lower than LBM's present ratio of 1.65. How will LBM's ratio change after the acquisition?

Exhibit 8: The WACC after the Acquisition of Global Manufacturing

Global Manufacturing has \$1,048 million of total capital and a 50% debt to total capital ratio. The total debt would now include both companies' existing debt plus the \$550 million used to finance the acquisition.

WACC: $[E \div V] \times R_e + ([D \div V] \times R_d \times [1 - T_c])$

E = the value of equity of the company = 2,865 + 524 = 3,389

D = the value of the company's debt = 2,092 + 355.50 + 524 + 550 = 3,521.50

V = E + D = 3,389 + 3,521.50 = 6,910.50

E ÷ V = the percentage of financing that's equity = .49 (3,389 ÷ 6,910.50)

D ÷ V = the percentage of financing that's debt = .51 (3,521.5 ÷ 6,910.50)

R_e = the cost of equity = .085

R_d = the cost of debt = .056

T_c = the corporate tax rate = .14

WACC = $[\text{.49} \times \text{.085}] + [\text{.51} \times \text{.056}] \times \text{.86} = \text{.042} + \text{.025} = 6.62\%$ Weighted Average Cost of Capital

It should be recalled that the WACC of LBM was 6.81%. The acquisition of Global Manufacturing would reduce the WACC of LBM slightly.

The return on invested capital of the combined companies must be considered. The formula for ROIC as previously described:

ROIC =	$\frac{\text{Net Profits After Taxes}}{\text{Invested Capital}}$
Invested Capital =	$\text{Total Assets (9)} - \text{Non-Interest-Bearing Current Liabilities (11 + 12)} - \text{Excess Cash}$

In 2015, the net operating profits after taxes of Global Manufacturing were \$183,850,000. The companies invested capital was \$2,438 million.

The NOPAT of the company after acquisition would have been \$656.50 million (LBM) + \$183.85 million (Global) = \$840.35 million for 2015.

The new invested capital of the companies would be \$5,849.54 million + \$2,438 million = \$8,287.54 million.

The ROIC: $\$840.35 \text{ million} \div \$8,287.54 \text{ million} = .1014$

The ROIC for LBM in 2015 was .1122. The acquisition will negatively impact LMB. By what degree will the decline in the ROIC affect the ROIC-to-WACC ratio?

$.1014 \div .0662 = 1.53$. The ratio is virtually the same (1.51) without Global Manufacturing.

Is the Acquisition Worthwhile for LBM?

Recall the reason for the proposed acquisition of a company. The additional debt issued by LBM should have allowed the company to create value at a higher rate than had been the case in 2015. In this example, the use of free cash flow and the issue of debt illustrate that all growth is not good growth. Growth through acquisition is not the best plan for a company if the wrong acquisition is made. LBM has historically bought and sold smaller companies. The acquisitions have not always improved the financial status of the company. Because the possible purchase of Global Manufacturing doesn't significantly increase the ROIC-to-WACC ratio, it doesn't add value; therefore, it may not be the best use of LBM's capital.

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